

Math 110 The Fundamentals of Mathematics: Trigonometry

Numbered Transcript

MATH 110 · Dr. Robert Snellman

Contents

Lecture 1. Trigonometric Functions

Lecture 2. The Unit Circle

Lecture 3. Trigonometric Equations

Lecture 4. Complex Numbers

Lecture 5. Graphing in Trigonometry

Lecture 6. Inverse Functions

Lecture 7. Laws of Sines and Cosines

Lecture 8. Useful Applications

Lecture 1: Trigonometric Functions

At a glance — Snellman introduces trigonometric functions.

- 1.1 All right, so welcome to the Fundamentals of Mathematics, Trigonometry. My hope in this course is to convince you that trigonometry is relevant in the real world and that we can use it for all sorts of interesting applications and how we can model the world using these particular things. For example, one thing that you may or may not be aware of, when I think of trigonometry personally, I start thinking about angles, I start thinking about triangles and geometric shapes and things along that nature.
- 1.2 But one thing that you might not be aware of is that this is, trigonometry is the basis of things such as noise-canceling headphones, which is an interesting thing to think about. Now, why is that the case? Well, everything that we do creates a wave associated to that. Even the words coming out of my mouth right now, they're creating a particular wave. How do noise-canceling headphones work? You have a particular sound that creates a wave, and what you want to do is you want to cancel out the frequencies or the noises, if you will, associated to certain types of waves. You can do in a very precise way, but it all has its basis in trigonometry and trigonometric functions.
- 1.3 Let's start off just with the basics overall. For trigonometry, what we're going to have is, we're going to be interested in triangles. If I just give you an arbitrary triangle here, a triangle is simply something that just has three sides associated to it, connected by three straight lines. Associated to this, I've got angles. I've got an angle here. Typically, angles are denoted with Greek letters, so this one's just theta. Let's say this one is a different one. We can say that's phi, and then this one's something else. We can say that this is equal to psi, something along those lines.
- 1.4 We just give them names. We just give them names. It doesn't matter what you call them. You can call them A, B, and C, alpha, beta, gamma. It doesn't matter. We just have to create some language by which we can say, OK, this is what I'm talking about. These are the angles I'm talking about. Now, one of the tenets of trig in this sense is that the sum of the angles, so the sum of the angles must equal 180 degrees. So 180 degrees. Well, that's great.
- 1.5 Then it begs the question, well, what types of triangles are we going to study? Are we going to study just arbitrary triangles that look like this? Eventually, the answer will be yes. But what we want to start off with first is by making an assumption, we're going to assume one angle is 90 degrees. We're going to assume that one of our angles is 90 degrees. Why? Well, you have to start somewhere. You don't start learning to read English by opening a book on Shakespeare. You start with, what is the letter A? What is the letter B? What is the letter C? You have to start somewhere. Once you start there, then you build, and eventually you get to a place where you start reading Hamlet and you start reading things like that.

- 1.6 OK, so let's start off with right triangles. A right triangle is one where an angle is 90 degrees. One of its interior angles, one of its interior angles, the interior angles are the theta, the phi, the psi, things like that. One of the interior angles is 90 degrees. We can say, OK, so if I draw that, I can have something along those lines, something like that. Then to indicate the angle that is 90 degrees, what we do is we give it this little symbol, this little right angle symbol, if you will, to indicate that that's 90 degrees. This is the one here that has 90 degrees from there.
- 1.7 Now, the other two angles, they can be anything, but what we know is that because the sum of the angles all has to be 180 degrees, then I know that whatever this angle is, let's call it theta, whatever that angle is, immediately I know what this angle has to be. Why? Well, think about it. This one is 90, so I already have that 90 degrees plus whatever this one is, theta, plus, if I don't know what this is, I could just call it a variable X. This is a very common thing. In an algebra course or something along those lines, you learn about variables, and then you assign variables and solve for them accordingly.
- 1.8 If I didn't know what that angle measure was, I could say, well, that's equal to X, whatever that is. What do I know? Well, I know that the sum of the angles is 180 degrees. I know that 90 plus theta plus X is equal to 180. I can say, OK, what does that mean? I could subtract 90 on both sides, subtract 90. I get that theta plus X is equal to 180 minus 90 is 90. Then I can solve for X in this case. If I move theta over, so subtract theta on both sides, I get X is 90 minus theta.
- 1.9 What's nice about right triangles is you really just need to know one of the angle measures because you know that one of them is already 90. If I know theta, then I know what X is. X is simply 90 minus theta. When I add all of them up, 90 minus theta plus theta is 90, 90 plus 90 is 180, so I know what my angle measure is. Now, one of the probably most important theorems, if you will, associated to right triangles has been known for many, many years, is the Pythagorean theorem. We're going to start there.
- 1.10 What does the Pythagorean theorem say? Well, I could give you this statement that's completely fine, and that's great. I don't necessarily like that, personally. I have this philosophy that if you're learning a particular topic, you should be able to convince yourself that what I'm saying is true. I joke around and I say, "You should never believe anything that I say. You should just prove to yourself that what I'm saying is actually true." What I want to do is, I could just give you the Pythagorean theorem. That's fine. I just give you the statement and you just believe it and have— fine, that's great.
- 1.11 But what I want to do is I want to show you why it's actually true. Mathetically, we would call this a proof. We would say this is actually a mathematical proof as to why a certain statement is true. But let's try and just discover this organically on some level. Here's what I'm going to do. The proof of this is geometric in nature, and it's actually fairly intuitive, which is nice. The first thing I'm going to start with is the following. If I just give you a right triangle, just an arbitrary one, and I ask you, what is its area? What

is its area? In some sense, how do you think about area? You think about how much space does something take up?

- 1.12** If I give you a square, if I just start with a square, and I say that this length is L and this side length is L , then I would say that its area is length times width, in this case, which is L times L . I would say that its area is equal to L squared. Then I can pull that aside and say, OK, well, what if the side lengths aren't equal to each other? What if I have, the length is 2 inches and the width is 5 inches or something like that? How would I find the area associated to that? If I gave you a rectangle, let's say, here's my length and width, then I would say that my area is equal just to length times width.
- 1.13** What's nice about that is, I can then say, well, that's incredibly useful because if I start at this corner and I decide to cut this in half, I decide to cut this in half that way, then I can immediately say, well, here's a triangle. It happens to be a right triangle because these are rectangles. And so these are all 90-degree measures on the corners. And so then if I cut this in half, I can say, oh, well, then the area of the triangle that looks like this, here's the length, here's the width, should be what? Well, it should be the area of the rectangle just cut in half. That's what we did. So this should be one half the length times the width.
- 1.14** OK, so if I have that the area of a right triangle is half times the length times the width, or sometimes you'll hear people say half base times height, the base of the triangle times the height of the triangle, how would I use that to actually get at the Pythagorean theorem? That's the idea. Here's what we're going to do. I'm going to give you a square, and I'm going to declare that its side lengths are some measure, C . I'm going to give you a square, and I'm going to draw it tilted on its side. I'm going to do my best, but this is definitely a little tricky. Just bear with me.
- 1.15** The side lengths are going to be C , and it's a square all the way around. All of these are 90-degree measures. These are all 90 degrees. All right. Then I'm going to make another square on top of that in a clever way. The way I'm going to do it is the following. If I drop this down and then come across here and then come up and then over and then down. That's the idea. All right. Now, if I follow this, notice that all of these are giving me right triangles. These are all right angles. These are all right angles. OK, great.
- 1.16** Now, I'm going to declare that the sides of these have certain side lengths as well. I don't know what those are, but what I'm going to get is, if this one is equal to A , and this one is equal to B , then what I'm going to get is, let's see, this one, this will be a B , an A , B , A , this'll be B and A . The key is that the side length of this side is A plus B . It's this length A plus this B . What do I mean by A ? I mean that this length here is A , this length here is B . I get a side length of A plus B . I get a side length of A plus B here. Same thing here, same thing there.

- 1.17** Now what I want to do is, I want to try and compute the area of the big square. But I'm going to do it in two ways. I'm going to do it in two ways. Now, remember, the area of a triangle here, the area of our triangle is half base height. Half times base times height. Let's see what we can do. The area of a square is just the side length squared. What's going to happen? The area of the large square is what? The area of the large square, what is that going to be? Well, this side length is $A + B$. According to my formula, the area should be whatever the side length is and then just square that. If the side length is $A + B$, I square that. The area of the large square is equal to $A + B$, the whole thing squared. OK?
- 1.18** But then you tell this to somebody and they're like, well, I could give this to my 6.5-year-old son in terms of shapes. I could cut out a little shape that's this triangle, a little shape that's this triangle, the same shape triangle and a square, and I could arrange them in a square and I could say, "Rocco, what's the area of this thing?" He'd be like, "Well, I could just find the area of this one, and then the area of this one, and the area of this one, this one, and this one, and I could just add those up." Do we agree? Because those, what is it, one, two, three, four, five shapes, make up the entirety of the big square.
- 1.19** I can compute the area by saying, well, the side of this is $A + B$, the side of this is $A + B$. I have an area for that. But then I could say, well, what's the area of each individual piece? What's the area of this triangle here? Well, its base is A , its height is B . So according to my area, this should be a one-half A times B . And then this one is, OK, one-half base B , height A . This is, again, a one-half A, B . This one, same thing, base A , height B . So this is a one-half A, B . Same thing here. This is a one-half A, B . That's great. I have four one-half A, B s. Then I also have the square in the middle. That's a C -squared.
- 1.20** Alternatively, I could say, well, look, the area of the large square, on the one hand, it's this, but the other hand, what is it? Well, it's four copies of half A, B . I'm going to have four copies of one-half A, B , plus the square in the middle. I get plus C squared. We step back and we're like, OK, we're lost in the weeds a little bit. It's like, wait a second. You were telling me you were going to give me this Pythagorean theorem, and then you're going to justify why it's true, and now you're just adding triangles and squares together. What the heck is, what's the point?
- 1.21** Well, the point is, now let's just do the algebra and see what it gives us. Let's see what it gives us. If I work out the algebra on the left, what do we have? 4 times one-half A, B , plus C squared, is equal to this thing on the right-hand side. I get $A + B$ squared. What do I do? The 4 comes in. This is all multiplication. So 4 times one-half, this is 4 over 2. A, B plus C squared. What is this? 4 divided by 2 is 2. So this is $2AB$ plus C squared. But now I have to work out the right-hand side. Now, the right-hand side, remember, we have this distributive law. I have an $A + B$. What does it mean to square something? It's really that number times itself.

- 1.22 So whatever $A + B$ is, if $A + B$ were 5, then $A + B$ squared would be 5 times 5, which is 25. That's great. But we don't know what $A + B$ is. It's just symbols. It's just algebraic manipulation here. How do I actually do this? I've got the distributive law. It says start with A , move it over, multiply by A . So A times A is A squared. Then take this A , move it into B . A times B is A, B . Take the B , multiply into A . B times A is the same as A times B because these are just real numbers. I'm just going to write this as A, B . And B times B gives me B squared. I end up getting that. OK.
- 1.23 Now, I can combine the two. AB plus AB , that's just two copies of AB . I get a $2AB$. This becomes A squared plus $2AB$ plus B squared. What's on the left-hand side? That's $2AB$ plus C squared. Can I simplify anything further? Well, what I notice is that $2AB$ is common on the left and it's on the right. So what happens if I subtract $2AB$ on both sides? Those cancel. I'm left with, let me reverse the order here. I'm left with what? I'm left with the Pythagorean theorem, which states that A squared plus B squared is equal to C squared.
- 1.24 That's the point. That's what the Pythagorean theorem says. It gives us a relationship for right triangles that says the following. Look, if I come back and say, all right, let's look at these right triangles. This is a perfect example, this guy right here. If I have this side length of this triangle is B , and the side length of the triangle on this side is A , this side length, whatever that is, I call it C , it says that the relationship between A , B , and C in a right triangle satisfies this relationship, that A squared plus B squared is equal to C squared. That's the Pythagorean theorem. That's been known for ages. Super useful. We're going to take advantage of this a ton in this course. But that's the starting point.
- 1.25 Let's just do some silly examples. Let's do some silly examples. I like to, whenever you get a new result, whatever that is, so in this case, the Pythagorean theorem, I like to illustrate this with just simple examples to try and make things a little more concrete overall. I used to be a firm believer that if you understood the theory of something, then you would understand the applications of it. I realized that I was incredibly naive and that that's not true. I think, as an example of that, you can have a complete understanding of how to walk, for example. But that doesn't mean that you know how to walk.
- 1.26 You could have a complete understanding of how to run a marathon. That doesn't mean that you can go and run a marathon, if you're not trained for a marathon or something like that. You could understand, oh, I'm going to run 26.2 miles. That's fantastic. You would have a complete theoretical understanding of what the Boston Marathon looks like. But until you actually put on your shoes and run, you have no idea what it's like. Does that make sense? I can have a theoretical understanding about the Pythagorean theorem. That's great. But do I actually know how to use it? Do I actually know how to apply it? And things like that.
- 1.27 Let me give you, let's just do some examples. If I gave you a right triangle, and I tell you that this side is 3 and this side is 4, and I ask you the question, what is X ? I'm going to

tell you X is the other side of it. Now, these things have names associated to them. I'm calling these things the sides of a triangle, the 3, the 4. I'm calling this guy the third side of the triangle. It's got a name, and we'll get that language in just a minute. But for now, we can simply say, OK, if this side is 3, if this side is 4, what's that length? Why would this be useful? Maybe it's not. Maybe it's not.

- 1.28** This is the age-old question. When am I going to use mathematics? When am I going to use this? Maybe the answer is never. But what's the the historical context of this? Why is this useful? Well, this really came about from a standpoint of geographical subdivision, more or less. If I have a particular plot of land and I want to subdivide it in a particular way, can I do that? How would I do that? How would I do that? This is like, to some extent, this is the basis of, I guess, what we would call in modern day, surveying, in terms of trying to map out a particular geographical location.
- 1.29** Why would I want to know what this length is? Well, maybe I already have, this side is 3 miles, this side is 4 miles, and I want to know how far is it going to be to connect this point to this point in a straight line, all right? What could I do? I could say, well, Pythagorean theorem tells me what I should do. It says if I have the side lengths of my triangle, then they're related by this. I have to remember, OK, the side that is opposite my 90 degree, the side that is opposite my 90 degree, that was my C . That was C here. That was this guy.
- 1.30** That means that X squared is equal to what? Well, I've got this side. That's my 3. That's my 4. That's going to be my A and B . That should be equal to 3 squared plus 4 squared. I've got my 9. I've got my 16. 9 plus 16 is what? That's 25. Now I'm asking myself the question, what solves that? Now, you have to be a little bit careful here because if I'm asking myself the question, what solves this equation? In the algebra class, we talk about how to solve quadratic equations. I could say, well, X is plus or minus 5. 5 squared is 25 and negative 5 squared is also 25. But I have to put it in the right context. What does negative 5 mean?
- 1.31** Well, this is talking about a length of a side of a triangle, for goodness' sakes. What does it mean to say negative 5? That doesn't make any sense. I'm going to say that my X is equal to 5. That tells me that the side of this triangle, X equals 5, satisfies that. Satisfies that by the Pythagorean theorem. That's great. This is what's called, people will typically call this a 3, 4, 5 triangle. And that's one of—so if you hear that language, oh, it's a 3, 4, 5 triangle, they mean it's a right triangle with side lengths 3, 4, and 5, satisfying the Pythagorean theorem.
- 1.32** All right, what if I wanted a different side, though? What if I didn't want the side directly across from the 90? What if instead I had said something like this? What if I had said, OK, I'm going to give you two other sides. And this is also another very common right triangle that you'll see quite a bit. What about this one? Here's another example. Instead of giving you these two sides, I'm going to give you this side and this side. I'm going to say, what's the other one? What's the other side?

- 1.33 Again, let's call this, maybe we should give it another letter. Let's call it Y . The question is, find Y . We look at this, we say, OK, well, Pythagorean theorem still holds. This is a right triangle. By the Pythagorean theorem—that's my abbreviation for solution, by the way, so SOL upper N is just saying this is my solution. Then what do I get? By Pythagorean theorem, Pythagorean theorem, we get what? Side length 5 squared plus other side length Y , which is what we don't know, squared, is equal to the side across from the 90-degree angle, 13 squared. Then the hope is maybe we can solve for Y .
- 1.34 Or remember in algebra, the beautiful thing about the Pythagorean theorem is that it's always talking about quadratics. As we talked about in algebra, the quadratic formula gives me a concrete formula to actually solve for these. It's like, I can always solve for this, which is a beautiful thing. What do I get? I get 25 plus Y squared is equal to 13 squared. What is that? How do you figure out 13 squared? You have 13, 13, multiplication-wise, this is 9, 3 times 1 is 3, 0, 3, 1, 169. OK, so that's 169. However you do arithmetic, it's totally fine.
- 1.35 What do I get here? I subtract off my 25 on both sides, minus 25. I end up getting Y squared is equal to—OK, how do you do this? Now, again, however you do arithmetic, I do arithmetic really weird, but that's how my brain works. If it works, it works. The way I look at this is, I say, OK, I'm going to take 169, I'm going to subtract away 20. That's only going to take away the 6 down to a 4. Now I have 149, and then that 9, I'm going to take away a 5 from it. I'm going to get 144, whatever. Whatever works for you. I don't care how you do arithmetic. That's completely fine.
- 1.36 But I get Y is equal to 12. That tells me I have this side is equal to 12. This is another special right triangle. It's called a 5, 12, 13. So this is a 5, 12, 13 triangle. That's what we have for right triangles and the Pythagorean theorem. All right. One of the most probably classical theorems in all of mathematics. What's interesting is that it shows up everywhere. It shows up everywhere. Classically, it comes from geometry, but surprisingly, it shows up in all sorts of branches of mathematics and analysis and statistics and in various things like that. OK.
- 1.37 One thing that's very important is we need to decide on, what are we going to talk about in terms of our angle measures? Are we going to talk about degrees? Or is there another way that we can talk about the angle measure? I need to be able to convert from units from one to the other, from degrees to whatever this other thing is. To motivate that, it makes sense. Because, for example, in the US, we measure things with miles or feet or inches or things along those lines, which is what it is. That's great. Whereas in other places, you measure things with meters or kilometers or things along those lines, which is great. I personally like that a little bit more, but that's fine.
- 1.38 But the question is, well, if I give you 1 mile, how do I convert that into kilometers? Or if I give you 12 inches, how do I convert that into centimeters or things along those lines? These are called unit conversions. How do we convert different units? Well, we need a way of interpreting those particular units in terms of the other units. I need to be able

to say 1 kilometer is a certain number of miles, or 1 mile is a certain number of kilometers, or things along those lines. Let's do it from a time perspective. Time unit conversions. Time unit conversions.

- 1.39** What do we know? Let's suppose I want to understand how many seconds are in, let's say, 2 hours. How many seconds are in 2 hours? Or scratch that. Let's say, how many seconds are in a day? We have 24 hours in a day. We have 24 hours in a day. The motivating question is, how many seconds in one day? How could I do this? Well, one way to do this, which is not one that I actually recommend, is that you can literally sit there all day long and you can just say, oh, OK, I'm going to count every single second that goes by and I'm just going to count them. You start at 12 AM and you start counting 1, 2, 3, 4, and then you get through your first minute, you get bored, and you realize that this is a complete waste of time, and there's a better way to do this.
- 1.40** How could I do this? I could say, OK, well, I know that in one day, I know that one day has 24 hours. OK. One day has 24 hours. Then I can say, OK, well, if I know that there's 24 hours, can I subdivide hours even further? Then I say, OK, well, I know that one hour has 60 minutes. OK. I can subdivide hours into minutes, and then I can take minutes and I can subdivide those into seconds. I can say that one minute has 60 seconds. It's got 60 seconds. That's great because then I can say, oh, now I have some chained relationship between a day, hours, minutes to seconds. I have some connection between the objects at hand. That's the idea.
- 1.41** Then how can I do this? I can say, well, one day, in one day, how many hours are in a day? This is how I like to think about it, because when we think about fractions, when we think about fractions, over here, when I was doing the proof of the Pythagorean theorem, I had something that was like 4 times one-half. We said 4 times one-half is really 4 over 1 times one-half. How do fractions multiply? I get 4 over 2. Then I was able to simplify. I was able to simplify by writing 4 as 2 times 2 over 2, and I can cancel one of the 2s, so I ended up getting a 2. That's how that works.
- 1.42** The same thing holds, but with our units in this case. What do I mean by that? Well, one day, how many hours are in a day? Well, I have 24 hours per one day. I'm going to say 24 hours in one day and think of the units as being fractions. What I can say now is this one day, like this, the days cancel, and what do I have? I have 1 times 24 hours. So one day is equal to 24 hours. But now I can do the same thing with minutes. I can say, OK, with minutes, what do I know? I know there are 60 minutes per one hour. I'm going to have 60 minutes per one hour, and I notice that hours now cancel.
- 1.43** What does that mean? That means that I have one day has 24 times 60 minutes. 24 times 60 minutes. Now, I'm almost there. I know that I have 60 seconds in one minute. That means I have 60 seconds per one minute, and the minutes cancel. Then I look at the units. What are the units of this particular expression? The units are, OK, days canceled, hours canceled, minutes canceled. This is in seconds. Then it says that one

day is what? 24 times 60 times 60 seconds. Whatever that is. Whatever that is. The principle I'm trying to illustrate is how you can do unit conversions from this.

- 1.44** Now, if you knew how many kilometers are in a mile, I think it's like 0.6 kilometers in a mile or something like that, then you can say, oh, if I want to run a marathon, that's 26.2 miles, how many kilometers is that? You can do the unit conversion on that and so forth. The reason why I mentioned that is because there's another way to measure angles, and that's in what's called radians. That's in what's called radians. Here we go. An alternate, alternate units for angles. These are called radians. These are radians.
- 1.45** The conversion between the two is the following. We say that there are π radians in 180 degrees. This little upper circle in the right-hand side, I'm not saying 180 to the 0th power. It's just an indication for degrees. It's an angle measure. This thing here really means 180 degrees. That's what it means. One way to think about 180 degrees, well, you can think about it as the sum of the interior of a triangle. They have to add up to 180 degrees. That was the first thing that we talked about. You could also think about it as saying, if I give you a line and I say, what is the angle measure from here to here? That's 180 degrees. I just give you a straight line.
- 1.46** That makes sense because if I think about, I go from here to here and then from here to here, I would say, well, I did a 360. I did a 360-degree rotation if I were to just spin around on the spot. If I have 180 degrees here and I have 180 degrees here, I have a total of 360 degrees. I just went completely around. But you have this conversion for radians in terms of degrees. So now what we could say, we could say things like, OK, we know that the sum of the interior angles of a triangle is 180 degrees. Therefore, I could say the same equivalent statement of saying the sum of the interior angles of a triangle is π radians. Because 180 degrees is π radians.
- 1.47** I could think about it as either in degrees or in radians. I could say, if I want to do a complete 360, what am I really doing? Well, I'm doing π radians and then π radians. I would say that 360 degrees is 2π radians. But you never hear this. Let's say you watch the Winter Olympics or whatever it is, and some snowboarder is doing some crazy acrobatic thing and they pull a 1080. They're talking about degrees. They're not saying, what are they saying? They did a 1080, they went around 3 times 360s. If 360s, if 360 is two π radians, you don't hear the announcer saying, "Whoa, they threw a 6π radian." They don't say that. They say they did a 1080 or something like that.
- 1.48** But you can talk about these in terms of two different things. I would say typically the more standard mathematically is to think about things in terms of radians as opposed to degrees. We're going to bounce back and forth between these particular objects. Let's talk about the trigonometric functions, how they arise, and some of the language associated to right triangles. Let's talk about those. All right. Here we go. Here's the language of, let's say right triangles.

- 1.49** Again, I give you a right triangle. You're going to see that my right triangles typically always have this shape. I always seem to draw them that way. I don't know why, but that's the case. Now, whenever you have an angle, whenever you have an angle and you refer to the opposite side of that angle, the opposite side of that angle, what I'm referring to is, if I fix this angle, whatever it is, and I say the opposite side, I mean, if you imagine just taking a line to this side here, this is called the opposite side of theta. That's the opposite side of theta. Typically, we're just going to denote this as OPP for the opposite, just to abbreviate it.
- 1.50** This side here, this side here, it's the side that's right next to theta. This is called the adjacent side of theta, typically just denoted ADJ. That's the adjacent side. But once you fix an angle, once you fix an angle, this is a very important thing, you have to fix the angle in order for opposite and adjacent to actually make sense. For example, if I had fixed this angle up here, when I say opposite, what do I mean? Well, I mean this side. When I say adjacent, what do I mean? I mean this side because it's in relation to the fixed angle. Once you fix the angle, opposite and adjacent makes sense.
- 1.51** The only angle for which opposite is in some sense not ambiguous is the 90 degree. The 90 degree is the only one because the 90 degree is already fixed. It's already fixed. If I talk about, what is the opposite side of the 90? This has a name and it's called the hypotenuse. It's the hypotenuse. That one's fixed. That one's fixed because the 90 degree is automatically part of the triangle. It's whichever one is denoted by this little box symbol. If I look at this particular triangle, I'm just going to draw the same triangle. Here's theta. Here is the opposite. Here is the adjacent, and here is the hypotenuse.
- 1.52** The trigonometric functions come from ratios of these particular sides. That's where they come from. That's how they're defined. Here are the trigonometric functions. Here are the trigonometric functions. OK. Trigonometric functions. OK. The first one I'm going to give you is the sine of the angle. This one is sine. Here's its English spelling. It's not the S-I-G-N in terms of the sign as a sign on a billboard or something like that. It's S-I-N-E, and its notation is sin, S-I-N, of whatever angle you're talking about. Again, you have to fix an angle. I'm going to fix this angle theta. I take the sine of theta. That's how that's read. What is this equal to? It's equal to the opposite side of the triangle divided by the hypotenuse.
- 1.53** So trigonometric functions are ratios. They're just ratios of sides of triangles with respect to other things. I have the sine of theta is equal to the opposite divided by the hypotenuse. That's what we get there. The cosine, this isn't like you're taking out a loan and somebody cosigns that loan with you. This is very different from that. This is not that. The cosine is cos of theta. So cosine of theta. And what is that? That is simply the adjacent divided by the hypotenuse. So this is the adjacent divided by the hypotenuse.
- 1.54** And then the last one. The last one is the tangent. Last one is a tangent, so that's T-A-N of theta. And what is that one? That is the opposite divided by the adjacent. That's the

opposite divided by the adjacent. Now, if you stare at this for a little while, some natural questions come up. The questions are, OK, if we're simply just talking about ratios, opposite over adjacent, opposite over hypotenuse, adjacent over hypotenuse, things like that, well, why am I not doing things that are like hypotenuse over opposite, hypotenuse over adjacent, or adjacent over opposite?

1.55 After all, those are all the possible ratios I can have. I can have opposite over hypotenuse. I could also have hypotenuse over opposite. If I have 2 divided by 3, then I could have 3 divided by 2. If I have adjacent divided by hypotenuse, let's say that's 1 divided by 4, then I could have, well, hypotenuse divided by adjacent, 4 divided by 1. Then the question is, what are the other three? What are the other three? What is the one that says, for example, if I go hypotenuse over opposite, what is that one? That one is called the cosecant. So that's C-S-C of theta. Cosecant is hypotenuse over the opposite, otherwise known as 1 divided by sine.

1.56 So it depends on who you ask. It depends on who you ask. Sometimes people say the trigonometric functions are simply the sine, cosine, and tangent. Those are the only ones, those three. And then the other three, you don't call them trigonometric functions, you call them the reciprocals of the original. So like 1 over sine, 1 over cosine, 1 over tangent. I personally just err on the side of saying the six trigonometric functions are the six things I'm going to give you now. I don't delineate between saying that, oh, cosecant doesn't qualify as a trig function because it's 1 over the trig function sine. I'm just going to call them all trig functions.

1.57 Similarly, I get the secant function. That's S-E-C of theta. What is secant? Well, that's going to be 1 over cosine, so that'll be the hypotenuse over the adjacent, and that's 1 divided by cosine. Then the last one, what have I done? I've taken cosecant, that was 1 over sine. I've taken secant, that's 1 over cosine. Then the only other one on the left is 1 over tangent. These ones have been cosecant, secant. The last one is cotangent. That's C-O-T of theta. That's what? That's going to be, OK, 1 over tangent will be, flip this, so this will be adjacent over the opposite. That's 1 over the tangent of theta. That's what I'm going to call the 6 trigonometric functions. The six trigonometric functions.

1.58 Now, what's nice is that there's a little mnemonic to remember those three trig functions, sine, cosine, and tangent. Because how do you remember this? That sine is the opposite over the hypotenuse, or that cosine is the adjacent divided by the hypotenuse, or tangent is opposite over adjacent. And that's this, let me write it this way. There's SOH-CAH-TOA. So, SOHCAHTOA, you'll hear people say. What does that mean? It means that the sine, the sine of an angle, is what? The opposite. It's the opposite over the hypotenuse. So that's the SOH. Sine is opposite over hypotenuse.

1.59 CAH, you can guess what that's going to be. The C is going to stand for cosine. That's cosine. What is cosine? Cosine is the adjacent over the hypotenuse. This is going to be the adjacent over the hypotenuse. Then TOA, T is going to stand for my tangent. So tangent is equal to the opposite over the adjacent. So SOHCAHTOA is the way that you

remember these. So that helps me to understand what is sine, cosine, and tangent. But then the question is like, OK, well, that's great. That's only half of the picture. What are the other three? How do I remember those?

1.60 I have a very ridiculous way of remembering this. So if this works for you, that's great. If it doesn't, that's fine. Find a way that works for you. But for me, this is how I remember it. So don't laugh at me. This is embarrassing. How do I remember cosecant? Well, the first thing I have to remember is the cosecant, the secant, and the cotangent are all reciprocals of things. So they're one over. The cotangent is very easy because it says, OK, the cotangent, tangent is in the name. And so what is cotangent? It's going to be 1 divided by tangent. So that one's easy to remember, in some way.

1.61 How do I delineate between cosecant and secant? It's like, oh, shoot, I don't know. Cosecant is either 1 over sine or cosine. I don't remember. Secant is 1 over sine or cosine. I don't remember. Here's how I remember. Cosecant starts with C, which means that it's going to be 1 over the trig function that doesn't start with C. It doesn't make any sense, but for whatever reason it sticks in my brain. So cosecant, it starts with C. It's going to be 1 over the trig function that doesn't start with C, 1 over sine. Versus secant starts with S, so it's going to be 1 over the thing that doesn't start with S. So it's going to be 1 over the cosine.

1.62 I don't know why that works for me. If it works for you, that's great. But that's how I remember this. So cosecant and secant, cosecant is 1 over sine, secant is 1 over cosine. What we can do now is just to get some examples on how to actually play around with these things. Again, we have our mnemonic for sine, cosine, tangent, and then now we have our ways of remembering cosecant, secant, and cotangent. Let's suppose I just give you a right triangle and ask you to fill in all of the angles. Or not the angles, the sides, the ratios. Here's a silly example just to give us a sense of computing sines, cosines, and tangents.

1.63 Again, let's just take a 3, 4, 5 triangle. That's our basis. It's a Pythagorean triple. Let's say this is a 3, 4, 5. I want you to find all six trig functions of theta, associated to theta. Remember, trig functions, it doesn't make sense to say, just find the trig functions. It's like, they don't exist in and of themselves. They exist once you fix an angle. Once you fix an angle, then it makes sense to say, oh, OK, that's the angle. Find sine, cosine, tangent, secant, cosecant, and cotangent of that angle. That makes sense.

1.64 We'll talk about the sine, cosine, tangent, and cosecant, secant, and cotangent as functions, just abstractly, what they are, independent of triangles. Independent of triangles. We'll talk about that, but that'll come later. OK. Let's see, what do I have here? I have to identify, OK, once I fix theta, again, theta is fixed, so now it makes sense to talk about opposite, adjacent, hypotenuse. If you want, you could say, OK, what is the opposite? The opposite is going to be, that's 4. What is the adjacent? The adjacent is this guy, so that's 3. And what is the hypotenuse? Well, the hypotenuse is always whatever's across from 90, so that's 5.

- 1.65** Now, I need to go through. If I have these memorized, that's great. Otherwise, I can use my mnemonic, the SOHCAHTOAH. SOHCAHTOAH, I could say, all right, the sine of theta. Again, SOH, S-O-H, that's equal to O over H, opposite over, this is opposite over hypotenuse. The opposite I've declared is 4, and the hypotenuse we've declared is 5. OK, so sine is done. That's great. What is cosine? The cosine is the adjacent over the hypotenuse. OK, so adjacent we've declared is 3. That's going to be my 3 over the hypotenuse, which is 5. That's good.
- 1.66** Then the tangent, that's the opposite over the adjacent. That's the TOA part of it, T-O-A. So tangent is opposite over adjacent. The opposite is 4, and the adjacent is 3. So this is 4 over 3. But now I'm golden because what do I have? Well, I have sine, cosine, tangent. Well, how do I get cosecant, secant, and cotangent? Cosecant is 1 over sine. So what does that mean? All I need to do is take 1 over sine. How do you do 1 over A divided by B? How do we actually do that algebraically? Well, you think about this as the numerator as being 1 over 1 divided by A over B.
- 1.67** And what is that? You flip the denominator, so you take its reciprocal, so that's B over A, and you multiply by that, which is B over A. You literally just take the fraction and flip it at the end of the day. That's really all you're doing. I get sine, cosine, tangent. What is cosecant? Cosecant of theta, the definition of cosecant is 1 over sine, so I find sine and I flip it. That's 5 over 4. What is secant? Secant is 1 over cosine. Cosine is three-fifths, so that's going to be five-thirds. And cotangent is 1 over the tangent, so that's equal to 3 over 4. There's the sine, the cosine, the tangent, and so on. Is that making sense, how we can find those corresponding ratios?
- 1.68** Let me give you the following example. I want to ask, I want to tie together the Pythagorean theorem and the corresponding sine, cosine, tangent. Let me give you the following example. All right. Example number one. I'm going to give you a right triangle, but I'm not going to tell you what the side lengths are. I'm not going to tell you what the side lengths are, except for two of them. Then I'm going to specify an angle and ask you to find all of the trig functions associated to that angle. It's a two-part problem.
- 1.69** I'm going to say, given the triangle that looks like this, let's say, this is 7, this is, let's say, 10. Let's just give this a name. Let's call that X. The first part is going to be, find X. The second part is, I'm going to specify an angle, theta, and I'm going to say, find all six trig functions, functions of theta. That's the goal. I'm going to find all the trig functions of theta, and I'm going to find X. How do I solve this? How do I solve this?
- 1.70** Well, the first thing is to identify, well, OK, I have an angle theta. I don't know what the angle is. Frankly, it doesn't matter what the angle is. It doesn't matter if the angle is 30 degrees. It doesn't matter if it's 45 degrees. It doesn't matter if it's, whatever it is. I just denote it by theta. I can simply say, just based on this being a variable, I could say, oh, sine is what? It's the opposite over this. I could say sine of theta is X over 10. Great. Cosine of theta is what? 7 over 10. Tangent of theta is what? Opposite over adjacent.

So tangent is X over 7. I could just walk through and compute all those things. Sine, cosine, tangent, secant, cosecant, cotangent. That's fine.

- 1.71** But if I don't know X , then, well, what am I actually doing? I need to know X first. How do I find X ? My solution for number one is my Pythagorean theorem. Pythagorean theorem says, look, this is a right triangle, so I know what? 7 squared plus X squared is equal to 10 squared. I know 7 squared plus X squared is equal to 10 squared. That's my Pythagorean theorem. That's great. 7 squared is 49. X squared is, I don't know. That's what I'm trying to figure out. 10 squared is 100. I can subtract 49. I get X squared is equal to, what is this, 51. That tells me that X is equal to the square root of 51. Goodness knows what that is. I have no idea.
- 1.72** What is that? 17 times 3, something like that. It doesn't look like I can take any numbers out of this. There's no squares in it. It's 17 times 3. Those are both primes. There's nothing I can do on that front. The square root of 51, it is what it is. Now, I can go use some calculator to figure out what it is as a number. It's something in between, how would I figure what this is roughly, start squaring. Here's a trick. Start squaring numbers and see what you get. If I start, here's a little trick for computing, for getting a sense as to how big a number is.
- 1.73** I see a lot of students who do the following. They fall into this, maybe trap is too strong of a word, but they fall into this pattern, if you will, of saying once they have something like this, they have to immediately have a calculator. That's fine. There's nothing wrong with that. But if I don't have a calculator, I don't carry a calculator. I'm not just going to pull out of my wallet like, oh, let me get my calculator and figure this out. No, that's not a thing. I don't carry one around with me. Well, that's a lie. I have my phone on me which has a calculator, so that's fair.
- 1.74** But X is equal to square of 51. How do you actually figure this out? Well, look, what this is really coming from is, it's coming from this. X squared is equal to 51. Just start taking numbers and squaring them. Just start taking numbers and squaring them. 1 squared is 1. OK, that's great. 2 squared is 4, 3 squared is 9, 4 squared is 16, 5 squared is 25, 6 squared is 36, 7 squared is 49, 8 squared is 64, and I can stop. I can stop. Why? Well, because look, somewhere in between 7 squared and 8 squared, well, 7 squared is 49, 8 squared is 64. So somewhere in between 7 and 8, there has to be a 51. There has to be a 51.
- 1.75** So what do I know about the square root of 51? I know that it's somewhere in between 7 and 8. And I notice that 7 squared is relatively close to it. So I would probably guess that the square root of 51 is 7.1 or 7.2, somewhere in that range. Now I have some sense in my head as to, oh, 51 is around 7 point something. It's probably not 7.9 because 7.9 is close to 8, and 8 squared is 64. It's probably closer to the low end of 7.1, 7.2. I don't know. But we can have a calculator to check that out. But at the end of the day, I know that this is now the square root of 51. Great.

- 1.76** Now that I have all the sides, I can just cruise through and find the sine, cosine, tangent, secant, cosecant, cotangent. The solution for number 2 is that if I have the sine of theta, so remember our mnemonic, sine of theta is opposite over hypotenuse. So sine of theta should be what? Sine is the opposite over hypotenuse, so that's the square root of 51 over 10. And what I'm going to do is I'm going to just take its reciprocal and write it right next to it, so that I can just keep it straight.
- 1.77** Remember, sine, OK, what's the one that's a reciprocal? It's the one that doesn't start with S, so that's cosecant. So that's cosecant of theta. That's 1 over sine, so that's 10 over the square root of 51. If I look at cosine, cosine is the adjacent, 7 over the hypotenuse, so that's 7 over 10. Therefore, the one that doesn't start with C, so secant, secant of theta will be 10 over 7. Then tangent, that's my opposite over adjacent, so the opposite is the square root of 51 over the adjacent 7. Then the cotangent of theta is equal to 7 over the square root of 51.
- 1.78** There we have it. That takes care of the example. Just to recap, Pythagorean theorem, ancient theorem, super useful for right triangles, gives a relationship between the sides, A, B, and C, $A^2 + B^2 = C^2$. The trigonometric functions coming from ratios of sides of right triangles, but we're just getting started. What comes next is, we want to try and understand sine, cosine, tangent as functions themselves. We want to try and understand, what do their graphs look like? Can I invert them? Can I do anything along those lines? Are there other theorems and other laws that are associated to these particular things? The answer is yes, and we're on our way.

Lecture 2: The Unit Circle

At a glance — The unit circle.

- 2.1 OK, so now that we have the basics of Pythagorean theorem, which gives us a relationship between the sides of the triangle, $A^2 + B^2 = C^2$, and the six trigonometric functions, sine, cosine, tangent, secant, cosecant, cotangent, we want to try and understand, what are some of the relationships that we have between these objects? What we do today is, we give a little bit of, I guess, an introduction, if you will, into the relationship between sine, cosine, tangent from triangles, as the ratios that we have there, and circles. That's the goal for today.
- 2.2 Let me just remind you, we have the SOHCAHTOA, and that was telling us that our sine, so let's give ourselves a corresponding triangle over here, so it's a right triangle. I have the opposite, I have the adjacent, and I have the hypotenuse. Again, remember, the opposite adjacent don't make sense unless you fixed a particular angle. If I look at the sine of θ , that's going to be what? That's the SOH part. So sine is opposite over hypotenuse. And the cosine of θ is equal to the adjacent. So that's the AH. So that's the adjacent over the hypotenuse.
- 2.3 So the question I want to ask, and this is actually just a general, I don't know, a general information, is— a question I get asked a lot is, how is mathematics done? How is math research done and all of that? I think there's this misunderstanding that math is done where you just have a mathematician sitting in a room with a pen and a pad, and they're just sitting there thinking about like, "Oh, this is the next brilliant idea that comes into my mind." To some extent, that is true. There is some aspect of that, but I would be willing to argue that that's a very small percentage about how math is done.
- 2.4 Math is very much done much like science is done. I have a question. I want to explore that question. I'm going to run some experiments to see what data I can generate based on that question. Then based on that data, I want to see if I can actually prove that something is correct or not. That's the idea. One thing that we can do is, we can say, all right, well, what happens if I take—what's the relationship between trigonometric functions and a Pythagorean theorem, if there is one? If there is one? After all, it's a natural thing to ask, the sine, cosine, and tangent are talking about the ratios of sides of a triangle. OK, that's fine. And I know what?
- 2.5 Well, Pythagorean theorem tells me something about, well, what? It tells me that if I take the adjacent side and I square it, and I add that to the opposite side and I square it, then I should get the hypotenuse squared. That's what the Pythagorean theorem says. I can ask, the motivating question is, is there a relationship between the trig functions and the Pythagorean theorem? The answer is yes. How do I see that? Well, a natural place to start would be to say, well, look, these things are being squared. I'm

taking the opposite side squared, I'm taking the adjacent side squared, I'm taking the hypotenuse squared, and there's a relationship between them.

- 2.6 Well, sine and cosine, they only involve the opposite and the hypotenuse, the adjacent and the hypotenuse. Nothing is squared. I could ask the question, OK, well, let's just square them and see what happens. If I take the sine of theta, and I square it, and I square it, what do I mean by squaring it? I really mean, this is just sine of theta times the sine of theta. That's all I mean. Just like when I say, what is three squared? It's just 3 times 3. Sine squared is given by this. Then I can say, well, what is sine of theta times sine of theta? Sine of theta is opposite over hypotenuse. That's opposite over hypotenuse times opposite over the hypotenuse, and what am I going to get?
- 2.7 Well, how do I multiply fractions? I just multiply numerators, multiply denominators. I'm going to get the opposite squared over the hypotenuse squared. OK, fine. One thing I should note is that, as a notational thing, this is a frustrating thing when you first learn it, is that, notationally, if I want to write sine of theta quantity squared, notationally, the 2, you put it right there. So notationally, this is going to be written as sine squared of theta. All I mean by this is that this is sine of theta, the whole thing is squared. It just cuts down on the number of parentheses that I need to use. It's a little cleaner, but it seems weird the first time you see it.
- 2.8 It's like sine upper 2 of theta. It's just a notation. That's it. That's all we're getting at. Then what happens if I take cosine of theta and I square it? Again, that's cosine times cosine. But cosine was what? That was the adjacent over hypotenuse. This is adjacent over hypotenuse times adjacent over hypotenuse. This is what? Adjacent squared over hypotenuse squared. That's good because, again, remember, the motivating question was, is there a relationship between the Pythagorean theorem and the trigonometric functions?
- 2.9 So far, I can say Pythagorean theorem says that I have adjacent squared plus opposite squared is equal to hypotenuse squared. I see that I have an adjacent squared here. I have an opposite squared here. They're divided by hypotenuse squared, which is a little annoying, but we can figure that out. What happens if I add these two quantities together? What happens if I add these two quantities? So if I take, I'm going to use that notation going forward. If I take cosine squared plus sine squared, what do I end up getting? Well, cosine squared, we know is, this is adjacent. Sorry, this is equal to adjacent squared over hypotenuse squared plus, what do I have?
- 2.10 Sine squared is opposite squared over hypotenuse squared. Just symbolically, that's what we have. Now, how do you add fractions? Well, you need a common denominator. Thankfully, the denominators are literally the same thing, hypotenuse squared, so I don't have to modify anything. This is equal to what? This is equal to adjacent squared plus opposite squared all over, hypotenuse squared. Then I stand back and I say, oh, that's exactly what I like because adjacent squared plus opposite

squared via the Pythagorean theorem is equal to the hypotenuse squared. This is just equal to hypotenuse squared over hypotenuse squared, which is what? That's 1.

- 2.11** What this is getting at is, it's getting at what's called the Pythagorean identity, at least one of three of them. There's technically three Pythagorean identities, but this is the most commonly known one, is the Pythagorean identity. And what is that? It's given simply by a cosine squared plus sine squared is equal to 1. This is my cosine squared of theta plus sine squared of theta is equal to 1. Now, as I said, that's one of three Pythagorean identities. What do I mean by the other two? Well, this is only involving cosine and sine, and I have six trigonometric functions altogether.
- 2.12** I'm missing tangent, secant, cosecant, and cotangent. I'm missing those four. If you think about it, look, you have six trigonometric functions. I have one identity taking two of those trig functions. It stands to reason that, if I have one identity per every two trig functions, then I should have two other identities here. One involving tangent and secant, that's the other one, and another one involving cosecant and cotangent. Let's see what those are. How do we figure those out? What are the other trigonometric Pythagorean identities? Excuse me.
- 2.13** Well, there's two ways that you can think about them. You can think about them in the way that we just generated this one, is we could choose, say, secant and tangent, cosecant and cotangent, and write them in terms of their ratios and see what we get. That's completely fine. Alternatively, we can make the following observation. We can make the observation that, if I look at tangent of theta, again, that is my, so, TOA— this is tangent is opposite over adjacent, this is opposite over adjacent— let's think about this in terms of fractions really quickly. In terms of fractions, I could really think about this as the opposite over the hypotenuse, divided by the adjacent over the hypotenuse.
- 2.14** Why is that true? Well, remember, how do you take a fraction and divide it by a fraction? What happens? You take the lower fraction, you take its reciprocal, so you flip it, right? You flip it and you multiply it by the numerator. What is this really telling me? I'm saying opposite over hypotenuse times hypotenuse over adjacent. Hypotenuse is canceled, I get opposite over adjacent. That's exactly what I'm expecting. But why am I writing it this way? Well, what is the opposite over the hypotenuse? That's OH. What's the one in SOHCAHTOA that has the OH? That's the SOH part, S-O-H. So this is sine. This is sine of theta.
- 2.15** And what's the part that has AH? That's the CAH part of it, the C-A-H, SOHCAHTOA. So this is cosine. We get a relationship between tangent, sine, and cosine. Tangent is equal to the sine divided by the cosine. In particular, what that means, this little implication symbol, all that means is that, well, if I square tangent, so again, the same notation when I'm squaring tangent is when I square any of the trig functions. The 2 is in a goofy spot. By the way, the reason why the 2 is here, the reason why it's there is because some people, myself, I put parentheses around whatever I'm plugging into the tangent function.

- 2.16** Some people don't do that. Some people don't do that. That's totally fine. It doesn't matter. Some people would write tangent of theta like this. They don't put parentheses around the theta. So why, when I square it, do I put the 2 here? Well, think about it. If you're using this notation, if you're using this notation and you just put the 2 here, it's confusing. Why is it confusing? Because, well, what am I actually saying here? Is this equal to tangent of theta squared? Is it equal to that? Or is it equal to tangent of theta times tangent of theta? Which one is it equal to?
- 2.17** But if I put the 2 here, well, then I can just say, oh, notationally, what do I mean? I mean tangent times tangent, whether I put parentheses or not. If I have a tangent squared of theta, and I don't use parentheses around theta, it's clear, this is tangent times tangent. There's no ambiguity that, oh, am I talking about tangent of theta squared, or am I talking about tangent times tangent? Anyway, tangent of theta is sine over cosine, so when I square it, I'm going to get sine squared over cosine squared, like that. That's good. Let's go back to our Pythagorean identity here. Let's go back to this.
- 2.18** What would happen if I just took both sides of this and I divided by sine squared? What would happen symbolically? Well, if I have cosine squared of theta plus sine squared of theta equals 1, and I were to divide both sides by sine squared, whatever I do to one side, I have to do to the other side. That's part of our algebraic side of things, where you have to balance the equation. Whatever I do on one side, I do the other. What do I get here? How do we do fractions like this? Well, remember, if we have something that looks like $A + B \text{ over } C$, we're reverse-engineering how we add fractions.
- 2.19** Normally, we're given things like, what is $A \text{ over } B \text{ plus } C \text{ over } D$? We say, how do I combine the fractions? Oh, I get a common denominator, and then I add, once I get the common denominators, I add the numerators. That's the idea. What if I wanted to go in reverse? Well, what would I do? I'd say, well, this is $A \text{ over } C \text{ plus } B \text{ over } C$. The denominator is the same, the C is the same. Then how would I combine those two fractions? Well, the denominator is the same. I add the numerators, I get $A + B$. OK, that's fine.
- 2.20** Then let's apply that here. Cosine squared plus sine squared, that's like my $A + B$, and I'm dividing it by C. What does this give me? This gives me cosine squared over sine squared plus sine squared over sine squared is equal to $1 \text{ over } \sin^2$. Yes? I'm just splitting it per this rule that I have up here. All right? But then what do I notice? Well, sine squared over sine squared, that's 1. OK, so that's fine. So I get a plus 1. Cosine squared over sine squared, I can really think of as cosine of theta over sine of theta, the whole thing squared.
- 2.21** Again, that was something we addressed when we talked about the algebra of things. In the fundamentals of algebra, we're talking about the cosine over sine, and how do we handle exponents. Exponents, you square the top, square the bottom. Same thing over here. I could write this as $1 \text{ squared over } \sin^2$, the whole thing squared, like that. Then I stepped back for a second and I was like, oh, well, I actually have a definition of

this. If tangent is equal to sine over cosine, then what would be the flip of that? Will it be 1 over tangent? But 1 over tangent is cotangent. This guy becomes cotangent squared of theta plus 1.

- 2.22** And I say, 1 over sine, I'm like, oh, I know what 1 over sine is. That's the one, that's the one that doesn't start with an S, so that's cosecant in this case. This is cosecant squared. That's our second Pythagorean identity. So cosecant and cotangent are related to each other, just like sine and cosine are related to each other in precise ways. Then it stands to reason, look, what trig functions have we taken out of the match? We have six players. We've taken cosine and sine out. Cosine and sine, those are out. We've taken cotangent and cosecant out. What are the only two players? We have secant and tangent. That's the only two left.
- 2.23** It stands to reason there's probably going to be another Pythagorean identity involving secant and tangent. Well, let's think about it for a second. How did I get this identity? I started with this identity by starting with the one that I knew, cosine squared plus sine squared equals 1. I just randomly made the choice to say, let's take this and divide by sine squared on both sides. Well, why did I choose to divide by sine squared? Why wouldn't I divide by cosine squared?
- 2.24** The answer is, that's where the third one comes from. Start from here and divide by cosine squared on both sides, and what's going to happen? You can see what's going to happen from here. If I start with cosine squared and sine squared, what do I get? So cosine squared plus sine squared is equal to 1. Divide both sides by cosine squared. The algebra is exactly the same as we did over here. We're going to split it into the two terms. I'm going to get cosine squared over cosine squared plus sine squared over cosine squared is equal to 1 over cosine squared. Cosine over cosine, that's 1. So I get a 1. What's sine squared over cosine squared? Well, that's sine over cosine.
- 2.25** Again, using our properties of exponents, the whole thing squared plus 1 over cosine, or sorry, equals 1 over cosine, the whole thing squared like that. Sine over cosine, I go back and I was like, oh, I know that one. That's tangent. And 1 over cosine, cosine starts with C, 1 over the thing that starts with C, that's going to be the one that doesn't start with C, so that's secant. This is 1 plus tangent squared of theta is equal to secant squared of theta. There you have it. There are the three Pythagorean identities associated to that. That's where they are.
- 2.26** 1 plus tangent squared equals secant squared. Cotangent squared plus 1 is cosecant squared, and cosine squared plus sine squared is equal to 1. Now, to be brutally honest, this one is by far the most important, and that's the one that shows up the most frequent. OK? Cosine squared plus sine squared is equal to 1. That's the one that shows up all the time. In my opinion, which really doesn't account for much, but in my opinion, cosine squared plus sine squared equals 1 is really the only one you need to know. Because if I know this, and I know the technique of generating the other two, the technique is what?

- 2.27** Take this one, divide by cosine squared. What do I get? This one. Take this one, divide by sine squared. What do I get? This one. You can generate the identities simply by knowing this and knowing the technique, divide by cosine or divide by sine. That's personally how I like to remember them. I have a hard time remembering different identities and things like that, so I prefer to think about it that way. We're going to focus on this one, the cosine squared plus sine squared is equal to 1, and make the following observation. We're going to make the following observation.
- 2.28** What's the relationship between trigonometric functions, triangles, and circles? That's the motivating question. Is there a relationship between those things? Maybe there is, maybe there isn't. If there is, then we want to understand, what is the relationship? What is the relationship? Let's think about that. Let's think about that. All right. Here we go. Let's consider a circle centered at the origin, 0, 0, of radius 1, OK, of radius 1. Why am I doing this? Well, it'll become clear in just a second.
- 2.29** Let's consider a circle centered at the origin of radius 1. What do I have? We've got our circle. That's not too bad. I like that. All right. I'm going to put this on the X and Y coordinate system. The circle is simply just a collection of points that are all distance 1 away from the origin here. This is my 0, 0. Now, if they're distance 1, then I can classify these points. This point is going to be 1 in the X direction, 0 in the Y. This point is going to be 0 in the X, 1 in the Y. This point here is going to be minus 1, 0, and this point here is going to be 0, minus 1. That's what I have. That's the idea.
- 2.30** Now, we have to make sense of, we have to set a notation and a standard of saying, how do you measure angles? How do you measure angles? The standard is going to be, you take your positive X axis, so that's this line right here, and you declare an angle to be positive if you measure it counterclockwise from that side. You imagine this as being angle 0, and then I start going angle counterclockwise from there. Now, if instead I had decided to go clockwise from there, I declare those angles to be negative. I declare those angles to be negative. It's literally just a standard. There's reason for it, and that's fine. It doesn't really matter right now.
- 2.31** We're going to say that if I go counterclockwise, the angle is positive. If I go clockwise, the angle is negative. If I decide to go out an angle of theta, whatever that is, if it helps, you can think about theta as being a very specific value. You can say, like, theta is 20 degrees or 30 degrees or 45 degrees or whatever helps try and get the ideas through. But you go counterclockwise some angle of theta. Then what do I know? Well, this particular point here, it lives on the circle. It lives on the circle. So whatever its coordinates are, let's call it an X, Y. I don't necessarily know what its coordinates are yet. But what I know is that if I drop this down, I get a right triangle. Yes? I get a right triangle there.
- 2.32** Well, that right triangle, I know what its hypotenuse is. Its hypotenuse is the radius of the circle. Well, the radius is chosen to be 1. So I know what the hypotenuse is. That's great. Now, I know what this side, let me move this to 0, 0. I know what this side of the

triangle is. That's X . I know that that's equal to X . I know what this side of the triangle is. That has to be Y because this point, after all, is in the XY position, saying you move over X , you go up Y , and I'm at that point. So if I go over X , that has to be the length of this triangle. I go up Y , that has to be the length of this side of the triangle. That's Y .

2.33 Now, what do I really have? I have a right triangle with side length X , Y , and hypotenuse 1. I know that this angle is θ . Now I can say, oh, sweet. I know relationships between the trig functions and those sides. I can say, well, what is the sine of θ ? I can say the sine of θ is equal to what? Opposite over hypotenuse. So that's Y over 1. So that means that Y has to be the sine of the angle of θ . That's what that has to be. And then cosine of θ has to be the adjacent, so that's X over 1. So that means that X is equal to cosine of θ .

2.34 So now here's the relationship. Here's the relationship between these two quantities. What is this point X , Y ? X is cosine of θ , and Y is the sine of θ . That's what I get. So that's why sometimes you'll hear people refer to, when you talk about the circle and its relationship to trig functions, they'll say, oh, the X coordinate on the circle is the cosine, and the Y coordinate is the sine. That's something that we can document. We can say, on the circle of radius 1, the X coordinate is equal to the cosine, and the Y coordinate is equal to the sine. That's a very common thing. You'll hear people say, oh, X is cosine, Y is sine. And this is Y . This is Y .

2.35 Here's our relationship between our trigonometric functions, sine and cosine. Those are coming from right-angle triangles, right triangles, and circles with respect to this. That's our relationship. The question then comes as the following. If I now have a relationship between sine and cosine and the circle, I can ask myself the following question. If this is my angle θ , and corresponding to that angle θ , I land at this particular point X , Y , which is now cosine θ sine θ , what happens if I go 360 degrees or 2π radians? What happens?

2.36 Well, let's think about it. If I go here θ and I go around 2π , where am I at? Well, I'm at the same point. Do we agree? If I have this line and it points at θ , it points at this, and then I just allow that to go around 2π radians, I'm still pointing at the same place. Do we agree? But the angle now is different. The angle is different. Even though I'm at the same point, the angle is different. I've now gone θ plus 2π all the way around. What do I notice? I notice that, if you go around, the circle, 2π radians, what do we get? We get that cosine of that new angle, θ plus 2π , that new angle. I start at θ , I go all the way around 2π again. What do I get? Well, I'm at the same place. I'm at the same place.

2.37 Well, the original place was cosine of θ . What do I notice? This has to equal cosine of θ . Those two things have to be the same. Similarly, I'm at the same place in the Y coordinate. If I go θ , I'm at Y , which is sine of θ . I go around another 2π , I end up back at sine of θ , I get the same thing. I get that sine of θ plus 2π is equal to the sine of θ . That's confusing. That's like, well, wait a second. You're telling me

that if I start at some angle θ , I go around another circle of 2π radians, and I evaluate cosine at that point, I get the same value as if I was at the original θ ? I'm saying, yes, that's exactly what I'm saying.

- 2.38** That's the relationship between the circle and our trigonometric functions here. It's hard to envision saying, look, let's start with this triangle, which has angle θ , whatever that is. If we think of θ as being, say, I don't know, 45 degrees. And I go around 360 degrees, I'm saying the value of cosine is going to be the same. It's like, well, what triangle looks like that? You have an angle of 45 degrees, and then you add 360 to it. It's like, you're what, 405 degrees? It's like, that's not a triangle. You told me the sum of the interior angles of a triangle had to be 180, not 405. What the heck does that mean?
- 2.39** You don't really have a geometric intuition for what the triangle looks like, which has this angle of 45 degrees plus 360. That's why the circle is so much more useful in terms of interpreting this, because I start with θ , I then go around 360 degrees or 2π radians, and I end up back at the same point. Well, if these points are given by sines and cosines, then that's telling me these relationships. That's the importance of having the intuition of what the sine, cosine, tangent are in terms of triangles, versus what are the sine, cosine, tangent in terms of circles.
- 2.40** But then there's no stopping this at this point. If I take $\theta + 2\pi$, so I went around the circle once, a 2π , it's like, well, what's to say I don't do it again? Why don't I just spin in a circle again or spin in a circle again or keep doing this over and over and over again? What we notice is that we get cosine of what? $\theta + 2\pi$ times any integer K . If K is 1, I've gone around in circles once. If K is 2, I've gone in circles twice. If K is 3, I've gone in circles three times. What should I get? Well, I should get the same value of cosine of θ . I should always get this. OK? Cosine of θ .
- 2.41** Similarly for sine, if I take sine of $\theta + 2\pi K$, I should get sine of θ , where K is equal to an integer. What is an integer? Well, they're the numbers that we count with. Negative 2, negative 1, 0, 1, 2, 3, 4, 5, things like that. That's what the integers are. I'm not allowing K to be a fraction. K is not allowed to be like one-half. If K were one-half, then that's saying, go around the circle half times. Well, that's not going to be the same value. You imagine if I start here, that's θ , and I go around the circle half, that means I'm going π radians. I'm looking at this point here. I'm not the same point up here. You have to go a full 2π . That's why K has to be an integer.
- 2.42** But why is it allowed to be negative? Well, why did I decide to go this way around the circle? What if I instead went this way around the circle? Well, remember, angles are measured negatively going clockwise. If I go negative 2π , I still end up at the same place. Versus if I go positive 2π , I end up at the same place. That's why K is allowed to be negative 1, negative 2, negative 3. You're just going around the circle the other direction. That's it.

- 2.43 But what's nice is, what this is telling me is that cosine and sine are what are called 2π periodic, 2π periodic. It just means that after every 2π , they repeat themselves. That's all it says. We're going to see this. We're going to graph these functions as abstract functions in and of themselves, and we'll see that the function does something, and then after 2π , it does the exact same thing. And after another 2π , it does the exact same thing. That's what we mean by 2π periodic. It just repeats itself every 2π . Every 2π , we just get periodicity of that particular function.
- 2.44 That's our relationship between trigonometric functions and right triangles, and the circle. Now, this circle is special. It's centered at the origin and it has radius 1. It has a name, the name nobody likes. It's called the unit circle. We're going to have an entire class based on the unit circle. We're going to have a full period time of the unit circle, and how do we find values on there. Now, one thing that we can do is the following. We can say, let's consider certain angles that are deemed valuable, certain angles that are deemed valuable.
- 2.45 That gets into these certain types of triangles, which are called 30-60-90 triangles and 45-45-90 triangles. We want to ask ourselves, can we figure out what the coordinates are of sine and cosine at these particular angles? But first, before we do that, let's figure out, how do we convert from degrees to radians? Just like we had that conversion, how many seconds are there in a day, we want to understand how do we convert from radians to degrees, and then how do we go from degrees back to radians. We want to understand converting from degrees to radians, and radians to degrees.
- 2.46 Because after all, there's lots of different ways to describe the same thing. If I'm talking about an angle of 45 degrees, but somebody doesn't like to use degrees, they like to use radians, then we need to be able to have a systematic way of making sure that we're talking about the same thing. 45 degrees is what radians and vice versa. If somebody says, "I only like talking about radians," I want to understand what they are in terms of degrees, then we want to be able to go both ways on that front.
- 2.47 Remember, we have that there are π radians in 180 degrees. I'm going to write them out as degrees verbally so that we can see how the units are going to cancel, how they're going to cancel. On this side, we're going to go from degrees to radians. We want to understand how to go from degrees to radians. Here's a silly example. The example, I want to convert, let's say, 30 degrees to radians. How do I do that? We'll think back to the example on how we converted days into seconds. We needed a ratio regarding the units. What I notice is I have 30 degrees. I start with that quantity.
- 2.48 Do I know a relationship between radians and degrees? The answer is, well, yes, I do. I know that there are π radians, π radians, per 180 degrees. Can we agree? I have π radians in 180 degrees. If I look at my units, I see that degrees cancel, and I'm given with that. I have 30 times π over 180 radians now. That's my conversion factor. Just like we had in the example of saying one day, we said, OK, there's 60 minutes or there's

24 hours in one day. There's 60 minutes in one hour. There's 60 seconds in one minute. That's how we're doing this.

- 2.49** The degrees cancel and I'm left with radians. Now what do I end up getting? This is like 30 divided by 1. So 30 times pi, I get 30 pi all over 1 times 180, I get 180. What am I left with? The zero cancels, so I get 3 pi over 180. Sorry, over 18. Three goes into 18 six times. This is pi over 6 radians. So my 30 degrees is pi over 6 radians. What we've taken away from this is, how do I go from degrees to radians? We now have a systematic way to do this. You give me an angle measure in degrees, I need to do what? I need to multiply it by pi over 180. That's our systematic approach to it. So multiply degrees by pi over 180. By pi over 180. That's how I convert from degrees to radians.
- 2.50** So how would I go in reverse? How would I go from, you give me a radian measure, I want to give that to you in terms of degrees? Well, how do I do that? I'm going to flip this. Rather than multiplying by pi over 180, I'm going to multiply by 180 divided by pi. OK? Let's see how that works. Let's do the same type of example, but now we're going to convert from radians to degrees. All right. How do we do that? OK. Let's do convert. What's a good example? Let's say pi over 4 radians to degrees. I want to convert pi over 4 radians to degrees. So this is converting radians to degrees.
- 2.51** What can I do? Well, again, I have this relationship between their pi radians in 180 degrees. OK. So what do I start with? I start with, I have pi over 4 radians, and I want to convert that into degrees. Well, what's my relationship between the two? I know that there are 180 degrees in pi radians. Now, again, this is like pi over 4 radians is like radians per 1, if you want to think about it that way. The radians now cancel. And what are the units that I have? Well, degrees are the only thing that I have left.
- 2.52** So if I look at this, I end up getting—what is this? Pi times 180. I get 180 times pi, divided by 4 times pi, so I get 4 pi. My units should be degrees. Now, again, pi is a symbol here, but it's representing the number 3.14159, whatever. Pi cancels here and here. Four goes into 180. At the very least, if I don't know that 4 goes in a certain number of times, I can say 2 goes into both, as a first approximation. I'm going to get what? 90 divided by 2 degrees, but 90 divided by 2 is 45, so this is 45 degrees. A 45-degree angle is pi over 4 radians.
- 2.53** You have this dictionary in your head of, OK, if you give me degrees, I can give you radians. If you give me radians, I can give you degrees. It's like any other language to some extent. You give me a phrase in English, I can translate that into Spanish. You give me a phrase in Spanish, I can translate that into English. OK, things like that. That's what we're getting at here. We're just translating between one language and the other. Radians versus degrees. There are certain triangles, right triangles, that have measures that are of interest. This is what's going to govern the unit circle, ultimately. But let's see how this works. OK.

- 2.54** There are certain types of triangles. There are the 30-60-90 triangle, and then there's the 45-45-90 triangle. There's those two. Now, what do I mean by that? I mean, those are the angle measures in degrees of these triangles. That's really what I mean. I have a triangle over here. It's a right triangle, and it looks something like this. I've got my 90 degree. That's this part. I've got my 60 degree. Now, based on how I've drawn it, 60 is bigger than 30. So this is probably my 60 degree, realistically, and this one would be my 30 degree.
- 2.55** Remember that the sum of the interior angles has to be 180. I get 30 plus 60 is 90, 90 plus 90 is 180. That's a legitimate right triangle. Then the 45-45-90, this triangle looks like, OK, what do we have here? It looks something like this. Here's 90, here's 45 degrees, here's 45 degrees, and there we go. OK. So one thing that you can imagine that seems plausible, you can think about this kind of—you imagine you have a laser pointer or something like that. And if I'm standing at this particular point and I shine the laser pointer on the wall and I mark that point, and then I somehow measure that I move the laser pointer 30 degrees up and I see what is the distance that it traces, it's going to be the length of this side.
- 2.56** Similarly, if I was standing over here and I shot the laser pointer here and then I swept out 60 degrees, I would trace out this. You can imagine that the side is going to be longer corresponding to the wider the angle. If you just point the laser pointer and then move 30 degrees, it's going to have a certain distance associated to it, versus if I point the laser pointer and then I open up 60 degrees, it's going to have a larger distance associated to it. That's panning out here. The 60-degree angle, I have the larger side here. The 30-degree angle, I have the smaller side here.
- 2.57** We want to try and understand what we have for 30-60-90. Let's suppose I give you something like this. Let's say I have 4 square root of 3 of 4 here, and I want to know by the Pythagorean theorem, what does my hypotenuse have to be? Well, the Pythagorean theorem says that this side has to be what? 4 square root of 3 squared plus 4 squared is equal to my hypotenuse squared. Do we agree? That's Pythagorean theorem, $A^2 + B^2 = \text{hypotenuse}^2$. Well, 4 squared of 3, how do I do that? This will be 16 times 3 plus 16 is equal to hypotenuse squared. 16 times 3 is 30 and 16, so that's 48, plus 16 is hypotenuse squared. This is 50 and 14, so that's 64. 64 equals hypotenuse squared.
- 2.58** That means that my hypotenuse has to be 8. It's got to be 8. That's my hypotenuse. Now I can ask myself the question, what are the sine, cosine, tangent of 30? What are the sine, cosine, tangent of 60? Try and understand what those corresponding values are. If I say, OK, the sine of 30 degrees, sine of 30 degrees. Now, remember, the sine of 30 degrees is the opposite over the hypotenuse. It doesn't matter what the angle is. The sine is always opposite over hypotenuse. The fact that theta is equal to 30 degrees is irrelevant here. The sine is always opposite over hypotenuse.

- 2.59** The sine of 30 is 4 divided by 8. OK, so that's $\frac{4}{8}$. That's one-half. OK, that's fine. OK, what is the cosine of 30? Cosine of 30 degrees. Cosine of 30 is the adjacent divided by the hypotenuse. OK, so that's $\frac{4\sqrt{3}}{8}$. So that's $\frac{\sqrt{3}}{2}$. OK, so I'm given that. Now, but what if I asked what is the sine and cosine of 60 degrees? What if I change the angle now? If I say, what is the sine of 60 degrees versus what is the cosine of 60 degrees? If I say the sine of 60 degrees, now I'm saying, OK, the opposite is this one, $\frac{4\sqrt{3}}{8}$. So this is $\frac{4\sqrt{3}}{8}$. OK, well, we just found that. That was the cosine of 30, so that's $\frac{\sqrt{3}}{2}$.
- 2.60** And cosine of 60, the adjacent now is 4, 4 divided by 8, and that's one-half. OK, so that's weird. There's this relationship between the cosine of 60 and the sine of 30, those are the same number. And the sine of 60 and the cosine of 30, those are the same number. So somehow I'm getting this relationship between those. So that's strange. They just switch. That makes sense in terms of the triangle, if you think about this, because when I take sine of 30, that's the opposite over the hypotenuse, whereas the sine of 60 is the opposite of this. Well, what's the opposite here in relation to 30? That's the adjacent.
- 2.61** What's the adjacent of 60? Well, that's the opposite of 30. It's like, oh, of course, it makes sense that the sines and cosines are going to be related in that way. We're given that. Now, what about over here in the 45-45-90? Let's say, if this is 45 degrees, and that's 45 degrees, then if you go back to the laser pointer example, those two things have to have the same length. If you imagine yourself standing here, you point the laser pointer on this point of the wall and you sweep out 45 degrees, whatever this length is, it has to be the same as that length.
- 2.62** Choose whatever number you want. Let's say this is 5 and this is 5. That's fine. These are triangles that are called isosceles, meaning that they have two sides that are the same length. So isosceles triangles. What does the Pythagorean theorem say here? Well, it says, OK, if I take 5 squared plus 5 squared, I should get the hypotenuse squared. OK. Well, 5 squared plus 5 squared, this is 25 plus 25 is equal to hypotenuse squared. OK, so what is this? 50 equals the hypotenuse squared. That means that the hypotenuse is equal to the square root of 50.
- 2.63** OK. Well, that's all great, but can I simplify that somehow? Can I simplify the square root of 50? Well, the square root of 50 is really the square root of 25 times 2. We agree? 25 times 2 is 50. But 25 is 5 squared. So this is 5 squared times 2. And the square root, as we talked about in the algebra section, the square root is fine when it comes to multiplication. What do I mean by that? I mean that, if I take the square root of X times Y, that's the square root of X, the square root of Y, when those two things are defined.
- 2.64** Then what is this here? This is the square root of 5 squared times the square root of 2, which is just 5 times the square root of 2. So it simplifies it a little bit. You can take squares outside of the square root. So my hypotenuse then becomes 5 times the square root of 2. OK, well, great. Well, what are my sine, cosine of 45 degrees? Now,

notice over here, it mattered which angle I started with. If I start with 30 and I take the sine, that's going to be opposite over hypotenuse, versus the sine here, that's going to be a different side over the hypotenuse.

- 2.65** Notice over here, it doesn't matter which 45 I start with. Because if I take sine of this 45, what do I get? 5 over the hypotenuse. If I take sine of this 45, what do I get? Oh, another 5 over the same thing. It doesn't matter which angle because they're the same angle. I'm not going to have four things to write here. I'm only going to have two. I'm going to have one sine of 45 and one cosine of 45. They're going to give me the same values.
- 2.66** We've computed our hypotenuse. I'm going to erase this, and we need to figure out what are the sine and cosine. If I take sine of 45 degrees, what do I get? Sine of 45. Let's suppose I take this angle. That's my opposite 5 over my hypotenuse $5\sqrt{2}$. The fives cancel. The fives cancel, and I get 1 over the square root of 2. OK, that's fine. So 1 over the square root of 2, that's great. What about cosine of 45? Well, cosine is, take the adjacent. Take the adjacent. Well, that's still 5. OK. Divided by the hypotenuse. And that's, well, again, 1 over the square root of 2.
- 2.67** I claim that that shouldn't be too farfetched. Why? Well, let's go back to the example of the circle. Remember, when I had a circle—see if I can pull that off. It's actually not too bad. I always get in my head when I try and draw a circle on the board because you never know if it's going to turn out to actually look like a circle or some squiggly craziness. If this is 45 degrees, we said that this X and Y, those are the same thing. Do we agree? Those lengths are the same because why? That's the length of this side of the triangle, and that's the length of this side of the triangle. X and Y are the same. This point is X, Y. But remember, X and Y are the same. Those are the same.
- 2.68** If this is, again, the unit circle, well, X is cosine of 45. Cosine of 45 is 1 over the square root of 2. The Y coordinate is sine of 45. That's 1 over the square root of 2. Well, that makes sense that those should be the same. Because this 45-degree angle says that X and Y are the same, the same values. The fact that I'm getting 1 over square root of 2, 1 over square root of 2, actually tracks. OK, great. These are the angles that we care about. 30 degrees, 60 degrees, 45 degrees, 90 degrees, that's another one, and zero degrees. Zero, 30, 45, 60, and 90. You have five angles that you care about. Five angles that you care about.
- 2.69** Now, what I want to do is, I just want to make a table of all of these values. What are the sines, cosines, of these things? Then what are their radian measures associated to that? Because these are going to form the fundamental basis of the unit circle when we go after finding all of the values on the unit circle. But they all start here. Let's just tabulate all of this. The reason why the unit circle causes so many problems in the sense of, let me say, fear, when people go through the unit circle and they learn the unit circle, is because there's so many values associated to it. There's so many values associated to it.

- 2.70** Then you ask yourself the question, well, how in the world am I supposed to remember this? The answer is, you only have to remember five things. Then you just need two or three tricks on top of those five things to generate the entirety of the unit circle. You don't need to memorize all 30-plus things on the unit circle. You need to memorize five things plus two or three tricks, and that's it. Once you have that, you can generate the entire thing.
- 2.71** Let's look at our table. Here's what I'm going to do. I'm going to say θ in degrees. I'm going to then have θ in radians. I'm going to have those. Then I'm going to have the sine, cosine, tangent of those values. I'm going to have the sine of θ . I'm going to have the cosine of θ . Then I'm going to have the tangent of θ . Well, and why not? Let's just take the other three as well, secant, cosecant, and cotangent. If we know sine, cosine, tangent, we know what the others are because they're just reciprocals. That's fine. But why don't we just take those as well?
- 2.72** Corresponding to sine, I'm going to have cosecant. Corresponding to cosine, I'm going to have a secant. Then corresponding to tangent, I'll have cotangent of θ . We're just going to generate our table. All right. What do we end up getting? Well, in degrees, I told you the degrees that were of interest were zero degrees, 30 degrees, 45 degrees, 60 degrees, and 90 degrees. We need to understand what are those in terms of radians. Remember, how do I go from degrees to radians? I need to multiply by π over 180. To go from degrees to radians, if I take zero and I multiply by π over 180, it's zero. Zero times anything is zero. I get zero radians here.
- 2.73** If I have 30 degrees, this is the one that we did. We had 30 degrees, and I multiply by π over 180. 30 over 180 is 6, so it's π over 6. I get π over 6 radians. 45 degrees. How many times does 45 go into 180? It goes in 4, so I'm going to get 45 times π over 180 is π over 4. There's π over 4. 60 goes into 180 three times. So 60 times π over 180 is π -thirds. And then 90 times π over 180, 90 goes into 180 twice. I'm going to get π -halves. OK? I'm going to get π -halves. OK. Good.
- 2.74** Now, if we go back to what we had here, let's think about this for a second. What is the sine of zero degrees? Let's go back to the unit circle. I've got this unit circle. Zero degrees is right here. That's θ equals zero. I haven't measured any degrees from there. It's just my X axis. If I say this point here, remember, points on the unit circle, the X coordinate is cosine of that angle, and the Y is the sine of that angle, this is a 1, 0. Well, that means that sine of θ , which is the Y coordinate, should be zero. So sine of zero is zero.
- 2.75** The cosine is the X coordinate. So cosine of zero should be 1, in that case. All right, let's keep moving. So this one is 30 degrees. So here's θ equals 30 degrees. The 30 degrees, we already had that one. Remember, when we just did the 30-60-90 triangle, we said, oh, we know what sine, cosine, sine and cosine of those things are. Sine of 30 degrees was our one-half. That's where we got our one-half. Cosine of 30 degrees was equal to the square root of 3 over 2. We got that. The 45 degrees, 45 degrees is up

here. Here's theta equals 45 degrees. We said that sine of 45 was 1 over the square root of 2. Cosine of 45 was 1 over the square root of 2.

- 2.76** Then we saw this amazing thing that the sine and cosine flipped with the 30 degrees. Those flipped, right? I get sine of 30 degrees was one-half, and the cosine of 30 degrees was root 3 over 2. That means the sine of 60 is going to be the square root of 3 over 2, and the cosine of 60 is going to be one-half. Those two things switch. Those two things switch. That's it. Theta is equal to 60 degrees. Now, if theta equals 90 degrees, I'm on the Y axis. This is theta equals 90 degrees. This point here is 0, 1. Remember, the X coordinate is cosine, the Y coordinate is sine. So the X coordinate, so cosine of 90 needs to be 0, and sine of 90 needs to be 1. So sine of 90 is 1, cosine is 0.
- 2.77** Now we can fill out the rest. Now we can fill out the rest. So what is tangent of 0? Well, that's sine divided by cosine. That's 0 over 1, that's 0. So, tangent of 30, one-half divided by the square root of 3 over 2, that's 1 over the square root of 3. Tangent of this, those are the same, so tangent is 1 there. Here, root 3 over 2 divided by one-half, that's just square root of 3. Then 1 divided by 0, you're never allowed to divide by 0, so this is undefined. This is undefined. That's what I get for tangent at those values.
- 2.78** Now, what is cosecant? Cosecant is 1 over sine, so what happens here? Well, 1 over sine, 1 over 0, this is undefined, so that's not good. 1 over a half is 2. 1 over this, 1 over root 2 is root 2. 1 over that is 2 over the square root of 3. What do I get here? One over 1, that's just 1. Then we get the other two columns as follows. Secant of theta is 1 over cosine. I go to the cosine column, 1 over this is 1. One over this is 2 over root 3. One over this is root 2. One over this is 2. One over 0, undefined.
- 2.79** The last one is cotangent, which is 1 over tangent, so 1 over 0, undefined. One over 1 over root 3 is root 3. One over 1 is 1. One over root 3 is what it is. And 1 over undefined, how do we do that? Well, where did this happen? This was at 90 degrees. Remember, tangent of theta is equal to sine of theta over cosine of theta, which means cotangent of theta is equal to cosine over sine. If I say what is cotangent of 90 degrees, that's cosine of 90 over sine of 90. Sine of 90 was 1. Cosine of 90 was 0. So cotangent of 90 is 0 over 1, which is 0.
- 2.80** That's where we get that particular value. These are the main values that we're going to need when computing the unit circle and deciding what are the other values based on some symmetries that we'll take advantage of. But these are the ones that we're going to need going forward.

Lecture 3: Trigonometric Equations

At a glance — Trigonometric equations.

- 3.1** Now that we have the trigonometric functions, sine, cosine, tangent, secant, cosecant, cotangent, the Pythagorean theorem, and now we have a relationship between the right triangles from which these things are defined and circles, we want to actually fill out the rest of that circle. This circle is called the unit circle. It's nobody's favorite thing. I'll be brutally honest, nobody likes this circle. It's got too many values associated to it. It's too hard to remember. It's just a difficult object in general.
- 3.2** My hope today is to be able to get rid of some of those concerns or fears about the unit circle, give you techniques for being able to generate values on the unit circle without actually having to memorize every single value on this thing, so that really all you need to do is understand what are the values in the first quadrant. What we decided or what we computed last class when it was zero, 30 degrees, 45 degrees, 60 degrees, and 90 degrees, we're going to do everything in terms of radians. Zero, pi-sixths, pi-fourths, pi-thirds, pi-halves.
- 3.3** We're going to fill out the rest of the values of the unit circle using some tricks. That's the goal for today. Let's start out with our unit circle. Here we go. The first thing with the unit circle that we have to be aware of is, what is its defining equation? Number one. That comes from the Pythagorean identity. Remember, the unit circle is given in terms of coordinates as $X^2 + Y^2 = 1$. What I remember is that X was cosine, Y was sine, for a given angle. That's giving me the Pythagorean identity, cosine squared of theta plus sine squared of theta is equal to 1. That's my Pythagorean identity. OK, great.
- 3.4** Now, the values that we're going to be interested in, that's going to be our 30 degrees, 45 degrees, 60 degrees, we want to fill out what are the corresponding X and Y coordinates because we know that cosine and sine give me the X and Y coordinates, respectively. Cosine is X , sine is Y , and we want to evaluate the circle at those points. All right. At the very least, I can say what these values are. This is my 1, 0. This is my 0, 1. This is my negative 1, 0, and down here is my 0, minus 1. So I have those points. That's completely fine.
- 3.5** Now, what we're going to do is, we're going to say, the positive X axis is given as my theta equals zero. My angle is zero. Then angles from there are going to be measured counterclockwise from the positive X axis. Really, what I'm going to do is each one of these lines, I'm going to write a theta value. What I mean by that is, for this one, I'm going to say theta is equal to zero. What I really mean is that the angle measured from the positive X axis to whatever this is is zero.
- 3.6** When I put this one as saying theta is equal to pi over 6, what I really mean is that the angle from here to here is pi over 6. That's what I really mean. It's like I'm saying this angle

right in here is the π over 6. But because I'm going to have to do this for so many different values, I'm simply just going to mark the angle as corresponding to these different rays. This is going to be my θ is equal to π over 4. This one will be θ is equal to π over 3. This one is θ equals π over 2, and we're going to march our way all the way around. That's the idea, and that's the notation for this. Here we go.

- 3.7** We did this last time, where we found out what are the cosine and sine of each of those corresponding points. Remember, we said that for angle 30 degrees, or π over 6 radians, we said that the X coordinate here was square root of 3 over 2, and the Y coordinate was one-half. We had those particular values. Now, again, this is coming from saying what was the cosine of 30 degrees or π over 6 radians? That was equal to the square root of 3 over 2, and that the sine of π over 6 was equal to one-half. That's what we had.
- 3.8** Then we marched our way around. θ is equal to π over 4. That was 45 degrees. If I'm at 45 degrees, remember, the X and the Y coordinates are equal to each other. This is our 45-45-90 triangle when we were using degrees. What did I get here? I had a 1 over the square root of 2, 1 over the square root of 2. Then we had this nice phenomenon that said, well, at 30 degrees, I had cosine with something, sine with something. But then when I went to 60 degrees, they flipped. Remember that cosine of 30 was equal to sine of 60, and sine of 30 was equal to cosine of 60. So they just flip in this case.
- 3.9** If these coordinates are square root of 3 over 2, one-half, then what are these coordinates? These ones have to be one-half square root of 3 over 2. That's what has to go there. OK. All right. Then I keep going. I have θ is equal to zero, θ is equal to π -sixths, π -fourths, π -thirds, π -halves. Well, we know what π -halves is, what it is, 0, 1. But the question we want to ask is, how do I get the values over here? How do I get the values down here? How do I get the values over here? This thing over here is just called the first quadrant. You can think about this as being separated into four different infinite squares.
- 3.10** You have this quadrant, so this is quadrant one, and then you work your way around this direction to get the remaining quadrants. You have quadrant one, quadrant two, quadrant three, quadrant four. Those are the particular quadrants that we're looking at. What we want to understand is the values in quadrant two, the values in quadrant three, and the values in quadrant four. The first thing that we're going to do is the following. If I have the following picture, so let's suppose I have my circle and I have an angle of θ that's there. I have an angle of θ .
- 3.11** What I can do is I can imagine taking this ray that originally starts from the origin and goes out to infinity, and I can imagine it just being this infinite line that goes through. It's going to intersect the circle in a third quadrant. I have one intersection point in the first quadrant, one intersection point in the third quadrant. How can I think about its angle? Well, what do I do? If I want to think about, what is the angle corresponding to

that point, what did I really do? I went to theta, and then I went 180 degrees, so I went pi radians. To go from here around is really, what? It's really, I go theta, but then I add pi to it.

- 3.12** It's like you imagine yourself, point yourself in the direction of theta, and then just go 180 degrees or pi radians, I'd be pointing in this direction here. Does that make sense? You start with theta, you add pi to it. That's what's going to give me this measure for this particular point. Now, what's beautiful about this is that, whatever this point is—I don't know what it is, but I'm going to call it A, B—this point over here has coordinates minus A, minus B. That's what it is. I can take advantage of that now, because what does it say? It says, look, if this angle is $\pi/6$, well, then I can add pi to it and extend this line through there, and I can get the corresponding coordinates of this point.
- 3.13** Similarly, if this angle is $\pi/4$, then I can extend this down, and I can get the coordinates of this. What are those coordinates going to be? It's just going to be negative this, negative that. Similarly, what are the coordinates of this going to be? It's going to be negative this, negative that. Then I can keep doing that with this one. This one's $\pi/3$, so this one's going to come down straight through and go to right there. Now I have a systematic way of filling out quadrant three.
- 3.14** What do I need to know? I only need to know the values in quadrant one and the fact that when I go all the way through to the other side, all I do is, I negate the coefficient or I negate the coordinates. I get quadrant three, in some sense, "for free." I just need to know quadrant one, and the trick that if I go through the origin, I just negate the coordinates. OK. What does that mean? Well, I don't yet know what these values of theta are. I don't yet know what these are. OK. We'll handle that in a minute. But at the very least, I know what these coordinates are.
- 3.15** These coordinates are given by what? This one came from here, so I negate both of those. This is going to be minus the square root of 3 over 2 minus one-half. That's what that one's going to be. This one came from where? It came from $1/\sqrt{2}$, $1/\sqrt{2}$. This one's going to look like minus $1/\sqrt{2}$, minus $1/\sqrt{2}$. That's that point. This one came from here, so that's one-half $\sqrt{3}$ over 2. This one is minus a half minus $\sqrt{3}$ over 2. I know what those coordinates are. That's big.
- 3.16** But now the question becomes, what are the angle values? What are the angle values? Well, let's think about it. What am I really doing? I'm going theta, and then I'm adding pi. How did I get to this angle value? I went $\pi/6$ and then I added pi. This angle value should be what? This angle value should be, take $\pi/6$ and add pi to it. That's what that angle value should be. I have to do the arithmetic and actually simplify that and figure out what that is. So how do I do this? Well, this is like $\pi/1$. How do we combine fractions? I need a common denominator here. This is a $\pi/6$, plus what do I need? I need $6\pi/6$, to make that a fraction that has the same denominator.

- 3.17** I then have 1 pi that I add to 6 pi, so I have 7 pi, all over 6, and that's what the angle value is. This angle value here is theta equals 7 pi over 6. Now we can play that same game for this angle, which is pi-fourths. What am I going to do? I'm going to take pi-fourths, I'm going to add pi to it, and that's going to land me down here. I want to figure out what is that value. Just like we did over here, let's do that simplification. I start with pi-fourths, and I add pi, which is like pi over 1.
- 3.18** Common denominator is going to be a 4, so I need to multiply by 4, divided by 4. I'm going to get pi plus 4 pi all over 4, which is 5 pi over 4. All right? That's that angle. Theta is equal to 5 pi over 4. Now, another way you can see this, another way you can see this is, look, this is one quarter of pi. I have one quarter, two quarters, three quarters, four quarters, five quarters. You can see how that's working, right? I'm taking one quarter, two quarter, three quarters, four quarters, five quarters. OK, so that's matching up. Good.
- 3.19** Similarly for the sixth. How do you see that? You have one-sixth, two-sixths, three-sixths, four-sixths, five-sixths, six-sixths, seven-sixths. Then you get your 7 pi over 6. All right? OK, so how do we get this one? How do we get this one here? We take pi-thirds, we add pi to it. OK, so what is that going to get me? If I take pi-thirds, pi over 3, and I add pi, which is pi over 1, again, I get a common denominator, so that's going to be 3 over 3. I'm going to get pi plus 3 pi all over 3. That's 4 pi-thirds. That's what this angle is right here. Here's theta equals 4 pi over 3.
- 3.20** OK. Now I'm like 50% done. I have quadrant one filled out. I have quadrant three filled out. Now I need to figure out what are quadrants two and quadrants four. Now, there's a couple of ways that you can do this. The way that I personally like to do it is, I like to say, OK, well, look, if I am measuring angles positively by going counterclockwise, and negatively by going clockwise, then what I can do is say the following. If I imagine taking this point and dropping it straight down, do you agree that it's going to intersect the circle at some point directly below it?
- 3.21** You imagine just taking a vertical line. You can think about this. I'm a big proponent of keeping these things real world-type examples and intuition from the real world. If you put a hula hoop on the ground and you take a yardstick or something like that and you just lay it across one side of the hula hoop, it's going to intersect in these two points. You're going to have it intersect here, you're going to have it intersect here. But what's beautiful about this is, I can say, well, look, if this angle from here to here was pi over 6, then the angle from here to here should be what? Well, I'm measuring it in the other direction. So that's minus pi over 6. That's minus pi over 6.
- 3.22** Similarly, if I take this point and I drop it all the way down, this way from here to here was pi over 4. Therefore, this way from here to here is going to be negative pi over 4. This was pi over 3. This will be negative pi over 3. But the one thing that I have to be very careful about is the following. Think about it. It doesn't make a whole lot of sense to say, I'm going to start at theta equals zero. I'm going to go pi-sixths, pi-fourths, pi-

thirds, pi-halves, don't know, don't know, don't know, pi, 7 pi-sixths, 5 pi-fourths, 4 pi-halves, 3 pi-halves, negative pi, negative pi, negative.

3.23 What are you talking about? Why in the heck are you going from zero to positive, and then suddenly over here, you're going negative? That doesn't make any sense. You don't count numbers by 1, 2, 3, 4, negative 2, negative 3, negative 5. That doesn't make any sense. You go 1, 2, 3, 4, 5, 6, 7, you go positive. You don't suddenly become negative at some point. How do I interpret negative pi over 6? Well, remember, the negative is just an indication of your direction. You're measuring clockwise. You get a negative pi over 6, that's clockwise.

3.24 The question then becomes, well, what if instead of going clockwise, I wanted to go counterclockwise? I want to go counterclockwise. Well, let's think about that. Remember, adding 360 degrees changes your angle, yes, but it doesn't change the point that you're staring at. Do we agree? It's like, if I'm staring in this particular direction and I do a 360, what direction am I staring in? The same direction. I'm at the same point. I haven't changed. If I go down negative pi over 6 and I point myself in that direction, well, that's going to be the same direction as if I go around 2 pi, do a 360, and stare at the same point.

3.25 Really, this value of theta is what? I take minus pi over 6, I do a 360, that's 2 pi, and I simplify that. What does that simplify to? We get a minus pi over 6 plus, OK, what do I need? I need a 6 over 6, so 6 times 2 is 12, so this is 12 pi over 6. That's really my 2 pi. I get negative 1 pi plus 12 pi is 11 pi. I get 11 pi over 6. That's the angle I want to take. Rather than viewing this as minus pi over 6, I'm going to view it as 11 pi over 6. They're different values, but they're pointing in the same direction. One is measured clockwise from the X axis, the other is measured counterclockwise from the X axis, but they're pointing at the same point.

3.26 Does that make sense, how that works? This will be theta is equal to 11 pi over 6. Similarly, how do I get this point? If I drop this one down, so let's erase this, if I drop this point down, again, everything going clockwise is going to go negative values. If I drop this one down, I get that point, that one there. This angle from here to here is pi over 4. Therefore, this one from here to here should be minus pi over 4. But minus pi over 4 is what? Do a 360. If I take minus pi over 4 and add 2 pi to it, that's really what? Minus pi over 4 plus 8 pi over 4, which is 7 pi-fourths. That's 7 pi-fourths. Great.

3.27 So there's theta. It is 7 pi over 4. Similarly, if I drop this point all the way down, and I take that line all the way through, what do I get? This will be theta equals—if this one was pi-thirds, then below that should be minus pi-thirds. Minus pi-thirds plus a 360 is what? Minus pi over 3 plus, what do we get here? 6 pi over 3. OK, so that's 5 pi-thirds. It's 5 pi-thirds. That's what this angle is here. This is 5 pi over 3. Now we know what the angles are. That's fantastic. But what are the coordinates of the points? What are the coordinates of the points?

- 3.28** After all, I know what this theta is. I know what that theta is. I know what that one is, but I don't know what the coordinates of that point is. Here's the beauty of it all. Because of the symmetry of the circle, what I notice is, look, the square root of 3 over 2 is my X direction. That's the distance from the origin out to right there. That's the square root of 3 over 2. Has that changed for this point? This point, the X distance, is still this distance. In particular, it's the same value. It's the square root of 3 over 2. I can say this point has coordinates, root 3 over 2, what? Well, what is the one-half representing? It's representing this distance here. That's positive one-half.
- 3.29** But by the symmetry of it, this distance and this distance are the same. It's just this one's negative. So this is going to be minus one-half. So the moral of the story is, all of these points are going to be the same as these points, except the Y coordinates are going to be negative. That's it. So this point is going to be 1 over the square root of 2 minus 1 over the square root of 2. And this point will be, let's see, one-half and then minus root 3 over 2. That's what we get. Now we're 75% done. What did we need to understand? We needed to understand the first quadrant for sure. We needed to know this.
- 3.30** We then needed to know the trick that, if I go across the origin, these points become negative. You negate the coordinates. I then needed to say, well, what if I take these points and I reflect them down here? Well, then I can get everything in the fourth quadrant. The only thing I did is, I negate the Y coordinate. That's it. But now you can see the game that we're going to play. Now I have points in the fourth coordinate. How am I going to get them in the second coordinate? I'm going to do the exact same thing I did in the first coordinate to get them into the third.
- 3.31** I'm going to take the line given by this and complete it through. I'm going to take the line given by this and complete that one through. I'm going to take the line given by this and bring that one through. What do I have? Now, remember, this value down here was 11π over 6, but the way we originally got to it was negative π over 6. We had negative π over 6. If I take negative π over 6, how am I going to get the angle that comes back up to this one? I'm going to add π to it. What do I end up getting there? Let's find the angle values first. Let's find the angle values first.
- 3.32** If I start with minus π over 6 and I add π to it, I need a common denominator, 6. It's π over 1. It's the same game we've been playing this whole time. This is what? Minus π over 6 plus 6 π over 6, that's 5 π over 6. That's my angle value right here. Theta equals 5 π over 6. What is this angle value going to be, straight up through here? Well, I originally go down negative π over 4, and I add π to that. What do I get? Minus π over 4 plus π , that's minus π over 4 plus 4 π over 4 is equal to 3 π over 4. I have three quarters of 1 π . This comes up here, and I get theta is equal to 3 π over 4.
- 3.33** What is this one going to be right here? OK, well, this is π -thirds, so this is minus π -thirds. If I take minus π -thirds plus π , it's minus π over 3 plus 3 π over 3, which is 2 π over 3. I take, this one is theta equals 2 π -thirds. Now I have the angle values. I have

the angle values, but what I notice is that now there's a symmetry this way. Now, there's a symmetry this way. Here, it's a miracle that this worked out. It's nice. Now, there's a symmetry there. What do I notice?

- 3.34** Well, if the coordinates of this point have the X coordinate is one-half, so that's the distance from here to here, and the distance from here to here is root three over two, well, notice, look, the distance there hasn't changed in the Y coordinate. The only thing that's changed is the X coordinate. Just like we had over here, the Y coordinates are changing. Here, the X coordinates are changing. What do I get for this point? I get negative one-half and then the same Y coordinate. What do I get for this one? I get negative 1 over the square root of 2 and the same Y coordinate.
- 3.35** For this point, what do I get? I get negative square root of 3 over 2, the same Y coordinate of one-half. There you have it. There's the entire unit circle and all of the values that you need to know. What's tricky about this is, if you just hand this to somebody and you say, "Oh, you need to know all these values," you look at it and you're like, "Where do you even start to memorize this? This is insane." There's way too much stuff going on here. You've got all of the angles. I mean, how many angles do we have? We have 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 angles. You've got 16 angles.
- 3.36** Corresponding to each one of those 16 angles, you have a tuple with an X and a Y coordinate that have two values. You have 32 things on top of that that you have to remember. You have 16 plus 32, that's 48 things that you have to remember. I can't do that. That's terrible. I can't remember any of that. What I can remember, though, is I can remember the first quadrant. I can remember those. How many angles are there? 1, 2, 3, 4, 5. Theta is zero, pi-sixths, pi-fourths, pi-thirds, and pi-halves, so five angles. And then corresponding to each of those angles, I have two points. So I have 10 things to remember there. I have 10 things to remember there.
- 3.37** Now, I can simplify that by saying, look, really, the only things I need to remember are what? One, zero, root 3 over 2, one-half, and 1 over the square root of 2. That's only five things. 0, 1, square root of 3 over 2, one-half, and 1 over root 2. I have five angles to remember, that's five things, and five values to remember, that's an additional five things, so 10 things total, as opposed to 48 things that we had to remember in the first place. Then as long as I remember how to go from a point here to the corresponding point here, you add pi. What does that do? You just negate the coordinates.
- 3.38** As long as I remember what? Maybe two or three more techniques, how to go from that point to this point, how to go from this point to this point, how to go from this point to this point, I maybe have to remember 13 total things as opposed to 48. That's a huge win. The unit circle is not this scary monster that's given by all of these different things that you have to memorize. You just have to remember the first quadrant and then a couple of tricks, and that's it. That's pretty amazing.

- 3.39 But what is the circle actually telling us? What is it telling us? It's saying the following. It's saying, look, if I want to say, "What is the sine of 11π over 6?" If I want to say, "What is the sine of 11π over 6?", then I have to step back and say, "OK, think unit circle." I know that sine is what? This is the Y coordinate on the circle, $X^2 + Y^2 = 1$. It's the Y coordinate on the circle at the angle 11π over 6. OK. Then I draw my circle. I've got my circle. I find the angle 11π over 6. I have to remind myself, OK, I start at the positive X axis. I go around counterclockwise until I hit π over 6. Boom. π over 6.
- 3.40 I'm saying, what is the sine? I'm saying, what is the Y coordinate? What's my answer? Negative a half. It's equal to negative a half. How do I remember this personally? Because if you just came up to me and said—no one ever does this, by the way, there's always this hypothetical "if someone comes up to you on the street and asks you this question, what would you say?" No one asks you math questions on the street. That's not a thing, OK?
- 3.41 So sine of 11π over 6, if somebody just came up to me and said, "Hey, what's sine of 11π over 6?" I don't immediately just rattle off, "Oh, it's negative a half." I don't. Because frankly, I don't remember all of that. What I remember is, I say, OK, 11π over 6, where does that occur? OK, that occurs all the way down here. OK, 11π over 6, good. Whatever this point is, I know that there's a corresponding point up here in the second area in the first quadrant, which is going to be given by the angle, if this is 11π over 6, which is really minus π over 6, this is going to be positive π over 6.
- 3.42 I say, oh, what is the sine of π over 6? It's one-half. But because I'm in the fourth quadrant, I need to negate it. So I get a negative half. That's how I remember this. You remember the first quadrant and then the tricks to get from the first quadrant to the second, or the first to the fourth, or the first to the third. If you remember those tricks, then you can just find out what the values are accordingly. It's a little slick that way. All right. Another one. OK, another one.
- 3.43 This one's kind of mean, but here's another one. OK, so here's an example. I want you to find the cosine of, let's see. Let's do one like—how would we do this? Let's do 22π over 3. OK? 22π over 3. Now, to be brutally honest, how would I do this, personally? I would just plug it into a calculator, get my answer, and be happy with that. I'm not going to sit there and be like, "Oh, here's the, let me think through it. Let me do this." I'm actually just going to figure out, what is the value on the nose, plug it into a calculator, get my answer, I'm happy.
- 3.44 But if I don't have that, if I don't have a calculator, then I have to look back and I say, OK, well, let's think about this for a second. I don't see 22π over 3 as one of my possible angles. I don't see that. Then what am I really being asked? I mean, is this even solvable? Can I even say, like, what cosine of 22π over 3 is? The question is, OK, well, how many times does 3 go into 22 without going over? How many π s am I talking

about here? How many pis am I talking about? OK. What I can say is, well, look, if 3 goes into 22, how many times? This is how I would think about this.

- 3.45** If I say 22 divided by 3, because I want to know how many pis are we talking about? Well, 3 goes into 22 seven times. 3 times 7 is 21 without going over. What is this? This is 7 plus what? Well, it goes in 7 times, that's 21, and then I need 1 leftover. I need 7 plus one-third. Really, all I'm doing is this, how do we do this arithmetic? How do we teach people to do this? We say, OK, 3, and then we draw this goofy little symbol, and we put a 22 here, and we do the long division. Then we say, OK, how do we do this? 3 goes in 7 times without going over. 7 times 3 is 21, and then I subtract and I get a 1. There's my remainder, and that's where the 1 is coming from here. Then that's 1 over 3.
- 3.46** Just as a sanity check, I can say, OK, what am I really doing here? The 7 plus one-third, if I combine those, this is really, what, 21 over 3? If I get a common denominator, plus 1 over 3, that's my 22 over 3. So as a sanity check, I can be like, OK, I got it right. It's 7 plus one-third. OK? OK, so let's get rid of this. And what can I say from there? Well, that means that 22 over 3 times pi, so 22 pi over 3, is really what? 7 plus one-third pi. The distributive rule tells me that 7 pi is plus pi-thirds. That's what I get. OK?
- 3.47** OK, so maybe that helps. I'm really saying, OK, what is the cosine? Let's eliminate this. Let's erase this. Now, I'm really saying, what is cosine of 7 pi plus pi over 3? That's really what I'm asking now. That's the same thing as cosine of 22 pi over 3. All right. Well, remember, we know that cosine is 2 pi periodic. Remember, we had this formula that told us, if I take cosine of theta plus 2 pi, that's the same thing as cosine of theta. So maybe I can somehow exploit that. I don't know.
- 3.48** We said that, even if you multiply this by an integer, if K is an integer, it's cosine of theta plus 2 pi K, if I go around the circle 2 pi or K times in one direction or K times in the other direction, the cosine is going to be the same. I'm pointing at the same location. The angle is just different. That's it. Can I somehow make this look like a 2 pi times something? Well, as is, not really. But I can write this as what? Cosine of 6 pi. I can take out 6 of the pis. If I have 7 pis, I can take 6 of them out and I have 1 leftover. Plus pi plus pi-thirds. You're with me?
- 3.49** I take the 7 pis. You imagine, I mean, literally, you can think of actual eating pies. I have 7 pies in front of me. What do I do? I take 6 of them, put them in one bin, and I have one pie left over. Now this looks nice. This looks like cosine of pi plus pi-thirds plus 2 pi times 3. There's my 2 pi K. K is equal to 3. Now I can actually say something here. What is this equal to? By this formula here, this becomes cosine of pi plus pi-thirds. Because I have the 2 pi times 3 on the back end of that, so that can go away.
- 3.50** If I simplify this, this becomes cosine of—I need 3 pi over 3, so this is 4 pi-thirds. Then I can go back and say, oh, sweet, 4 pi-thirds, that's right here. That's actually on my unit circle. I can say cosine of 4 pi-thirds is the X coordinate, so that's going to be minus a

half. This is equal to minus one-half. You can always simplify these things down. You can always simplify these things down. Now, that was kind of involved. OK, cosine of 2π over 3. We had to break it down into, how many 2π s do I have?

3.51 Basically, what this is saying is, start at cosine of 4π over 3. Start at the angle 4π -thirds and do what? Go around three times. Go around three times. That corresponded to my 2π times 3. That gave me 2π over 3. But if I just go around three times, it doesn't matter. I'm still going to have the same coordinate right here. It's still going to be my negative one-half. That's good. Then what can I say about this? What can I actually do here? What can I actually do? And what can I not do? Well, I can ask you what 2π over 3 was. That's fine.

3.52 But what about something to illustrate how fragile this is? To illustrate how fragile this is, what if I said something like this? As an example, I want you to find cosine of π over 10. Cosine of π over 10, right? π over 10 is a valid angle. π over 10 is a valid angle. What is it as an angle? It's π over 10 radians. Now, remember, how do you convert from radians to degrees? You know that there are 180 degrees in π radians. So the radians cancel. That's good. This is equal to, the π s cancel, the 10 cancel. This is 18 degrees. That's 18 degrees. It's a valid angle. That's completely fine.

3.53 I want to know what is the cosine of that angle. I can draw a triangle associated to it. I have an angle of 18 degrees. That's fine. It's a right triangle. The hypotenuse is 1. I want to say, what is this thing here? This is the cosine of 18. It's that side. I want to understand what is that. No clue. No clue. I can't look on the unit circle. I don't see a π over 10 anywhere. I don't see 18 degrees anywhere. 18 degrees is what? If this is 30, then 18 is, I don't know, somewhere in here. It's somewhere over there. I could say, the cosine, it's got to be bigger than $\sqrt{3}$ over 2. It's probably close to 1, but I have no idea what it is. I have no idea what it is.

3.54 That's what's kind of illustrating, this is super fragile and somewhat gimmicky to some extent, because there are only certain values of θ that I can evaluate. Things that have a 4 in the denominator, things that have a 6 in the denominator, things that have a 2 in the denominator, and things like that. If I throw a 7 in the denominator, π over 7, no chance. π over 10, no. Now, there are certain formulas that will allow us to evaluate certain types of angles, but right now, we're really stuck with the unit circle ones.

3.55 We'll develop a method a little bit later that you give me any angle whatsoever, I'll be able to solve for it. I'll be able to tell you what the cosine is, what the sine is. But we need some machinery in order to get that out of the way. We'll get there. But the question I want to ask is the following. Well, look, whenever we have a new class of functions, whatever those are, if you think back to algebra, we had polynomials, which included the quadratics, degree one things. Then immediately, once we had those, we said, OK, how do I solve equations involving those? How do I solve equations? How do I solve quadratic equations?

- 3.56** Well, there's a quadratic formula and there's factoring and there's complete the square. OK, how do I solve degree one terms? Well, you move everything to one side and try and isolate the variable. Great. How do I solve exponential equations? How do I solve logarithmic equations? That's everything that we addressed from an algebra standpoint. Now we have this whole class of functions, sine, cosine, tangent, secant, cosecant, cotangent. Well, it's natural to say, well, how do I solve equations involving those?
- 3.57** What if I have an equation involving sine? How would I solve it? What if I have an equation involving cosine? How would I solve that? Let's just explore that. Let's figure out how we would do that. If we can do it—maybe it's a completely worthless endeavor. Who knows? Who knows? You're not going to know unless you explore it. Let's figure that out. Let's ask a very seemingly straightforward question. Example, find all solutions to sine squared of theta is equal to 1. Let's start off relatively simple.
- 3.58** Let's just take sine, let's square it, and set it equal to 1, and see if that makes sense. Maybe it does, maybe it doesn't. Maybe there's no reason to even do this. Maybe this is completely worthless. I shouldn't do this at all. I don't know. Find all solutions to sine squared of theta is equal to 1. How can we approach this? What is the game here? The game is, give me an angle theta such that when I plug it into sine and I square that result, I get 1. That's the game. How can I approach this? Well, I'm going to show you a way to approach this, which uses algebra and the stuff that we covered in the algebra course.
- 3.59** How would we do this? Sine squared of theta is equal to 1. I don't like sine. Sine scares me. I'm going to give it a variable name. I'm just going to call it something else. That's my first strategy. It's just, let's pretend that sine of theta is not sine of theta, it's just a variable that I don't want to deal with. I'm going to call it X. Just a variable I don't want to deal with, I'm going to call it X. I'm going to say, let X equal sine of theta. Why is that useful? Well, because now what does this equation become? This equation now becomes X squared is equal to 1. Do we agree? Now it becomes the equation X squared is equal to 1.
- 3.60** Now, I'm happy with that. Why am I happy with that? Well, because I know what the solutions are. I know what the solutions are. I know that this means that X equals 1 or X equals minus 1. The variable X is squared and that equals 1, which means X is either negative 1 or it's positive 1. OK, that's great. Now, how does that help me? Well, that helps me because I know what X is. X is the sine of theta. In particular, X equaling 1 or X equaling minus 1 is telling me what? This is saying either sine of theta equals 1 or sine of theta is equal to minus 1. OK.
- 3.61** Now I have to go back and say, OK, do I know values of theta such that when I plug them in, I get a plus 1 or a minus 1 on the unit circle? That's the question you have to ask yourself. Part of becoming, let's say, more— how do you say? Part of becoming more confident with mathematics is knowing how to ask yourself the right questions,

asking yourself the right questions. Probably one of the most common things I get all the time is that students will tell me things like, "I just don't even know how to start a problem. I just don't even know how to start."

- 3.62** Well, here's how you start. Ask yourself a question. What is the question asking me to do? The question is asking me, when is sine squared equal to 1? What does that mean? Well, it's really saying, find a number such that when you square it, you get 1. That number we're going to call sine, so that's X , and find a number such that when you square it, you get 1. Plus or minus 1. But that number had a name. That number was sine of theta. So now I'm really saying, find a value of theta such that sine of theta is 1 or sine of theta is negative 1.
- 3.63** OK, well, that's great. Now I have to ask myself the question, what do I know about sine? Well, I know sine comes from right triangles, and it happens on the unit circle as well. There's a connection between the two. Oh, OK, great. So now I have sine of theta is equal to 1 and sine of theta is negative 1. So I go back and look at my unit circle. And I say, sine of theta is the Y coordinate. Oh, so it's saying, look at the unit circle and find angles where you have a Y coordinate of 1 or minus 1.
- 3.64** I come back here and I say, let's look at the unit circle. I need a Y coordinate of minus 1, and I need a Y coordinate of 1, positive 1. The question is asking me, find an angle such that the Y coordinate is 1 or minus 1. In particular, what is the angle corresponding to the Y axis? That angle is π over 2. When is sine of theta equal to 1? Well, theta is equal to π over 2. We solved that one. That's fine. Now, what is the angle down here? We didn't write the angle down here, but I have π over 2, 2π over 2, 3π over 2. So this angle down here is theta equals 3π over 2. And that happens to be where sine is equal to negative 1.
- 3.65** So over here, I can say, oh, and that's going to be when theta equals 3π over 2. That's great. Have I given you all the answers? The answer is, not really. Why not? Well, because if I want to go to π over 2, what if I'm standing at π over 2 and I go another 2π ? Now I'm at 5π over 2. What if I go another 2π ? Now I'm at 9π over 2. Another 2π , 13π over 2, 17π over 2, 21π over 2. I can just keep going over and over and over again. Or what if I'm at π over 2 and I subtract 2π ? Now I'm at negative 3π -halves. Subtract another, negative 7π -halves, and so on.
- 3.66** Really, what do I need to do? Theta is equal to π over 2 plus any integer multiple of 2π . I can go around 2π positively, or I can go around 2π negatively. That'll generate all of the solutions to this one. But then what about this one? Well, 3π over 2 plus, again, another 2π , some integer will generate all of the solutions to this. A lot of times you'll run into questions which will say things like this. Find all solutions to sine squared of theta is equal to 1 where theta is in between zero and 2π . It restricts the values of theta that you're allowed to consider.

- 3.67** In which case, what would my answers be? Well, $\pi/2$ and $3\pi/2$ would be the only answers that I can generate from there. OK, so we kind of can solve equations a little bit. Kind of. Except then you realize, OK, well, wait a second. What if I change the 1 to, like, $1/5$? Something like that. What if I changed it to that? I want to say, I want values of θ for which $\sin^2 \theta$ is equal to $1/5$. You're like, OK, do the same trick. Do the same trick. You're looking for values of θ such that $\sin^2 \theta$ is equal to $1/5$.
- 3.68** So I'm really saying, solve the equation $X^2 = 1/5$. I'm looking for X is—I'm looking for when $\sin \theta$ is $1/\sqrt{5}$, and I'm looking for when $\sin \theta$ is equal to $-1/\sqrt{5}$. You quickly realize you're stuck because you look around the unit circle and you're like, OK, where's my $\sqrt{5}$? Where's my $\sqrt{5}$? I don't have one. There's got to be something we're missing. There's got to be something we're missing here.
- 3.69** These things, do they have a solution? That's another question. Do these things even have a solution? Well, let's think about that. $1/\sqrt{5}$ is less than 1. Square root of 5 is like 2 and change. It's like 2-point something. I'm taking 1 divided by 2-point something, that's less than 1 for sure. The sine coordinate is the Y coordinate. I'm saying, let's take on the Y axis, let's take something like this, $1/\sqrt{5}$. Can I come up with solutions for which $\sin \theta$ is equal to $1/\sqrt{5}$? It's like, yeah, sure, I can.
- 3.70** It's going to give me this point here, and it's going to give me this point here. But they just happen to not be angles that I know. It's not going to be $\pi/6$. Maybe it's $\pi/8$, or $\pi/9$, or $\pi/10$, or something. I don't know what it is. Maybe it's not even a π over anything. So there's got to be a better way for us to be able to approach these types of questions. And there is. We need to be able to understand, how do you somehow start with an equation that looks like this, turn it into equations that say $\sin \theta = \text{blank}$, and somehow eliminate the $\sin$ so that you can solve for θ ? You can solve for θ .
- 3.71** That's the notion of an inverse trigonometric function. Can I somehow invert $\sin$? I don't mean invert by $1/\sin$. I mean, can I somehow eliminate the $\sin$ part so that I can isolate θ ? Can I do that? Is that even sound? Does it make any sense? The answer is, well, yes, but you have to be very precise. You have to be very precise into what you mean by, can you actually invert the $\sin$ function? If you go back to algebra, when we introduce functions, we say that you can invert a function, if the function satisfies what? It takes an input to a unique output, and it doesn't send any another different input to that output.
- 3.72** If you can do that in a precise way, then you can invert these functions and solve for what their corresponding θ values are. That's not easy. That's not easy whatsoever. It's the notion of inverse trigonometric functions. It requires us to talk about, how do you graph trigonometric functions— $\sin$, $\cos$, $\tan$? How do you actually graph

what they look like? Which we'll do, concretely. How do you graph them? Then how do you make it so that they're invertible? How do you invert them? How do I find the inverse of sine? How do I find the inverse of cosine? How do I find the inverse of things of that nature?

- 3.73** We'll get into that. We'll get into that, but it takes time. There's one more thing that I want to mention, and it's the following. Let me illustrate it with the unit circle. I know this is busy. I know this is busy, but let me illustrate it in the following way. What do I notice? Well, remember, the way we measure angles is we go positive from the counterclockwise direction, negative from the clockwise direction. In particular, what I notice is, look, if I take the cosine of π over 6, OK, cosine of π over 6, I know what the value is. It's $\sqrt{3}$ over 2. Great. That's fine.
- 3.74** But what if instead I plug in cosine of minus π over 6? Can I say that those two things are equal? Well, let's look at it from the unit circle standpoint. Cosine of π over 6 is the X coordinate of this angle, which is $\sqrt{3}$ over 2. If instead I go minus π over 6, I also get $\sqrt{3}$ over 2. That seems to be true. This left side is $\sqrt{3}$ over 2, the right side is $\sqrt{3}$ over 2. That's fine. What if I did that for cosine of π over 4, versus cosine of minus π over 4? Same reasoning. π over 4, 1 over $\sqrt{2}$, cosine of minus π over 4, 1 over $\sqrt{2}$. Those are equal to each other. You can keep playing this game.
- 3.75** The question is, well, is this just a general fact? Is that true? If you take cosine of a minus angle, is that the same as cosine of the positive angle? Well, based on this, it doesn't matter what the angle is. It doesn't matter what the angle is. If you give me the unit circle, you give me the unit circle, and you give me a value of θ , whatever that is, well, minus θ is right here, and cosine is the X coordinate. So in particular, what? Down here, this looks like cosine of minus θ , sine of minus θ . This point here looks like cosine of θ , sine of θ .
- 3.76** And what I notice is, by the symmetry, that X coordinate has to equal that X coordinate. So in particular, yeah, that's true. Moreover, what can I say? Does sine have something similar? Does sine have something similar? Well, the answer is like, it's similar, but not exactly. Why? Well, look, sine of θ is here, sine of minus θ is here. Those aren't equal. Those aren't equal to each other. Sine of θ is positive, sine of minus θ is negative. But what do I notice? Sine of minus θ , that's just negative of sine of θ . What I notice is that sine of minus θ is equal to minus the sine of θ . That's true.
- 3.77** We can check that on the unit circle. Look, what is sine of minus π over 6? It's negative a half. What's sine of π over 6? That's positive a half. Those are negatives of each other. If I say, what's sine of negative π -fourths? It's negative 1 over $\sqrt{2}$. Well, that's negative sine of π -fourths. So they're negatives of each other. These two notions, sine of minus θ equals minus sine of θ , and cosine of minus θ equals cosine of θ , these two notions are the function definitions for calling a function even versus calling a function odd. These are the definitions.

3.78 A function is even if, when you plug in negative its input, it doesn't change. It's cosine of the original θ . Whereas a function is odd if, when you plug in negative the input, you get negative the output, negative sine of θ . Those are definitions. They have geometric intuitions for what they actually mean in terms of their graphs, which is what we're going to explore going further.

Lecture 4: Complex Numbers

At a glance — Complex numbers.

- 4.1 All right, we're going to switch gears a little bit and talk about complex numbers, which seems completely unrelated to trigonometry, but they're, they're connected in a very precise way. What we want to introduce is, what are complex numbers? What are complex numbers? Well, sometimes complex numbers are called imaginary numbers. I'll put them in quotes because I actually don't like this name. But historically, that's how they were adopted.
- 4.2 To understand where they come from, you need to understand, to some extent, the history of arithmetic, or at least, at the very least, the basis of where negative numbers came from, where negative numbers came from. If you take yourself all the way back, you start asking yourself these questions like, what is a negative number? What is negative 5? What is negative 3? What is negative 2? It's like, if you start asking yourself these questions, you're like, well, what does that actually mean? What does it mean to have a negative 5 or a negative 3 or negative 2 or something along those lines?
- 4.3 It makes sense that we can actually make sense of saying what 1, what 2, what 3, what 4, what 5 actually mean. Historically, it's like, well, if I have 5 loaves of bread, then I can say there's 1, there's 2, there's 3, there's 4, there's 5. I have some quantifying notion of what the number 5 represents. It's 5 fill-in-the-blank objects, whatever it is. But then the question is like, well, OK, so then what does it actually mean to say I have a negative? Just basic arithmetic, I can have something like 5 minus 8. We would say, what's 5 minus 8? We'd say, oh, well, basic arithmetic says, oh, that should be minus 3. OK, minus 3. But what does that actually mean?
- 4.4 Well, I think probably one of the best ways to understand this is in terms of accounting, for example. It's in terms of accounting. It's like, if I have \$5 and I owe you \$8, right? I take \$8 away from \$5. Well, it's like you can take my \$5 away from me, and then I can interpret negative 3 as being like, I owe you three bucks. I owe you \$3, right? That's a natural way to understand negative numbers. It's like, OK, I have 5 of something. You want to take away 8 of those things. You can't take away 8 of those things because I only have 5 of those things. So what you can do is, you can say, I'm going to take your 5, and then you're going to owe me 3.
- 4.5 That's how we can make sense of negative numbers. You can think about them also in terms of directions, for example. If you say, I'm walking down the street and I get to an intersection and I need to turn north, for example, but instead of turning north, I turn south; and rather than walking three blocks north, I walk three blocks south, then I can think of negative 3 as being, I walked negative 3 blocks in the wrong direction. That's another way that you can visualize what negative numbers actually represent.

- 4.6 Then complex numbers, they came about surprisingly in the 1500s. There's a mathematician by the name of Girolamo Cardano, an Italian mathematician. What he was studying was quadratic equations. He was studying solutions to quadratics. If I give you a quadratic, if I give you a quadratic, so we did this in algebra. If I say $AX^2 + BX + C$ —that's the general form of a quadratic, so a quadratic is a degree two polynomial, where I'm assuming A is non-zero because if A is zero, then it's really only a degree one. It's not a quadratic anymore. I'm assuming A is non-zero.
- 4.7 I ask myself the question, what are solutions to this? Again, A is non-zero. I ask myself, what are solutions to $AX^2 + BX + C = 0$? The quadratic formula gives them to me. The quadratic formula gives me solutions to them, which is quite nice. I can say very simply that X is equal to $\frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$. That's what the solutions are. Fantastic.
- 4.8 But the question becomes, OK, well, that's all fine and good, except for the fact, what happens when that's negative? That's the natural question. We can take square roots of numbers. That's fine. The square root of 5 is what it is. It's 2 and change. The square root of 1 is 1. The square root of 9 is 3. The square root of 14 is goodness knows what. It's between 3 and 4. I don't know. But what's the square root of negative 7? What's the square root of negative 12? What's the square root of negative π ? What's the square root of negative 1, for that matter? Does that make sense?
- 4.9 The question is, what is the square root of a negative number? What is the square root of a negative? We can sit here all day long and we can say, OK, what's the square root of negative 6? What's the square root of negative 5? What's the square root of negative 12? Things like that. We can try and define what these objects are. We can say the square root of negative 6 is fill in the blank, or square root of negative 5 is fill in the blank. Maybe those are specific to those particular problems. There's not something general that I can pull out from this and say, oh, this is a common feature amongst all of these things.
- 4.10 Except for the following thing. Remember, the square root function behaves like this. The square root function is multiplicative. The square root of a product is the product of the square roots. OK, so what does every single one of these objects have in common? They're all negative. 5, 6, and 12 are all different numbers, but they all have a negative. In particular, I could write this as the square root of what? Minus 1 times 6. This one's the square root of minus 1 times 5. This one's the square root of minus 1 times twelve.
- 4.11 And now I see, ah there's a pattern: they all have a negative 1. And if I stick with this rule for square roots I can say OK well this is the square root of minus 1 times the square root of 6; this one's the square root of minus 1 times the square root of 5; this one's the square root of minus 1 times the square root of 12. Now the square root of 6, square root of 5, square root of 12, those make sense.

- 4.12 Square root of 6 is somewhere in between 2 and 3. Square root of 5 is somewhere between 2 and 3. Square root of 12 is in between what? 3 and 4, somewhere in there. It's that technique we did earlier. Start squaring numbers, 1 squared, 2 squared, 3 squared, 4 squared, and find where these numbers live in between. But what's the common feature? This guy. That's common to every single one of them. It's common to every single one. What Cardano came up with is, he said, oh, OK, I'm just going to define this thing to be a particular number. It's going to be a symbol. I'm just going to declare a symbol to equal the square root of negative 1.
- 4.13 That's what he did. That's what he did. He said, OK, we're going to define the complex unit "i" to be defined as the square the square root of minus 1. He defines a number, he calls it i, and he says that it's equal to the square root of minus 1. In particular, what he's really saying is, he's saying, look, when I square i, I get minus 1. That's what he's saying. He says, that's precisely what I need to be able to reconcile how to write these particular numbers in a systematic way.
- 4.14 Now, I could say this. I could say this is equal to i times the square root of 6. This one's i times the square root of 5. This one's i times the square root of 12. OK, so those things have a definition now. Now, I can handle in the quadratic formula when B squared minus 4 AC is negative, I have this symbol, i, that I can use. Now, why were they called imaginary? It's funny because if you actually read through Cardano's explanation as to, "Oh, I've come up with this symbol, and it satisfies this property," he's basically saying he's come up with i, which is, i is equal to the square root of negative 1.
- 4.15 You can even see in his writing, he's saying, "I've come up with this imaginary thing such that it satisfies these quadratic equations." Hence the name imaginary numbers. Now, again, this is the 1500s. Typically, numbers were things that they wanted to be able to write down and say, this represents the number 7. This represents the number the square root of 2. I can draw it on the ground. I can come up with an area that has that particular thing, pi, e, all of these different types of numbers. There was a geometric interpretation of it.
- 4.16 So, when you have something like i, you're not writing down, oh, this is what i is. It's this length of this particular thing. It's this area. It's this thing. People are going to think it doesn't exist. It's imaginary. It's just some fantasyland-type thing. But it turns out to be exactly what you need to solve these particular equations. That's what introduces what are called complex numbers. These two things are synonymous. Imaginary numbers are complex numbers, but I would say the most common form of explaining these things is now complex numbers, not imaginary numbers.
- 4.17 Let's get into those. What the heck does this have to do with trigonometry? We'll see. All right, so here we go. Here's your definition. Here's your definition. A complex number. A complex number is one of the form $A + Bi$, where A and B are arbitrary real numbers. They're arbitrary real numbers. Any complex number looks like an A plus Bi. We have names for these things. A is called the real part. A is called the real part.

And B, not Bi , B, the coefficient of i , is called the, unfortunately, the imaginary part. Why? Because it's the thing attached to the imaginary number i . So B is the imaginary part.

- 4.18** The beautiful thing about this is that A plus Bi , you can represent geometrically. You can write them down. How do you write them down? The graph of these things is that, let's just grab some points. Let's say 2 plus $3i$. Let's grab, let's say, minus i , and let's say 1 minus i . Those are three complex numbers. Why are they all complex? Well, this one is presented to you in the form A plus Bi , so A is 2 , B is 3 , so A is 2 , B is 3 . This one is confusing. The minus i , you're like, that's not the right form. It's like, well, it is. It's just, it's kind of goofy.
- 4.19** It's zero plus minus 1 times i . A is zero, B is negative 1 . So A is zero, B is minus 1 . And 1 minus i , this is better, but it looks like 1 plus minus 1 times i . So in this case, A is 1 , B is minus 1 . Does that make sense? So you can always make it look like an A plus Bi , for an appropriate A and B . But in order to graph them, the beautiful thing is that, if you understand how to graph points in the Cartesian plane, the XY plane, then you know how to graph these. Because the X coordinate is A and the Y coordinate is B .
- 4.20** How do we graph these? Here's my X axis, if you will. Here's my Y axis. What is the point 2 plus $3i$ look like? I go over 2 in the X axis and up 3 on the Y axis. That point there is what I call 2 plus $3i$. I go over 2 , up 3 . What do I call the point negative i ? I go over zero in the X direction, so I don't move in the X direction, and I go down 1 in the Y direction. Where is minus i ? It's right here. That's minus i , right there. Where is 1 minus i ? Well, I'm going to go over 1 in the X direction and down 1 in the Y direction. So I go over 1 , down 1 . That's this point here, 1 minus i .
- 4.21** There's a relationship between complex numbers and points in the XY Cartesian coordinate system. That's pretty amazing. Now we have these new objects. We have these complex numbers. They turn out to help solve quadratic equations. That's a relationship. OK, well, what can I do with them? Whenever you have a new mathematical object, you want to understand, what is the arithmetic you're allowed to do with these objects? Can I add them? Can I subtract them? Can I multiply them? Can I divide them? Can I do arithmetic operations on these new objects?
- 4.22** OK, and the answer is, sure. Yes, you can. How do you do that? The arithmetic of complex numbers. Let's talk about the arithmetic. It sounds crazy. It sounds crazy because the math that we do on a regular basis is typically addition, subtraction, things like that. I make X amount of dollars a month. I have Y amount of expenses a month. Therefore, I'm going to have X minus Y dollars at the end of the month after I pay my expenses. We typically do basic arithmetic.
- 4.23** It's weird to ask the question, well, does it make sense to multiply? Does it make sense to subtract? Does it make sense to add? It's like, well, yeah, of course it does, because everything you do on a regular basis does have those properties. It turns out the

complex numbers do as well, but there are mathematical objects for which those operations are not defined. We have to spell that out. If I give you a complex number A plus Bi , and I say I want to add that to C plus Di , how do I do it? You think about it like polynomials. You think about it like polynomials.

- 4.24** How do you combine polynomials? You add like terms. You take the pieces that look the same, you apply the operation to those pieces, and then you combine that into the whole thing. I look at this and I say, OK, look, A and C are not attached to i , so A and C are like terms. I'm going to add those. This becomes A plus C . Now, I see a Bi and a Di . Those are "like" terms. They both have an i , and I'm going to combine those. I'm going to get B plus D times i . That's how you do it. That's how you do it.
- 4.25** Then I ask the question, well, how do I subtract? I'm going to cheat here and show you how that works. If instead I said A plus Bi minus C plus Di , what am I going to do? I'm going to take A minus C , so it'll be A minus C , plus B minus D . I'm giving you two formulas here. I'm giving you two formulas. You go along the top level of the operations to do the addition, and you go along the bottom level to get the subtraction, so A minus C plus B minus D . Again, you subtract like terms.
- 4.26** The interesting one is multiplication. If I take A plus Bi times C plus Di , how do I do this? Well, again, if you think about them like polynomials, you think, oh, OK, well, polynomials and distribution in terms of factoring, how did I do that? I took this term, I multiplied it into that term. OK, so I'm going to get an AC . I'm going to take A and multiply it into Di , so I'm going to get a plus ADi . I'm going to get Bi times C , so that'll be a plus BCi . Then I'm going to get this times that. What am I going to get? BD , and then i squared, i times i . This is going to be plus BD times i squared.
- 4.27** That's scary because you're like, whoa, that's not a complex number. Complex numbers have to look like number plus number i . There's no i -squareds allowed. But then you say, well, what was i squared? i was defined to be the square root of minus 1, so i squared is equal to minus 1. I can replace this with minus 1. What do I get? I get AC minus BD plus AD plus BC times i , because this i -squared makes that a negative. That's what my multiplication rule is. That's what my multiplication rule is.
- 4.28** Now, is there division? The answer is yes. How do I do it? The answer is, it'd take us too far afield from where we're going right now. It's not super complicated. But yes, you can divide complex numbers, but that's not the point. The point is, we want to come up with a relationship between complex numbers and trigonometry. And trigonometry. Just to illustrate this, let's do a silly example. We've got these two perfectly good complex numbers here, 2 plus $3i$ and 1 minus i . Let's add them, subtract them, and multiply them, to show what we get.
- 4.29** Number 1, if I add them, 2 plus $3i$ plus 1 minus i , what do I get? Well, the 2 and the 1 combine, so I get a 3 , 2 plus 1 , so that's a 3 . Then $3i$ plus negative $1i$, that's going to be a $2i$. I have $3i$ s, I take away 1 , so I'm going to get a plus $2i$. There it is. If I subtract them, 2

plus $3i$ minus 1 minus i , what do I do? 2 minus 1 , so 1 . $3i$ minus minus i . Remember, minus minus goes plus. So this is really $3i$ plus i . I have three i 's and I add one more. I get four i 's. The last one, if I multiply, 2 plus $3i$ times 1 minus i .

- 4.30** I'll be brutally honest, you never remember the formula. You just do it. You just do it. What do I mean by that? Well, I'm going to take the 2 , I'm going to bring it in. So 2 times 1 is 2 . 2 times minus i is minus $2i$. $3i$ times 1 is a plus $3i$. And then $3i$ times minus i is what? I get a negative $3i$ squared. I get minus $3i$ squared. Now I have to go back and say i squared is negative 1 . OK, i squared is negative 1 . This is what? 2 minus, let's see, what do I get? Negative $2i$ plus $3i$ is a plus i . Those give me a plus i .
- 4.31** I get 2 . What do I get here? Minus 3 times i squared, which is minus 1 . That's really a plus 3 . Then I'm going to get 2 plus 3 is 5 . Then I have an i hanging off the back end of that. Multiplication is hard to work out, but we have a concrete way of doing it. We do the distributive law and then the i squared is negative 1 . That's our arithmetic. Again, where's the trigonometry? Where's the sine? Where's the cosine? Where's all the fun stuff? Well, quote, "fun" stuff. Sine and cosine. How do I do that? I'm going to leave that graph here. I'm going to leave that graph here. I'm going to do the following.
- 4.32** All right, so let's think about it. If I give you an arbitrary point, let's say it's right here, and I tell you this is A plus Bi . I give you that point. What do I know about it? Well, I know that the A was obtained by going out this distance, A . I know that the B was taking this distance, B . In particular, if I draw a straight line to that point and I drop this down, I have a right angle here. I have an angle here, and I need to explain what is this distance here. Well, if this is A and that's B , Pythagorean theorem tells me this is equal to the square root of A squared plus B squared, by the Pythagorean theorem.
- 4.33** Now I have a right triangle running around. Trigonometry is going to reign supreme. What do I know? Well, let's give this thing a name. Let's call this thing R . Let's call it R . From a complex standpoint, it has a name. R is equal to the square root of A squared plus B squared. It's equal to vertical bar A plus Bi , vertical bar. This thing has a name. It's called either the absolute value. It's either called the absolute value or what's called the modulus of A plus Bi .
- 4.34** It doesn't really matter what you call it. It's really just saying, what is the distance from the origin out to the point A plus Bi ? That's really all it is. But we give it a name. We call it the modulus. That's fine. I'm just going to call it R . Really, what do I have now? I have a right triangle, where I have, here is theta, here is R , here is A , here is B . In particular, if I take sine and cosine of the angle, what happens? I get sine of theta equals what? Sine of theta is the opposite over the hypotenuse. That's a SOHCAHTOA part of it.
- 4.35** Opposite is B . Over the hypotenuse is R . In particular, I can solve for B . B is equal to R sine of theta. B is equal to R sine of theta. Similarly, if I take cosine of theta, if I take cosine of theta, that's going to be what? That's my adjacent over the hypotenuse. Adjacent is A over the hypotenuse, so A over R . That tells me that A is equal to R cosine

of θ . Now, think about that. The complex number, $A + Bi$, can now be written as what? A is what? $R \cos \theta + R \sin \theta i$. We know what A and B are. Well, R is common to both.

- 4.36** I'm going to bring the i out in front. It doesn't matter. That's fine, whether the i comes first or the coefficient comes first; it doesn't matter. So I get that. $R \cos \theta + i R \sin \theta$. Now, this quantity right here is special. This quantity is special. It comes from a particular formula that's called Euler's formula. Euler's formula tells me that $e^{i\theta}$ is equal to $\cos \theta + i \sin \theta$. That's what Euler's formula tells me. $e^{i\theta}$ is equal to $\cos \theta + i \sin \theta$.
- 4.37** So e is just a number. It's 2.718 and change. It's just a number. I'm looking at the exponential e raised to the $i\theta$. That is given by $\cos \theta + i \sin \theta$. In particular, this complex number looks like $Re^{i\theta}$. Where θ is what? θ is the angle from the positive axis up to whatever the point is. R is the distance to the point given by the Pythagorean theorem. That's all fine and good, but let's actually figure this out. Let's actually figure this out. Let's look at the number $1 - i$, and write it in this form. Let's write it in this form. Let's see how that works.
- 4.38** Here's the example. Write $1 - i$ as $Re^{i\theta}$. Write it as $Re^{i\theta}$. Basically, what I'm saying is, find R and find θ . Find R and find θ . Let's think about this. If I graph $1 - i$, so let's graph it a little bigger. Here's 1, here's -1 . This is the point down here. This is $1 - i$. This distance is 1, that's -1 . R is the distance from the origin out to the point. That's R . How would I find that out? Well, Pythagorean theorem tells me how to do that.
- 4.39** If I think about this right triangle right here, I have this length is negative 1, but if I'm thinking about the side of a triangle, I shouldn't label it as negative 1. I should label it as its actual length, which is positive 1. This length here is 1. Really, I have a triangle that looks like this. I've got R here, here's 1, here's 1. Notice this is a 45-45-90 triangle because it's isosceles. These two things are the same. These two angles are 45. Pythagorean theorem tells me that's the square root of 2. $1^2 + 1^2 = R^2$, so $R^2 = 2$, $R = \sqrt{2}$. OK, cool. So this is equal to the square root of 2.
- 4.40** Now, I need to figure out, what is θ ? What is θ ? Now, there's a couple of ways I can think about this. I could think about θ as being what? θ could be that. That's θ . That's completely fine. Or I could think about it how? I could think about it as going around counterclockwise, or I could think about it as going clockwise. On the one hand, I could say θ is equal to what? I go around $\frac{3\pi}{2} + \frac{\pi}{4}$? Or I go down $\frac{\pi}{4}$. I could think about it as $-\frac{\pi}{4}$. That's how I'm going to think about it because it's a little simpler. It's a smaller number. This will become $e^{-i\frac{\pi}{4}}$.

- 4.41 Now, you're asking, you're like, OK, you give me a number, $1 - i$, and I'm fine with that. It's OK. $A + Bi$, great. I'm fine with that. I don't want to deal with it. Just leave it alone. Now you're telling me, no, you can actually write it this way. You're like, I don't like that. That's annoying. Takes too much time. I've got angles, I've got lengths, I've got all this. No, I don't like that at all. I'm fine with $1 - i$. Let me motivate for why this is relevant and why we care. Why we care.
- 4.42 Look, if I give you $1 - i$ in this form, and I ask you the following question, look, I want to know, I want to know the following thing. Let's go back to multiplication. I want you to compute $(1 - i)^2$. You're like, all right, I can do that. I know what that means. That means this is equal to $(1 - i)(1 - i)$. You're like, all right, I can do that. That's fine. You work it out. What do we end up getting? This is $1 \times 1 - 1 \times i - i \times 1 + i^2$, so $1 - i - i + i^2$. i^2 is negative 1, so this is really $1 - 2i - 1$. Those cancel, and I'm left with $-2i$.
- 4.43 OK, great. I can multiply. So then do this. Figure that out. You're like, oh, OK. I know how I'll do this. Oh, fantastic. This is equal to $(1 - i)(1 - i)(1 - i)(1 - i)(1 - i)(1 - i)(1 - i)(1 - i)(1 - i)(1 - i)$ —I'm going to have 100 of those. OK. Now I just go one by one and I go, OK, this times this, this times this, this times this, this times this. Then now that's done. Then I take that and I'd multiply in and do it to the next one. Then I'd multiply in and do this. I'm just going to do that 100 times. It's like, that's a joke. You're not going to do that by hand. That's terrible.
- 4.44 Why? Because raising things to powers does not play well with addition or subtraction. When we talked about the distributive law, $A + B$ to the M is not A to the M plus B to the M . I can't simply just take each one to the 100th power. That doesn't make any sense. That's not what we're doing up here. That's not what we're doing up here. I'm not saying this is equal to $1^2 - i^2$. No, that makes no sense. That doesn't make any sense. You have to use the distributive rule.
- 4.45 Raising things to a power is a pain when you have that is a plus or a minus, that operation. Now I say, OK, look, this guy can be written like this. Maybe that helps. Well, let's go back to this example. When I want to compute $(1 - i)^2$, $1 - i$ is equal to the square root of 2 times e to the $-i\pi/4$. So this is $\sqrt{2}e^{-i\pi/4}$, and I'm saying, square that. Now I'm happy because exponent rules say this: as long as the variables A and B commute, that's true. As long as $A \times B$ is equal to $B \times A$, which we're talking about real numbers, this is true.
- 4.46 In particular, this is equal to what? This is the square root of 2 squared times e to the $-i\pi/4$ squared. Using that rule, the square root of 2 squared is 2. How do you do A to the M to the N ? That's A to the MN . It's one of our algebraic rules. What do I get here? 2 times that is $-i\pi/2$ now. Euler's formula tells me how to handle this. This becomes what? 2 times cosine of $-i\pi/2$ plus i sine of $-i\pi/2$. The unit circle allows me to evaluate these values. I go to the unit circle. I go down to $-i\pi/2$.

- 4.47 What is the cosine there? That's the X coordinate. What is the Y value? The sine value? That's the Y coordinate. So this becomes what? $2 \text{ times zero minus } i$, which is $\text{minus } 2i$. That's exactly what we said it was, when we used the long way. Why is this useful? Well, because this only worked when this power was small. The second I jumped it up to something big, this blew up in my face. It doesn't work. But now, if I were to say, OK, I can actually do this and do it in a reasonable amount of time, because now I can say, look, $1 \text{ minus } i$ is the square root of $2e$ to the $\text{minus } i \pi$ over 4, and I'm taking the whole thing to the 100th power.
- 4.48 OK, so that's square root of 2 to the 100, e to the $\text{minus } i \pi$ over 4 to the 100. This is 2 to the 50th power, which is insanely large, but a number that I know. And this is e to the what? 100 divided by 4 is 25. So this is $\text{minus } i 25 \pi$. Well, what is 25π ? Think about the unit circle. What does it say? I go 1π , 2π , 3π , 4π , and so on. So what is the negative? It's just saying, go the other way. So it's saying $\text{minus } 1 \pi$, $\text{minus } 2$, 3 , 4 , 5 , 6 , 7 , 8 , 9 , 10 , 11 , 12 , 13 , 14 , 15 , 16 , 16 , until you get to negative 25. Where are you going to end up? You're going to end up right there.
- 4.49 Think about it. Negative 1π , negative 2π . The even π s end up on the right, the odd π s end up on the left. The odd π is negative 25π . OK, so what is this going to be? This is 2 to the 50th times cosine of $\text{minus } 25 \pi$ plus i sine $\text{minus } 25 \pi$. What is the cosine of $\text{minus } 25 \pi$? It's the coordinate here. The sine is the Y coordinate. So what is this number? It's $\text{minus } 2$ to the 50th power.
- 4.50 There's utility in this representation. This is called the exponential or the polar representation of the complex number. We've implicitly, I've kind of hidden in here what's called polar coordinates, which you'd run into in a calculus course. But look at the utility in this. This thing that we otherwise had no chance, it was completely intractable, of doing this by hand, in what, one, two, three, four, five lines, we have the answer. It's relatively basic arithmetic. But what's nice about this is that the relatively basic arithmetic is grounded in the fact that we have trigonometry underlying everything, sines and cosines. That's the beauty of the entire thing. That's the beauty.
- 4.51 What we want to do now is we want to say, OK, if I have this representation in terms of this complex exponential—that's what it's called. It's complex because its exponent involves a complex number. It's an exponential because it's an exponential function. If I now have this representation of sines and cosines in terms of this complex exponential, what can I pull from this? What is everything that I can pull from this representation? What can I say about it?
- 4.52 It turns out you can say pretty much any trigonometric identity that you've ever encountered, you can prove using these complex exponentials. It's one of the most annoying things about trigonometry is, remember the Pythagorean identities that we came up with? Cosine squared plus sine squared equals 1? Secant squared plus, let's see, tangent squared plus 1 is equal to secant squared and so forth? It's one of the

most annoying things about trigonometry is, it's, how do I remember all the trig identities? The answer is, you don't. You don't, but you remember how to derive them.

- 4.53 If you remember how to derive them, you don't need to memorize them. You derive them when you need them. How do you derive them? Complex exponential, in my opinion. People disagree with me, that's fine. But the complex exponential is how you derive these things. Let's derive a couple of them and see what happens. See what happens. All right. Here we go.
- 4.54 There's a theorem that we're going to cite, which is called, it's a theorem of De Moivre. It's very simple to state. It's incredibly simple to state. It seems very profound. Its proof is one line, which is kind of nice. It's very simple to prove. But the theorem says the following. If I take cosine of θ plus i sine of θ , and I raise that to the N power, which seems horrible, it seems absolutely terrible, I'm going to have to do all this distributive. If N is 100 just like we had, and I had $1 - i$, and I try and compute that to some power, whatever, that was miserable.
- 4.55 But the complex exponential was exactly how we attacked that problem. What De Moivre's theorem says is that this is equal to cosine of $N\theta$ plus i sine of $N\theta$. That's it. It just says, whatever this exponent is, bring it in and multiply the θ s and you're done. Why do I claim that this is a trivial proof? Well, if you believe me from Euler's formula, it was $e^{i\theta}$ is equal to cosine of θ plus i sine of θ . Yes? That was Euler's formula that we used when writing out the number in polar form.
- 4.56 What does it really say? $e^{i\theta}$, whatever's in the box, goes in, you replace the boxes with that. OK, so great. That means from De Moivre's theorem, it tells me what? Cosine of θ plus i sine of θ is $e^{i\theta}$. It says $e^{i\theta}$, raised to the N th power, is what? Well, properties of exponents tell me what? Something raised to a power, raised to a power, that N just comes in and multiplies. Yeah? That's what it is.
- 4.57 But now just apply Euler's formula with this being in the box. What is it? This is cosine of $N\theta$ plus i sine of $N\theta$. This is the right-hand side up here. This is the left-hand side right there. There's the proof of De Moivre's theorem. It simply exploits Euler's formula and exponentiation rules to get what we want. Then what can I say from this? What can I say from this? Well, let's see. Let's do cases.
- 4.58 What if N is equal to zero? What if N is equal to zero? What am I really saying? Well, if I take cosine of θ plus i sine of θ to the zero power, I get 1 because I'm raising it to the zeroth power. I get 1. I'm saying 1 is equal to what? On the right-hand side, replace the N s with zeros. I get cosine of zero plus i sine of zero. That's what I get. Big deal. What does that tell me? Well, when are two complex numbers equal to each other? If I give you a complex number, $A + Bi$, and I say that's equal to $C + Di$, that's if and only if A is equal to C , and B is equal to D , if and only if their corresponding parts are equal to each other.

- 4.59 This is a complex number on the right-hand side. It doesn't look like a complex number on the left-hand side, but I can make it look like one by saying this is $1 + 0i$, and that's cosine of zero plus i sine of zero. What do we see? Cosine of zero has to be 1, and sine of zero has to be zero. So De Moivre tells us that cosine of zero is 1, sine of zero is equal to zero, which is exactly what we saw on the unit circle when I have that. If I have an angle of zero, cosine, which is my X , is a 1; sine, which is my Y , is a zero. So De Moivre gives us part of the unit circle, which is pretty miraculous.
- 4.60 OK, what happens when N is 2? If N is 1, it's like I'm saying 5 equals 5. It's like, well, OK, yeah, facts. That's true. 5 equals 5. But if N is equal to 2, what am I actually claiming? Again, let's look at the theorem. If N is equal to 2, what am I saying? I'm saying that cosine of 2θ plus i sine of 2θ is equal to cosine of θ plus i sine of θ , the whole thing squared. That's what I'm claiming by De Moivre. Work out the right-hand side. The right-hand side is what? This is cosine of θ plus i sine of θ times cosine of θ plus i sine of θ . Like that.
- 4.61 Now work it out in gory detail. All right? Cosine times cosine, that's cosine squared. Cosine times i sine. I get i sine θ , cosine θ . i sine times cosine. I'm going to get another copy of that. So i sine θ , cosine θ . Let's get this out of the way. Now, I'm going to get i sine times i sine. I'm going to get plus i squared sine squared. But sine i squared is negative 1. When I combine everything together, what do I get? I get cosine squared of θ minus sine squared of θ plus i times 2 sine θ cosine of θ .
- 4.62 But that has to equal the thing on the left. If that's equal to the thing on the left, then what can I say? I can say, oh, that means that cosine of 2θ has to be equal to cosine squared of θ minus sine squared of θ , and sine of 2θ has to be 2 sine of θ , cosine of θ . Those are two trig identities which are very, very seldom remembered. Where does it come from? Complex exponentials. You can generate trigonometric identities based on these complex exponentials. That's what's pretty miraculous with this.
- 4.63 Let's do one more. I know this is painful. Trigonometric identities are nobody's favorite thing at all. But you need to at least have the toolbox at some point to be able to generate them. To generate them. All right, let's see. How do we do this? OK, the last identity I'm going to give you. Last identity. Question I want to ask is, how do I find the cosine of the sum of two angles? Sum of two angles. How do I do that? How do I find the cosine of, rather than using Greek letters for angles, I'm just going to call them A and B . They're just variables. That's fine. But how do I find cosine of $A + B$? How do I do that?
- 4.64 A natural thing to do would be to say, oh, this is just equal to cosine of A plus cosine of B . Turns out, totally wrong. Totally wrong. Not legitimate whatsoever. But what I can do is, I can say, all right, well, how do I tie this into complex exponentials? How do I tie that in? Well, remember, e to the i , $A + B$, is equal to cosine of $A + B$ plus i sine of

A plus B. OK, so that's great. At the very least, I know where to find cosine of A plus B. There he is. He's right here. But now here's the beautiful thing.

- 4.65 Properties of exponents, again, algebra, properties of exponents. What is this? This is e to the iA plus e to the iB . The i can distribute in. Properties of exponents tells me this is what? e to the iA times e to the iB . How do you multiply things that have the same base? The same base as e here, how do you multiply them? You add their exponents. Properties of exponents. OK. Then what is this? Euler's formula, e to the iA . That's cosine of A plus i sine of A. What about e to the iB ? Well, that's cosine of B plus i sine of B.
- 4.66 OK, let's work that out. Let's work it out. What do we get? Cosine times cosine, I get cosine of A, cosine of B, plus cosine of A times sine of B. So with an i , I'm going to get $\cos A$, $\sin B$, plus this term times that one, i , sine of A, cosine of B, and then this one times that one. The i squared gives me negative 1. I'm going to get minus sine A, sine B. What is this whole thing in total? It's equal to $\cos A$, $\cos B$, minus sine A, sine B, plus i times—now I get the mixed terms, $\cos A$, sine B, plus $\cos B$, sine A. That whole thing has to equal that.
- 4.67 That means what? That means that the real part here, cosine AB, has to equal this. So cosine of AB, cosine of A plus B, is what? $\cos A$, $\cos B$, minus sine A, sine B. What's nice is the original question we were asking for was cosine of A plus B, and it's like a twofer to some extent. We got cosine of A plus B was this. But because we ran this through, we also get, look, sine of A plus B is that. We get an identity for free. We get sine of A plus B is what? $\cos A$, sine B, plus $\cos B$, sine A. That generates our trigonometric identities.
- 4.68 Now, do you remember these things? I hope not, because you don't want to remember these things. You've got better things to remember. You've got better things to remember. What's important, though, what's important is that you understand how to utilize Euler's formula and complex exponentials to find the identities that you want. To find the identities that you want. As an example, to tie everything in that we've been doing, as an example, let's look at the following. Let's look at the following.
- 4.69 Here's a silly example. Suppose, let's do this. Zero is less A and B, and that's less than π over 2. So A and B are going to be angles. A and B are going to be angles. I want to say that they're in the first quadrant, so between zero and 90 degrees. A and B are angles in between zero and 90. Suppose this is true, and the sine of A is equal to, let's say, $\frac{3}{4}$. Let's do, the secant of B is equal to, I don't know, let's say $\frac{5}{2}$. OK? $\frac{5}{2}$. I'm going to give you sine and secant, sine of A and secant of B. I'm going to say, OK, I want you to find the sine of A plus B. That's what I want you to find.
- 4.70 This literally ties everything together that we've done, which is kind of cute. All right? How do we do that? Well, here we go. Look, A and B are angles in between zero and π over 2. They live in the first quadrant. That's the first observation we have to make. They live in the first quadrant. Sine of A is $\frac{3}{4}$. I can interpret that as saying, look,

there's some triangle out there that's a right triangle that has angle A, whatever that is. A is in between zero and $\pi/2$. Such that the sine of A, which is the opposite over the hypotenuse, is $3/4$. So the opposite is 3, the hypotenuse is 4. Yeah? OK.

4.71 There's also a triangle out there which has angle B, such that the secant of B is $5/4$. Now, think about this. OK, secant, it starts with S, so it's 1 over the one that doesn't start with S. So that's $1/\cos B$. This is really saying what? This is saying $1/\cos B$ is equal to $5/4$, which, if I multiply over and multiply over, what am I saying? I'm really saying $\cos B$ is equal to $4/5$. I just flip them. $\cos B$ is the reciprocal, so that's $4/5$. So $\cos B$, SOHCAHTOA, cosine is adjacent over hypotenuse, adjacent is 4, hypotenuse is 5.

4.72 How would I solve for the sides of these corresponding triangles? Well, Pythagorean theorem tells me how to do that. Pythagorean theorem says what? If this is X, it says that $X^2 + 3^2 = 4^2$. So I get $X^2 = 7$, what is that, $\sqrt{7}$? And then I'm going to subtract 9. So this is $16 - 9$, which is hopefully 7, if I did that right. So that says that X is equal to the square root of 7. Great.

4.73 Pythagorean theorem here with Y, this says $2^2 + Y^2 = 5^2$. So you get that $Y^2 = 25 - 4$, which is 21. So this is the square root of 21. So Pythagorean theorem, relevant. Sine of A plus B, I have a formula for that. It's right here. I've got a formula for that. So now I can say, OK, $\sin(A+B)$ is equal to what? $\cos A \sin B + \sin A \cos B$. Now I really just need to say, well, what is sine and cosine of A and B, respectively? Well, what is cosine of A?

4.74 I come over here and I say, OK, cosine of A is adjacent over hypotenuse. The adjacent is square root of 7, hypotenuse is 4. This is the square root of 7 over 4. What is the sine of B? Sine of B is opposite over hypotenuse. That's square root of 21 over 5. Plus, what is the cosine of B? Adjacent over hypotenuse, that's $4/5$. Sine of A, that's opposite over hypotenuse, that's $3/4$. If we simplify this up, what do I get? I get the square root of 7 times the square root of 21. Square roots can combine. $7 \times 21 = 147$. This is the square root of 147, all over 20, plus $6/20$.

4.75 If I combine them into one, I get this beautifully ugly number that I have no idea what it is. Square root of $147 + 6$ over 20, no clue, but that at least allows me to solve for what $\sin(A+B)$ is equal to. The question then becomes, can I somehow invert sine to figure out what A plus B is? Because maybe I only care about what A and B actually are. That's what we're going to get into going forward.

Lecture 5: Graphing in Trigonometry

At a glance — Graphing in trigonometry.

- 5.1** All right. So today we're going to talk about graphing the trigonometric functions that we've been talking about. So what do I mean by graphing these functions? Well, we need to understand what the graph of a function is in the first place. So the structure for today is basically going to be, let's walk through what is the graph of a function? How do I represent it?
- 5.2** And then we're going to use this nice free online tool that we can actually go through and see what they look like, get some intuition for what they are, try and capture some of the information that we already know from some of the things that we've done regarding values of trig functions and how can I use this tool to actually solve equations involving trig functions and whatnot? Let's start off with graphs of functions in general.
- 5.3** If I'm looking at graphs of functions, well, the first thing I have to understand is, well, what is a function? Now, functions are complicated objects, and they cause a little bit of, I don't know, maybe stress or things like that when you start running into them. But ultimately, I want you to think about a function in the following way. A function is something that takes an input, whatever that input is; it does something to that input, and then it provides an output. That's it. That's literally all a function is. Takes input, does something to it, produces the output. That's it.
- 5.4** For example, if I give you an input, whatever that input is, let's say for us, we're going to be dealing with numbers. If I give you an input of an integer, let's say the integer 5, and I say I want you to take that integer 5, and I want you to multiply it by 3. You give me the input 5, I'm going to then multiply it by 3. OK, 5 times 3 is 15. My output that I'm going to give is 15. In order to make sense of that, I have to put a notation to things. There's a notation which is standard, which is, you give your function a name. You typically, I mean, mathematicians are not very unique when we name things, so it starts with F, so why not use the letter F? That's great. That's completely fine.
- 5.5** Then I have to have a letter for my input, whatever my input is. Now, from an algebra standpoint, we typically think about arbitrary inputs as being variables of some sort. I could take my variable as being x in this case. The way I want you to think about this is that x is going to be my input, and it has to be valid. I'll explain what I mean by valid. But x is equal to my input. F is going to be my function. I take x , I feed it into F , and this is F of x is my output. That's my corresponding output. All right, so that's the notation associated to this.
- 5.6** Let's think about this in terms of the example that we just gave. If I give you the example of, I want you to give me an input of 5, and I want you to multiply that input by 3, so the output should give me 15. Can we come up with a function which actually does that? You give me an arbitrary input; let's just say, for the sake of argument, it's 5.

I want you to triple it. I want you to multiply it by 3. It stands to reason that that would probably be a good candidate for this. F of X is equal to 3 times X .

- 5.7** Why is that? Well, you give me the input 5, and what should 5 do? Well, I want it to be 3 times 5, and that's going to give me my 15 that I'm interested in. It looks a little strange up here, F of X is equal to $3X$. But now that I have this concrete formula for what this function actually is, I can now just start plugging in values. I can say, well, you gave me 5 and you said to triple it, so you give me 15. But now it's like, now that I have a concrete formula, I can do this for anything. I can say, OK, what is F of zero? Well, if I triple zero, I'm going to get 3 times zero, which is zero. That's not going to do anything.
- 5.8** It doesn't have to be positive numbers either. I could say, what is F of minus a half? That's going to give me 3 times minus a half, which is equal to minus three-halves. That's it. It gives me this concrete relationship on how to take any input that you give me whatsoever, 5, 0, negative a half, whatever it is. Based on this formula, all I have to do is take X , whatever it is, 5, 0, negative a half, and then just replace all of the X 's on the right-hand side with that corresponding value. That's it. That's the notion of a function.
- 5.9** But we have to be careful here. I have X as an input, and I need it to be valid. I'm going to put that in quotes. Because what do I actually mean by valid? Well, let's think about it for a second. We have things like F of X is equal to $3X$. You can sit there and you can ask, OK, think of all numbers that you possibly know. Are any of those numbers not allowed to be plugged in? Are any of those not allowed to be plugged in? You give me any negative number whatsoever. Negative 5, negative a million, negative 1 over 57, whatever the negative number is. Can I plug it in? Can I multiply it by 3? The answer is, well, sure.
- 5.10** You give me any positive number and plug it in. 1 over a million, 1 over 57, 742. Can I plug it in? Sure, I can multiply that by 3. OK, so what do I just show? All negative numbers are valid, all positive numbers are valid, and we show that zero is valid, too. We would say that all possible X values that you can think of are actually valid inputs. OK?
- 5.11** But let me give you another function. What if I gave you this one? What if I gave you—let's use a different letter. So G of X is equal to, let's say, 1 over X minus 3. You give me an input, and I want to simply just take that input, subtract 3 from it, and divide it over, 1 over that. You give me 5. I'm going to take 5, I'm going to subtract 3, that'll give me 2, and then I'm just going to invert it, so I'm going to get one-half. You give me 12, I'm going to subtract 3 from it, I get 9, and then I'm just going to flip it, so one-ninth. That's what this function is doing.
- 5.12** So now I want to ask the question, well, what are my valid inputs? You start asking, OK, if I choose any negative number whatsoever, if I choose any negative number, negative 7, negative 32, negative pi, whatever it is, can I plug that into this and get something

that makes sense? The answer is, well, sure. Why not? Can I plug in zero? Sure. If I plug in zero, I get 1 over negative 3. That's a legitimate number. That's negative one-third. That's fine.

- 5.13** But then when I start asking what kind of positive numbers can I plug in, now I have to be careful. Why? Well, I have to worry about, we're not allowed, so we're not allowed things of the form 1 over zero. You can't divide by zero. That's not allowed, mathematically. What do I have to worry about? I have to worry about X cannot equal 3 in this case. Because then if X were equal to 3, what would happen? Well, if I take X as equal to 3 and I plug it in, I would get, so the question is why? Why is that the case?
- 5.14** Well, what would G of 3 be? That would be 1 over 3 minus 3, which is 1 over 0, and that's not allowed. That's not allowed. I can't simply just say, write down some expression for my definition of my function, and then I can just plug in whatever I want. I have to be careful with what I'm plugging in. One of the cases that you have to worry about is what happens when the denominator is equal to zero. There's other things you have to worry about as well, but really when the denominator is equal to zero is what we're going to be worried about today.
- 5.15** This is getting at, what do I mean by valid? I mean, what is the domain? What is the domain? The domain is literally just the collection of all valid inputs that you're allowed to have. Up here, if I looked at my domain, if I call it domain, so $\text{dom of } F$, what would I get here? Well, we said we could plug in anything we want. What's the notation for that? We can say anything from minus infinity up to infinity. I'm thinking about that as the negative infinity is going as far negative as you can possibly go, the positive infinity is going as far positive as you can possibly go, and it's anything in between those two things.
- 5.16** Whereas this one, we're saying, well, the domain can be anything except for 3. The domain can be anything except for 3. What can I say there? Well, the domain of G will be what? This will be, one way we could say it is, we could say, all X not equal to 3. That would be completely fine. Any X which is not equal to 3, that would be a valid input. Alternatively, what you can do is you can write this using interval notation. I'm going to draw a little picture to illustrate this. But the idea is this.
- 5.17** Look, this one up here, you can think about the domain as just being the entire real line. Any point on the line you can plug in. Anything there. If I look here and I look at my line, I have to say, OK, I can plug in everything except for 3. So what I'm going to do is, I'm going to put a little hole here saying that I'm not allowed X is equal to 3. You think about your whole entire number line, and I just take away 3. I just remove it. How do I notate this? Well, I would say that the domain of G looks like what?
- 5.18** Well, I'm allowed to go as far to the left as I want. You can think about minus infinity lives over here somewhere and plus infinity lives over here somewhere. Where am I allowed to go? I'm allowed to go from all the way from minus infinity up until 3. I'm

allowed to go all the way up until 3. Then I have to jump over 3. How do I notate that? Well, with intervals, what you have to worry about is parentheses or closed brackets.

- 5.19** If I have a closed bracket, it means you're allowed to include that number. If I have a rounded bracket, it means you're not allowed to include that number. So since I'm not allowed to include 3, it's going to be a parentheses on the side. And then how do I jump to over 3, from 3 up to infinity? How do I somehow connect these? The symbol is just the U. It just means union. So literally, all it means is, this thing or this thing. You're in one of those two camps. You're either in this or you're in this. So it means I'm either over here or I'm over here somewhere, but it means that I'm not allowed to equal 3. That would be my corresponding domain.
- 5.20** Now that we have functions, we want to understand, what is the graph of a function? If I gave you this one, for example, if I gave you F of X is equal to $3X$, I want to say, how do I graph it? There's a very systematic way to do this. Graphs of functions. What you do is, you say, OK, look, what is my input? My input is an X variable. That's fine. What is my output? Well, it's whatever F of X is. What you do is you replace F of X with a new variable, typically called Y . You replace F of X with Y .
- 5.21** Now, rather than looking at F of X equals $3X$, I look at Y equals $3X$. Then what I do is, I say, well, I have an axis, which I call my X axis. That's my list of valid inputs that I'm going to be considering, my domain, if you will. Then I give myself the corresponding variable, which is just perpendicular to that. I graph points in the following way. I graph points in the following way. I'm going to say, let's make a little table. Here's X , here's Y equals $3X$.
- 5.22** Now you just start giving yourself all valid inputs. Let's say X is, I don't know, let's just do five values. Let's just do some symmetric ones. Minus 2, minus 1, zero, 1, 2, and so on. Those are my inputs, and I have to ask myself the question, are those valid? The answer is, well, my domain is everything, so it doesn't matter what you give me, it's going to be valid. Negative 2, negative 1, 0, 1, 2, they're all valid. OK, fine.
- 5.23** If I plug them in, X is equal to negative 2, what happens? 3 times negative 2 is going to give me a negative 6. That's a negative 6. 3 times negative 1 is going to give me a negative 3. Zero times 3 is zero. 1 times 3 is 3. 2 times 3 is 6, and so forth and so on. I obviously cannot do this because there's infinitely many points in here, and so I can't write down a complete table of every single value of this particular thing.
- 5.24** What I do now is I say, OK, I'm going to create another column which is going to be the combination X , Y , where I'm going to have my X coordinate, and then I'm going to have my corresponding Y coordinate. I'm going to say, OK, what do these look like? My X coordinate was minus 2, my Y coordinate was minus 6, X coordinate minus 1, Y coordinate minus 3, X coordinate zero, Y coordinate zero. Next one, 1, 3. Last one, 2, 6.
- 5.25** Now what I do is I graph those corresponding points in the XY plane where I have my X coordinate is negative 2. Let's say I'm going to go over one, over two, and then I'm

going to go over one, over two. If I go over two, I'm going to go up 1, 2, 3, 4, 5, 6. I'm going to go down 1, 2, 3, 4, 5, 6, clearly to scale. Not even close to scale. If these things are like an inch apart and these are like half a centimeter apart, this is not even close to scale. But we're going to do our best.

- 5.26** I'm going to show you how Desmos actually allows you to do this not by hand, which is how you should do this personally whenever you're graphing anything. It's completely free, just an internet-based tool. We're going to talk about that. I look at this and I go over negative 2, so I go left 2, and I go down 6, so down 6. That'll be someplace, someplace here. This is my minus 2, minus 6. Same thing here. I go over negative 1, down 1, 2, 3. I'm going to be someplace here. This one is minus 1, minus 3.
- 5.27** Then I go zero, zero, that's easy. That's just the origin. I don't have to move. I just stand at the bullseye, just at the crosshair right here. Then I have 1, 3, so I go over one, up to three. This one is 1, 3. Then I go over 2, 6. That's something like this, 2, 6. I see that I'm getting points on the corresponding graph of the function. The question I want to ask myself is, well, what does this look like? Well, based on this, it looks like, well, if I just connect the dots, it looks like it's going to be a line. Do we agree? It looks like it's going to be a line.
- 5.28** I can see that. I'll do my best here. Somewhere in there, roughly, is what I would guess this thing is going to look like. Turns out that's correct. Turns out that's correct. This is a line. But that's how we graph corresponding functions. I give you a function, F of X is equal to something. To find its graph, I set Y equal to F of X , so Y is equal to $3X$. Then I plot a whole bunch of X values, a whole bunch of Y values, plot the dots, and hope that I can somehow extrapolate like, oh, this is what it's going to look like. All right? That's the idea.
- 5.29** What we want to do is, we want to understand, can I do this for the trig functions that we've had? Can I do this for the trig functions that we've had? The answer is yes. The answer is yes. Let's go through that. How does this work for the trig functions? OK. Graphs of trigonometric functions. How does this work? Well, we've seen that the trigonometric functions arise fairly naturally from right triangles. That's where they were defined. Sine, cosine, tangent. SOHCAHTOA, opposite over hypotenuse, adjacent over hypotenuse, opposite over adjacent.
- 5.30** Secant, cosecant, cotangent were the reciprocals. One over cosine, 1 over sine, 1 over tangent. Then I had to be careful with how to remember secant and cosecant, and I gave you a little goofy way to remember that. Well, that's great. But the question I have is the following. Well, if trigonometric functions are motivated by right triangles, so if I give you an angle θ , corresponding to once I fix an angle, I can call that the opposite, I can call that the adjacent. The hypotenuse is always fixed. It's always whatever's opposite of the 90 degree.

- 5.31** But once I fix this angle θ , it makes sense to say, what are sine, cosine, tangent of that particular angle? It's like, oh, OK, well, it makes sense to talk about sine of θ in terms of the context of right triangles. But then a natural question would be, well, what if I just think about this as a function? Is that allowed? Am I allowed to consider that? What if I just decided to say, I'm just going to define this as F of θ and say, does that even make sense? Independent of the triangle. Now, independent of the triangle.
- 5.32** I'm just going to say, does F of θ equals sine of θ make sense? You might be like, well, no, of course it can't make sense because θ is an angle and we know that the sum of the interior angles has to be 180, so you can't just plug in F of 5,000. That's a valid argument. That's a valid argument. But I would say, well, no, no, no, no, we had a connection between right triangles where sine, cosine, tangent were defined and the unit circle, yes? We had a connection between saying, look, if I take the unit circle and you give me any value of θ whatsoever, I said that the coordinates of this point were given by cosine of θ , sine of θ .
- 5.33** That was the whole point. This is my unit circle. So these points are 0, 1. This is a giant circle. We drew with all the values π -sixths, π -fourths, π -thirds, π -halves, all of that. The argument is, well, look, θ is somehow restricted here because θ has to be an angle to a triangle, and therefore, θ has to be somewhere in between zero and 180 because of the sum of the angles being 180. But I would counter that by saying, well, no, θ can be any real number you want, because if I wanted to say, what is F of 5,000, I'm just going to go, OK, go around 5,000 degrees.
- 5.34** So go 360, another 360, another 360, another 360 until you get to 5,000, and then take the corresponding angle corresponding to that, and the Y coordinate of that will give you the sine of 5,000. It's like, OK, well, maybe there's something to this. The answer is yes, you can do this. Now, what are its valid inputs? i.e., what's its domain? Again, we start thinking about it. We're like, OK, well, if its input now is an angle, and you think about the angle as being somehow related to the unit circle, then is there anything that you're not allowed to plug in?
- 5.35** For positive values, it's at least pretty clear. You can plug in any positive number you want. Just like I said, if I want to say sine of 5,000, what am I going to do? I'm going to go around 5,000 degrees and then just take the corresponding ray and say, oh, the Y coordinate, that's sine of 5,000. Fine. For any positive number, that seems reasonable. Then you're like, OK, but negative numbers you're not allowed to do. It's like, well, remember, how do we measure negative angles? We just went, rather than going counterclockwise, we went clockwise instead.
- 5.36** If I want to say, what is sine of negative 5,000, what am I going to do? Well, rather than going around this way, 5,000 degrees, I'm going to go around this way, 5,000 degrees. That seems plausible. I can plug in now negative values. You're like, OK, fine. You can plug in positive, you can plug in negative, but you can't plug in zero. It's like, false. What

is the angle zero? The angle zero is just the positive X axis. And what is the Y coordinate on the unit circle on the positive X axis? That's just Y equals zero. So sine of zero is zero.

- 5.37 So my valid inputs are, again, anything I want from minus infinity up to infinity. I can plug in anything I want. So the question we now want to ask is, OK, all right, all right, OK. Now I'm a little skeptical. You're saying you can now think of sine of theta, cosine of theta, tangent of theta, secant, cosecant, cotangent, you can now think of them as functions. OK, fine. Maybe I believe that. Maybe I believe that.
- 5.38 But then what do they look like? Because you just told me, look, if I give you a function like $3X$, you can draw its graph, just set Y equal to $3X$ and draw its graph and see what happens. That was the game. OK. Well, so then the question becomes, what does Y equal to sine of theta look like? What the heck does that look like? Well, let's see what it looks like. I'm going to take it one step further. I'm going to say, well, I don't care about just sine of theta. I want to know what cosine of theta, tangent of theta, all of these other things look like.
- 5.39 That's why we're going to use this graphing tool. I can sit here all day long and try and pretend that I draw a decent graph, but the answer is no. None of them are going to look good. It's better to use something that actually does this properly, and that's what we're going to do right now. All right, so how do you do this? Here we go. Again, it's just a free online calculator, just Desmos.com. You click on that, it'll give you a calculator from that standpoint. You could do simple calculations in this, just in this little left bar right over here.
- 5.40 If I wanted to do some basic computations, if I want to do 2 plus 3, OK, great. There it is. It gives me 5. That's pretty straightforward. You're like, "OK, Robbie, let's do something real." We had a function that told me how to do 3 times X. If I just did 3 times 5, there's my 15. OK, great. 3 times, if I want to do negative 1, great. That's negative 3. OK, so if you give me something along those lines, like $3X$, how do I actually compute it? Well, I could just plug in the values, 3 times 5, 3 times negative 1, and so on, but what if I want something more concrete? What if I actually want to define the function F of X is equal to $3X$?
- 5.41 OK, well, what's nice about Desmos is that it's very intuitive. It's like, well, I want a function that looks like F of X equals $3X$. So let's just try that. F and then parentheses, X, parentheses, equals $3X$. And it's like, oh, sweet. There it is. So this is defining the function, F of X is $3X$. And what Desmos is doing is it's saying, hey, look, we're going to define it or we're going to draw this corresponding thing. We're going to draw this corresponding thing. Let's see if I can zoom in a little bit with this. We're going to draw this as the graph of that corresponding function, and we're going to give it the color purple as its label.
- 5.42 If you don't like this one, then you can change it. You just hold this cursor over. If you want it to be red, now it's red. If you want it to be black, now it's black. If you want it to

be a dotted line, you can change it to be a dotted line. That's fine. If you want to change it to dots, you can change that. If you want to change the thickness of it and all of that, you can do that as well. We're just going to stick with solid lines because we're drawing their graphs. There we have. The second I don't want to deal with it anymore, I can just click on the circle and it disappears, and then the graph disappears with it.

- 5.43 But if I want to have the circle with me, there it is. There's my graph. All right. Now, why is this useful? Well, because now, once I've defined F of X is equal to $3X$, now I can just go back and start doing these computations that I had up here. Rather than saying 3 times 5, all I have to do is say, oh, I want to plug in 5 for F ? I'm going to take F of 5, and it's going to say, oh, that's 15. What is this really doing? Again, it's taking the function F , which is defined above, F of X is equal to $3X$, and it's saying, replace X with 5. I replace all my X 's with 5.
- 5.44 What's that going to do? It's going to give me equals to 3 times 5, which is 15 from here, and it's giving me the same result. If I want to do what is F of minus 1, it's giving me minus 3 as well. That's giving me this. Now, you might argue, you're like, "Well, Robbie, that's not how you graphed functions. Because how you graphed functions was, you said, set the function Y equal to F of X ." I'd say, OK, that's fair. That's fair. So what if I turn this off and I say, all right, let's just do Y equals $3X$. It's like, well, it's going to do the same thing. It's going to give me the same thing.
- 5.45 So however you want to define it, Y equals something, Y equals $3X$, or F of X equals that same thing, Desmos knows, "Oh, they're trying to graph the function. There it is." That's what's nice about Desmos. Then you can clear all these out. Let's ask ourselves the questions, well, what if I just set something equal to Y equals sine of theta? You might say, well, I don't know how to do theta. You just told me how to do functions of X . It's like, OK, well, how do you do theta? Well, how do you spell theta? Theta, T-H-E-T-A, and it automatically populates the variable theta for you.
- 5.46 If you want to graph this, let's say Y equals sine—this is what I love about Desmos, is that it says, look, this is what you want to type in. There's no coding underlying all that you have to do for this. There's no special things that you have to type in for sine. It's like, S-I-N is what I need to do. OK, so S-I-N, great. OK, now it populates that. I want to put in the variable theta. OK, so T-H-E-T-A. Well, there it is. It'll graph your corresponding function.
- 5.47 Now, one thing that's nice about Desmos is that it doesn't care what your variable is or what you call it necessarily. There are certain variables that it doesn't like because it reserves those for certain types of mathematics. But ultimately, right now, if I had taken in the theta and I had replaced it with an X , it's going to give me the same thing. It doesn't care. If I replace it with, say, a Y , or sorry, not a Y , a Z , it doesn't care. If I replace it with an M , it doesn't care. Whatever variable I want to use. So let's just use theta because that's what we've been using this entire time.

- 5.48** I see this nice little wave. I see this nice little wave. But if I zoom out, I realize, oh, this thing actually goes on forever. It's just doing this back and forth behavior. It's this nice little wave situation. That's why these things are called sinusoidal waves. That's why they're called sinusoidal waves. Let's analyze this a little bit. Let's zoom into this and say, all right, well, what do I notice? I notice that I'm at this point here, at the origin, so I can just click on that point, and it'll say I'm at the origin 0, 0. Then I notice that I go up, I come back down, and then I go back up, and I notice I cross the origin again.
- 5.49** What happens when I cross that origin? Well, it looks like it just starts again. It goes up, it goes down, it goes up, and then it crosses the axis again, and then it goes up, and then down, and then up, and then crosses again, and then up, and then down, and then up. It keeps doing this. What the heck is going on? What is this actually saying? And what are these points? What is this point over here? Oh, that's nice. It highlights it for me. It says that, when I go out, theta is equal to 2π , my Y coordinate is equal to zero. Why does that make perfect sense? Well, let's go back to the unit circle. Let's go back to the unit circle.
- 5.50** So if I go something like this, remember this was the angle theta was equal to zero. And then I start moving counterclockwise around the circle and I say, OK, this is going to be theta is equal to $\pi/2$. And this is going to be theta is equal to π . This one is theta is $3\pi/2$. This one will then also be theta is equal to 2π . This one will be theta is equal to $5\pi/2$, and so forth and so on. I just keep going around and around and around and around.
- 5.51** What do I notice? Remember, the sine value is the Y coordinate. Sine is Y, cosine is X. Then what am I seeing here? Here's 0, 1. Here's minus 1, 0. Here's 0, minus 1. So what am I actually seeing? I'm seeing, look, I know that the sine of zero is equal to the sine of 2π is equal to the sine of 4π . I just keep going around and around and around and around. Those are all equal to zero because the Y coordinate is zero. I also see that the sine of $\pi/2$ is equal to the sine of $5\pi/2$ is equal to so forth and so on. That's equal to 1 because the Y coordinate is a 1 up here.
- 5.52** I see that the sine of π is the same as the sine of 3π , is the same as the sine of 5π , 7π , and so forth. That's zero. Then the sine of $3\pi/2$, and then all the way around again, sine of $7\pi/2$, and so forth. I'm going to get a negative one. What is Desmos telling me? It's saying, look, let's highlight all these points. There they are. It's saying when theta is equal to zero, the Y coordinate is zero. The unit circle tells me that. When theta is equal to $\pi/2$, the Y coordinate is 1. That's the top of the circle. The Y coordinate is 1.
- 5.53** When theta is equal to π , that's on the far left of the circle, the Y coordinate is 0. When theta is at $3\pi/2$, that's the bottom of the circle, the Y coordinate is negative 1. When I go all the way back around, now I've gone all the way around the circle, now I'm back at 2π , the Y coordinate should be zero. And what happens? It starts again. This is why we call this 2π periodic. Every 2π , from zero out to here, it repeats. It repeats.

That's the period of the corresponding function. If I stretch this out to be in between zero and 4π , and I start highlighting these values, it's nice that Desmos already knows that these are probably points of interest.

- 5.54** It gives you these little gray dots that you can just go over and highlight. It says, look, I go around the circle once. I go from zero to 1, to zero to negative 1, to zero, and then I repeat. I'm going around the circle twice. Now I'm going 5π over 2, 3π , 7π over 2, 4π , and I repeat, and I keep doing this over and over and over and over again. That's why this function is called 2π periodic. From zero, theta equals zero out to 2π , is one full period of sine. It's one full period of sine. Then between 2π and 4π , it's another period of sine. Then from 4π to 6π , another period of sine.
- 5.55** Then propagate it to the left as well. If I came back over here on the left-hand side, I see, look, there's these points, negative 2π , zero, negative 3π over 2, 1, negative π , zero, negative π over 2, negative 1, and zero, zero. What is that doing? This is saying, if theta is negative 2π , my Y coordinate is zero. Negative 3π over 2, it's 1. Negative π at zero. What is that saying? It's saying, look, if I start over here and I go negative 2π , that means, go this way by 2π , go clockwise. If I go around this way, my Y coordinate is zero.
- 5.56** What's negative 3π over 2? Go around here. Coordinate's 1. Negative 3π over 2, the coordinate is 1. Negative π , go around here. Zero. Negative π -halves down here. Negative 1. Go nowhere. Y coordinate is equal to zero. Desmos does a way better job than I could ever do to try and explain this. There's my corresponding function. Now, what's nice is, I can say, all right, well, let's see what happens then. If that's what Y equals sine of theta looks like, well, I've got five other trig functions to deal with. I've got cosine, tangent, secant, cosecant, cotangent. It's like, all right, let's just start plugging those in and see what happens.
- 5.57** If I turn off sine, I just click on that dot. The nice thing with Desmos is, if I just don't want it to display, I just have to hit the little dot. That means it'll just turn it off, so its graph doesn't show, and I can graph something else if I want, but I don't have to eliminate it. I don't have to erase it. I don't have to do anything like that. I go down and I say, OK, let's see Y is cosine of theta. Again, C-O-S, and then just spell theta, T-H-E-T-A. It says, all right, here it is. It's having a hard time figuring out what it looks like. There it goes. There's my corresponding cosine function. Again, just a wave. That's all it is. It's just this wave.
- 5.58** What are the values of that? Well, these should be what? These should be the corresponding X coordinates on the unit circle. I notice, look, I've gone a full period. I've gone from this top of the peak down to the valley, back to the next top of the peak. I've gone from theta is zero to theta is 2π . What happens? When theta is zero, I get 1. When theta is π half, I get zero. Theta is π , I get negative 1. Why does that make sense? Well, when theta is zero, cosine is the X coordinate. Theta is zero, cosine is 1. Theta is π -halves, cosine is zero. Theta is π , cosine is negative 1. Theta is 3π over 2,

cosine is zero. Theta is all the way around to 2π , cosine is 1. That's exactly what this is saying.

- 5.59** But notice that I just have this peak, trough, peak, and then I repeat over and over and over again. You can just keep doing this. That's just saying you're going around and around and around the circle as long as you want. Now, what's also nice about this is, remember, I had this silly example, so I'll do it right below this. I had this silly example of saying, well, what happens when I take sine and I evaluate it at, say, 5,000, whatever it is? I don't know what the value is, but Desmos does. There it is. That's what's nice about this, is I say sine of 5,000, great. That's negative 0.987, something I don't care about. Some number that I'm never going to remember, but Desmos loves this thing.
- 5.60** When you're doing these computations with these trigonometric functions, Desmos allows you to do this for free. It's great. It's an awesome tool. We'll get rid of that. Let's look at the other trig functions and see what they are. Then I want to show you what we can do with these. Let's turn cosine off. Y equals tangent of theta. What does that look like? Let's back out a little bit. Fix it. There it goes. That one's freaking me out a little bit because that's very different from the sine and cosine. This isn't just a nice little wave doing this. It still has this repetitive nature to it. I'm getting this candy cane stripe over and over and over and over again.
- 5.61** But it's like, well, what are these gaps? Why is there this giant gap in between this red line and this red line and this red line? What the heck is happening? And what's corresponding to those gaps? Sine and cosine didn't have those gaps. Sine and cosine had what? They just had this nice, chill, wave-type pattern. So the question is, well, what in the world is actually happening here? Well, remember, what was tangent? Tangent was equal to sine over cosine. Tangent was equal to sine over cosine.
- 5.62** Maybe that's what's wrong. Maybe the graph of tangent is a little wonky because it's a quotient, sine and cosine. Maybe. I don't know. That's a natural question to ask. Let's write this as sine over cosine. That's what I have. But now immediately I'm like, oh, that's why something is going wrong. My valid inputs, my domain, my domain matters. I'm not allowed to divide by zero. In particular, any theta values for which the cosine is equal to zero, those aren't valid. Those aren't valid. The valid domain must be theta such that what? Such that cosine of that theta is non-zero.
- 5.63** OK, well, let's think about that. When does cosine of theta equal zero? Well, let's go back to the unit circle. Cosine of theta is zero when the X coordinate is zero. OK, so that's going to be when theta equals $\pi/2$, $3\pi/2$, $5\pi/2$, and so on. But also, instead of going counterclockwise, I could go clockwise. I'm also going to have minus $\pi/2$, minus $3\pi/2$, and so on. I've got an infinite amount of points where these fail. Then I say, oh, OK. Well, if I start plugging in things like theta equals $\pi/2$, it's not going to like this because it doesn't like this angle. It's thinking about this in a different context, mathematically.

- 5.64 What I want to do is instead I'm going to put X's in here. Think about it as X and Y. Even though the variable up here is a theta, if I think of this as an X, what it's going to do is what happened. It drew this vertical line, X is equal to $\pi/2$, and that happens to be precisely where this break in tangent occurs, number one. What if I did X is equal to $3\pi/2$? I'll put that one in blue as well. I'll put that one in blue. Great. I get a vertical line here. I get a vertical line here.
- 5.65 Let's see how that works. I get vertical lines there. You can see that if I look at every single one of these, I have a $3\pi/2$, I have a $\pi/2$. If I move over to $5\pi/2$, that's going to be the next gap. $7\pi/2$, the next one, $9\pi/2$, the next one. If I go back the other way, I'm getting negative $\pi/2$, negative $3\pi/2$, and so on, which is interesting.
- 5.66 What do the other graphs look like? We have sine, cosine, tangent. Let's look at Y equals secant of theta. OK. So secant of theta looks like that. It's like, well, that's strange. Why is it doing that? Well, remember, secant was equal to what? That was $1/\cos$. So that's $1/\cos$. I need to worry about when is cosine equal to zero? Oh, that's the $\pi/2$, $3\pi/2$, $5\pi/2$, and so forth. That's exactly where these gaps are occurring. $\pi/2$, $3\pi/2$, $5\pi/2$, and so on. That's where those gaps are occurring.
- 5.67 What's beautiful about secant is that if I graph both cosine and secant on top of each other, you'll see there's the connection between the two. You've got cosine doing this nice, beautiful wave form, and then secant does what? It basically flips cosine up and introduces these infinite little—they're called asymptotes, vertical asymptotes. I notice that, once cosine crosses the origin, what happens? I've got a vertical line. I've got an asymptote. Once I come back down and cross again, what happens? Another one. Vertical asymptote. This is a beautiful symmetry with these.
- 5.68 Same thing with sine and cosecant. If I look at Y is cosecant of theta, Y is equal to cosecant of theta, and I look at sine on top of that, there's the beautiful symmetry there. You've got this little sine wave form in between, and then you have these corresponding pieces there, which is neat. All right, what's the last one? Last one is cotangent. Y equals cotangent of theta, and that's this one. It looks like theta only or tangent, but tangent is turned backwards. Remember, tangent did what? It came down and came down like this, and this one is coming down and going that way, and it's been shifted a little bit.
- 5.69 But that's what the graphs of the six trigonometric functions look like. Now, one thing that we're going to be really interested in is, we're going to be interested in trying to figure out, well, how do you figure out how to move these graphs accordingly? How do you move them? Why would you care to move them? Why would you care to move them? Let's think about it in the following way. I'm going to take one of these trig functions, and I'm just going to start goofing around with it. Y is equal to sine of theta. There's sine of theta, and I'm just going to start messing around with it.

- 5.70** What happens if I multiply theta by a 2 or a 3 or a 4 or a 5 or something? What's going to happen with that? If I multiply it by a 2, what happens? What do I end up getting? Well, that looks like it's sped up a little bit. What if I multiply it by a 5, what happens? If I multiply it by 5, it's getting more and more frequent. It's moving faster and faster. What if I take that to the extreme and I put 25 in here? Now it's getting crazy. It looks like a little slinky. It's just getting more and more compressed. It looks like I'm taking this wave form, and what am I doing? I'm basically just taking it, sticking my hands on both ends of it, and squishing it together. That's what it looks like.
- 5.71** I'm increasing the frequency of that corresponding function. That makes sense if you think about it, because look, if I look at the unit circle and I say, if I think about theta as representing time, and I say it takes me 2π units of time to go from here around back to here, well, if I multiply sine by 25, well, then I'm going 25 times as fast. Rather than taking 2π units of time to get from here to here, it's going to take me what? 2π over 25 units of time to get there. It scales it down by whatever I'm multiplying by and vice versa.
- 5.72** If I make this number smaller, so if I think of this number as like one-half, what happens? If I think of this number as one-half, it's starting to stretch it out this way. What if I make the number smaller? What if I make it one-fourth? It's like, OK, well, it's stretching out even further. One-eighth, what's happening? It's even further and further. Now, when I multiply the theta by a small number, positive but small, what am I doing? It's like I'm taking the edges of the slinky and I'm just stretching them out this way. Versus multiplying by things like 5, 10, 15, 20, I'm taking the slinky and I'm compressing it this way, and so its frequency is getting higher.
- 5.73** That notion of frequency is its period. That's what it's called. The period of this function, and they all have different periods associated to them. The period is basically, how long does it take it to repeat? The period of sine of theta is 2π . That's why we call it 2π -periodic. Every 2π , it just repeats. Now, if I decide to say, what is the period of sine of $M\theta$, where M is anything, that's going to be 2π over M . When M is 25, the period is smaller, 2π divided by 25. But when M is like one-half, 2π divided by a half is 4π . That's bigger than 2π . If M is a third, 2π over a third is 6π . That's bigger than 4π .
- 5.74** What is it doing? It's saying it takes longer for it to repeat, i.e., I'm stretching it out like this. Versus if M is high, like 25, 2π over 25 is saying that's much smaller than 2π . It's saying, oh, I'm compressing it. It has to go through its cycle faster at 2π over 25. This coefficient in front of this has something to do with the period. Well, what else can I do to mess around with these things? Again, this is how mathematics is done. You start with a particular object, whatever it is, and you start messing around with it.
- 5.75** You start, OK, if I change theta a little bit, what happens? If instead of multiplying by theta, I say I move it. Let's say minus 1. This one's better to see if I draw a family of these on top of each other. Here's what I'm going to do. I'm going to start with Y equals

sine of theta. I'm going to look at $Y = \sin(\theta - 1)$. That's the blue curve. I stare at that for a second. I'm like, what the heck just happened? I've got the green curve. That's the original, $Y = \sin(\theta)$. Now I've got this blue curve that, like, what happened? It took the green curve, and now the blue curve is like, it's like I shifted a little bit. I moved it.

5.76 I'm like, all right, maybe that was a fluke. So $Y = \sin(\theta - 2)$. I don't want that one to be green, so I'm going to make this one red, let's say. So now I've got these three curves. I'm like, what the heck is happening? Somehow by putting $\theta - 1$, $\theta - 2$, I've taken the green curve. And what did the green curve do? By $\theta - 1$, I shifted it over by one unit. $\theta - 2$, I shifted the green curve over to the right by two units. So now I'm like, oh, OK, now I can take the graph and I can literally shift it to whatever place I want it to be. Simply by, rather than multiplying the theta, I incorporate a theta minus something.

5.77 And it's like, OK, well, why are you doing minus? Why don't you do plus? $Y = \sin(\theta + 1)$. Let's eliminate the minuses. What happened there? What about $Y = \sin(\theta + 2)$? What the heck is happening there? OK, here's the green curve. There's the green curve. And then plus 1, moved it to the purple curve. So that moved it to the left. And then I added 2, and that took the green curve and moved it to the left by two. So somehow there's this relationship of, if I subtract numbers, $\theta - 1$, $\theta - 2$, it moves the graph to the right. If I add numbers, $\theta + 1$, $\theta + 2$, it moves the graph to the left.

5.78 I have this transformational property. The period is governed by that particular quantity. We have horizontal shifts. Horizontal shifts are given by things like $\sin(\theta + C)$. Now, I'm going to write it as $\theta + C$. Actually, let's stick with $\theta - C$. But I'm going to let C be any number I want. For example, up here, C would be 1 for my $\theta - 1$. Here, C is 2. Here, C is minus 1. Here, C is minus 2, because $\theta - (-1)$ is $\theta + 1$. $\theta - (-2)$ is $\theta + 2$, and so on.

5.79 I have some horizontal shifting that's going on. It's like, all right. I've messed around with the interior variables enough. Now I want to ask myself the question, what happens if I don't mess with the interior variables? What if I mess with the output? What if I did something like this? $Y = \sin(\theta)$, I just leave it alone. But instead of doing that now, I'm going to say plus 2. Plus 2. What just happened? The green curve is my original sine of theta. I've got my original sine of theta, and I added 2 to it. What happened?

5.80 Well, I just moved the graph vertically by 2. I moved the graph vertically by 2. OK, well, that's great. What if I subtracted 2? $Y = \sin(\theta) - 2$. I take the green curve and I drop it down by 2. OK, so that means that, if I take my output, sine of theta, and I do plus or minus on the outside, I'm just vertically shifting it up and down. That's all I'm doing. It's just a vertical shift. I have vertical shifts. Vertical shifts. These are given by what? These are things that look like sine of theta, original function, plus some D on

the outside, where if D is positive, sine of theta plus 2, it moves the graph vertically, positively, in the upper direction.

- 5.81** If D is negative, like D is negative 2, it moves it down in the negative direction. That's all fine and good. That's modifying the output by adding it. But remember, up here, we originally talked about modifying the inputs of theta by multiplying it. That was the period, right? Multiply theta by 25, multiply theta by a half. What if I multiply the output? What happens when you multiply the output? Well, let's see what happens there. If I said Y is equal to 2 sine of theta—I don't want this to be a green curve. I want it to be different. So that's the black curve. What happens?
- 5.82** Well, I have the green curve, and then the doubled curve is the black curve. It looks like it got steeper. It keeps the same period. Notice that the peaks are occurring at the same place, the valleys are occurring at the same places, but their heights are different; their amplitudes are different. If I come back to this and I take this and multiply it by 10 to make it a little more extreme, and I zoom out a little bit, well, you can see the green curve is still just sitting here. Then the black curve has these massive amplitudes. Amplitude shifts.
- 5.83** The notion of amplitude is things that look like A times sine of theta. It's this thing in the front, and that's the height of it. What do I mean by the height? I mean, look, I have a peak that comes up and down, and then I have a peak that comes down and up, and there's some line in the middle that cuts it in half. It's called the midline of the graph. It's the midline of the graph. In this case, the midline is literally just the X axis because we're not shifting vertically or anything like that. The midline is literally just the X axis.
- 5.84** The amplitude, what is that? It's the height from the midline to the peak. In this case, what's the height from the X axis to the top? It's 10. The amplitude is 10, which was given by this coefficient, 10 times sine of theta. So that's what the amplitude is. So the question we want to ask, and this is what's amazing with Desmos, is you can do the following things. Let's clear some of these out. And let's do the following thing. I want to start with sine of theta, but now I want to do something like this.
- 5.85** I want to control its amplitude. I'm going to give that its own variable. I'm going to call it A and say sine of theta. And you see how Desmos puts this little thing saying, do you want to incorporate a slider? This is the one thing I love about this tool. You can say, yeah, I do, because they don't know what A is right now. Desmos is like, "I don't know what A is. You're defining a function in terms of theta and you're putting another variable in there. I don't know what that means." You say, OK, click A . Then it gives you this awesome little slider. It gives you this awesome little slider.
- 5.86** Then what happens is it gives you this little animation that you can click on and let it run, and it'll just bounce back and forth for what A is. It'll say, OK, so when A is running from zero to 10, I'm seeing what happens. These things get larger. When it goes negative, it just flips it, and it gives you this nice little simulation that you can use.

That's telling you, how does sine of theta behave when you let that be a little variable. You multiply it by that, and you see what happens.

- 5.87** All right, so let's let go of that. That was amazing. I stopped it right at zero. That was impressive. All right, so what if I instead say, Y is equal to—so let's do this. Rather than Y equals $A \sin$ of theta, what if I say, I want this to be sine of theta minus C . Now, again, Desmos is going to say, hey, do you want a slider for C ? I don't know what C is. You say, yeah, I do. You put a slider in and then you let it run and you say, oh, well, that's what it's doing. That's my horizontal shifting. It's moving me left and right, depending on what the values of C are.
- 5.88** When C is positive, it moves me to the right. When C is negative, it moves me to the left. That's great. I get rid of that. What about the period? What if I put the period we called M ? Let's say M . All right, so there's the period. Let's zoom in to see this one a little bit better. Because this is the one that slinkifies the—that's not a math term. That's just what I think, how I view it. That's why I say it gives you this slinky approach to it. It just compresses it when M is large, and then when M is negative and large, it then flips the graph and then compresses it and stretches it out.
- 5.89** So that's what we get. So which one haven't we done yet? We've done horizontal shifting. We've done the period. We've done amplitude. We haven't done vertical shifting yet. We haven't done vertical shifting. So let's do plus D , and they'll say, do you want a slider? Of course I do. I let D happen, and now it's just moving me up and down, up and down. That's how we can move these particular functions, sine, cosine, tangent, and whatnot.
- 5.90** Now, what's amazing with Desmos, and I'm not sure how well this is going to display, but we can do the following. We can say, well, what if I want to do all of them all together? What if I want the amplitude in there? What if I want the period in there? What if I want the C in there, and I want the D in there? I have the thing that controls the amplitude, the thing that controls the period, the horizontal shifting, and the vertical shifting. Desmos will say, oh, that's great. Which one of these do you want as being sliders? The answer is, well, I want all of them to be sliders. OK, so I'll let all of them be sliders.
- 5.91** My graph is this purple curve, and all chaos breaks loose when you just start letting these run. You start seeing these things moving all over the place. You have the slinky nature of it that's being controlled by the M . You have the amplitude version of it, which is the vertical stretching of it. You have the C values, which are moving it left and right, and then you have the D values, which are moving it up and down. You let all these play, and that gives you how these are moving.
- 5.92** That's simply just to illustrate one. Just to recap, one, we can think of sine, cosine, tangent, cosecant, secant, and cotangent as functions in and of themselves. f of theta equals blank. Solid. Number two, Desmos is your tool for graphing these things, which

is absolutely amazing. Free, nice calculating tool, where you can then run little simulations and animations to actually say what A, M, C, and D are going to be to move these things around.

5.93 Now, what we're going to do is try and figure out, can I somehow invert functions, sine, cosine, tangent, in a coherent way to allow me to solve equations involving trig functions? That's what we'll do next.

Lecture 6: Inverse Functions

At a glance — Inverse functions.

- 6.1** We have trigonometric functions. We have their graphs. We now want to understand, how can I somehow invert them? What do I mean by invert them? The motivating question for this, the motivation is going to be, if I give you something that looks like, I want to know what is a solution to sine of theta is equal to one-half, I have a shot of being able to actually figure out what that is because the unit circle has values for which the Y coordinate is equal to one-half. In particular, I could say π over 6 or 5π over 6. That would be my solution set for this, and then all 2π multiples of that.
- 6.2** OK, well, that's great. But really, what I want to do is think back to algebra. In algebra, we wanted to say, let's isolate the variable. Let's get variable on one side. In algebra, if I had $2X$ minus 3 is equal to 5, I would say, OK, add 3, add 3, $2X$ equals 8, divide by 2, divide by 2, I get X is equal to 4. What I didn't do in that situation is, I didn't say, oh, OK, let's go consult a geometric object, like the unit circle, to figure out what X needs to be. I had an algebraic procedure that allowed me to say, add 3, divide by 2, then I get X is equal to 4 on the nose. I was able to solve for X explicitly. I didn't have to go and consult some table or consult something or other.
- 6.3** The question that we want to understand is, OK, what if I modify the one-half? What if I say something like sine of theta is equal to 1 over the square root of 6? Well, there's no 1 over the square root of 6 on the unit circle, so I'm completely stuck. I have no way of approaching this now. That's the notion of inverting the function. Can I somehow get rid of this part in a concise, mathematically precise way, so that I can then say theta equals something? I want to know what goes there. That's the idea. That's the idea. That's the notion of inverse functions.
- 6.4** I'm just going to give it to you in general, in general. Remember, a function is a rule which takes something from its domain into its range. That's its range. What does it do? It takes a valid input, so something in its domain, which we call X . It does something to it, whatever that is; that something then maps to a point over here, which we call its output, F of X . F is the rule which assigns to X the value F of X . That's the idea. Here's the point. Look, if you give me X mapping to F of X , what I want to ask now is, can I take F of X back to X ? Can I somehow take F of X back to X ?
- 6.5** Well, the answer is, it depends. It's cheating to say something like, I'm going to take X is equal to 5. I'm going to send it to 15. So where should I send 15? Oh, I should send it back to 5. It's like, well, you know you should send it back to 5 because you knew where it came from. But the question is, what if you don't know where it came from? What if I don't tell you that 5 goes to 15? What if I just tell you, you have 15, send it back to something? How would you approach that question?

- 6.6** One thing that you immediately realize is, well, look, functions have the following property. Functions take a valid input to only one output. Functions do not take an input of, say, 5, and then give me this charcuterie board of choices that I get for outputs. I don't get X goes to 5 or X comes from 5 and it goes to 2, 3, 7, negative 14, and 32. It's like, no, you give me one input and you get one and only one output. OK? One thing that you run into very quickly is, if I want to somehow come up with a function that goes back this way, how would I do that? You quickly realize that you need the following to be true.
- 6.7** Look, if I gave you this function—let me motivate this with an example—if I give you the function F of X is, say, $2X$ minus 4, whatever that is, we can graph it. That's fine. If I want to graph it, this is down here at 0, minus 4. This comes up here at 2, 0, and it's a line. All right. These are our Y equals MX plus B . So these are our lines. I look at this, that's great. What I notice is that, if this is my function, where does X equal to 2 go? Let's say that's this point here. X is equal to 2 gets mapped to, what is F of 2 here? F of 2 is 2 times 2 minus 4, so that's equal to zero. So this point here is going to map to whatever the point is here, which is F of 2, which is equal to zero.
- 6.8** All right? OK, what that's saying is that I'm getting this point here in the XY coordinate plane. If my input is 2, my output Y is 0. That's all that's saying. So it would be natural to say, oh, OK, well, can I define a function that goes backwards, that goes this way to this way? I would say, let's define that function to be G . I would say G has to do what? It has to take as an input zero, and it has to output what? An output of 2. I could just define a function that's going to do that.
- 6.9** Now, maybe I don't know a formula for it, but I at the very least know that it has to send 0 back to 2, if it's going to somehow undo F , invert F . It's going to do the inverse operation of F . You think about it as functions. If a function moves from left to right, then its inverse should move from right to left. It should undo that motion of left to right. That's all fine and good. But then what if I—let's see. What if I give you another example, which is a little different? What if I give you a function, let's call it H of X , and I'm just going to give you X squared. X squared, so it's a quadratic.
- 6.10** It's what's called, its graph is what's called a parabola. We can graph that. Its graph looks like this, roughly speaking. Here's this graph. This is the Y equals X squared. There it is. Now, what I notice here is, I can say, OK, well, what happens here? If I look at the diagram here, where here's my domain, and here's my range, what I notice is, if I take the value which is X equals minus 1, where does that map? That maps to this point, which is H of minus 1, which is 1. OK, fine. Negative 1 squared is 1. Fantastic.
- 6.11** But then where does X equals to 1 map as well? Maps to the same place. That's confusing. That's OK. It's still a function. All that's saying in the picture is it's saying minus 1 is here. That means that positive 1 is here, and it's just saying that both 1 and negative 1 map to the same place, which is 1. That's all it's saying. But think about the question at hand. The question at hand is, if you give me something in the range, where

did it come from? Where did it come from? The question is, OK, I'm going to give you this value in the range. This value is 1. How would you define a function going from here back to here, which takes an input to one and only one output.

- 6.12** Which one is it going to go to? How would you define the inverse operation here? How would you go from the range to the domain? How would that even make sense? You would need a function which does what? You would need some function. I don't know what to call it. Let's call it J , such that J of 1 is either minus 1 or 1. Are functions allowed to have this picky-choosy nature to them? No. Functions say, look, you give me an input, I will give you one and only one output. It's not, you give me an input to the function J , and I'm going to give you your choice of negative 1 or 1.
- 6.13** That's not a function. Functions aren't allowed to do that. You don't want a function to give you something where you take as an input and you're expecting one output, and suddenly you get 20 different outputs, and you're like, what the heck is that? That's not a function. There's this notion of, look, I need that if two things are different on this side, if two values are different on this side, I need where they go, those need to be different. Those need to be different. Because if negative 1 had gone to this point, and 1 had gone to a point down here, then it's like, at least I have a shot of saying that, oh, this one should come from here and this one should come from here.
- 6.14** But the fact that both of them go to the same place, when I'm asking the question, where did this one come from? It's, I don't know. It came from multiple places. So I don't know where to send things back to. And that's the intuitive notion of what's called injectivity or a function is one to one. I'm going to keep this function. There's $2X$ minus 4. It's $2X$ minus 4. Then we're going to get into the notion that the characteristic that your function needs to have in order to be invertible.
- 6.15** OK, so here we go. Our definition. We say a function F is— the technical word for it is injective. The technical word for it is injective. That's fine. Another synonym, if you will, for it is one to one. It's one to one. I like that one a little bit more because the one-to-one nature of it means what? I take one thing on this side and I get one thing on that side. I take two different things on this side and I'm going to get two different things on this side in my range. It's one-to-one. It doesn't map two different things to one thing. That's a two-to-one type thing.
- 6.16** We say a function is injective if when X_1 does not equal X_2 in the domain. Then F of X_1 does not equal F of X_2 in the range. That's its defining characteristic. You give me two different inputs that are different, negative 1 and 1. Their outputs have to be different. F of X_1 doesn't equal F of X_2 . For example, this function here, is this function injective? Well, let's think about how we can view that graphically. Graphically, injective is the same thing as no horizontal line intersects the graph in two or more places. In two or more places. That's the way I want you to think about this.

- 6.17** This is sometimes called the horizontal line test for injectivity or determining whether or not a function is injective by looking at its graph and drawing a whole bunch of horizontal lines, and asking if any of those horizontal lines intersect the graph in two or more places. I just do the graph of $2X$ minus 4. It looked roughly something like this. It looked something like this. If I want to know, is this function injective or is it one-to-one, what do I do? I simply just draw a whole bunch of horizontal lines and ask the question, do any of these horizontal lines or are any of them going to intersect the graph in two or more points?
- 6.18** You can convince yourself that the answer is no. They're not going to intersect the graph in two or more points. What does that mean? That means that your function is one-to-one. If your function is one-to-one, then you can invert it. I'll give you a concrete way on how you can do that. But let's go back to the example of this, what was it? G of X ? Or no, we said H of X . Let's look at our H of X function, which was X squared. That was my parabola. You can see immediately just by the graph, if I start drawing horizontal lines, all of the horizontal lines below the X axis are irrelevant because they're not going to intersect the graph at all anyway.
- 6.19** But if I look at the horizontal lines above the X axis, I notice I start getting in trouble really quick. I get two intersection points there. I get an intersection point there and there, there and there, there and there. It seems like they're coming in pairs. Every time I have a horizontal line, I've got one on the left and one on the right. Well, that should make sense. Why? Well, what are these points on the X axis? What are these points on the X axis? This is things like H of minus 2 equals H of 2. They're both 4.
- 6.20** If I look at the line Y equals 4, which is a horizontal line, and ask, where does it intersect the graph? It's going to intersect the graph here and here, which corresponded to X is negative 2 and X is positive 2, both of those two values. Notice that this is failing the definition. It's saying, look, if I give you two different values in the domain, i. e., negative 2 and positive 2, I give you two different values in the domain, their outputs, F of X_1 and F of X_2 , should be different. But what are their outputs? H of negative H is 4, h of 2 is 4. It's literally the same number.
- 6.21** So H can't possibly be one-to-one because it's sending two things to one thing, not one thing to one thing. Sending two things to one. Similarly, what if I said, what is H of minus 3? Well, that's H of 3. That's 9. What's H of minus 1? That's H of 1. That's 1. So yes, the function is giving me, you give me an input, I give you one output, that's fine. But the problem is that you give me multiple distinct inputs, negative 1 and 1, and they map to the same place. Negative 2 and 2, they map to 4. Negative 3 and 3, they map to 9.
- 6.22** How in the world can I imagine just saying, give me a function which sends 4 to something? The question is, well, what does it send it to? Negative 2 or 2? Where does 9 get sent? Negative 3 or 3. You can imagine how crazy this would get. You can't define a function that takes a single input and gives me two outputs. It's like, a function needs

to take what? It needs to take an input to one output. That's the idea. One input goes to one output. But what this function is doing is, it's taking two distinct inputs to one output.

- 6.23** If I wanted to go backwards and my outputs are now inputs, what this is saying is, you're going to take one input to two outputs. That's not allowed. You're not allowed to have that because you can't give two outputs for a single input. OK. But let's not abandon this. Let's not abandon this. All right. This function is invertible, $2X$ minus 4. Fine. Invertible. Great. I haven't told you how to find its inverse yet, but we will. We'll figure that out, and we'll have to establish a notation for it.
- 6.24** This function H , not invertible, because the horizontal line test fails. But here's the point. The domain of H , so the domain of H , is everything. You can plug in any real number you imagine, negative infinity to infinity. You give me any number, I can square it. That's fine. What is its range? The range of H . Well, how low does it go? It goes to zero, and it goes all the way up to infinity. It goes from zero to infinity. That's its range. The question I want to ask you is, can I somehow restrict its domain and still get the same range? Can I somehow restrict its domain and still get the same range?
- 6.25** The answer is this, why do I need this part? Why do I need that? If I only allow myself to view things from here, I still get the same range. I still go from zero to infinity. I still get the same range, but now if I restrict the domain, if I restrict the domain to what? I only want to go from zero to infinity. I don't want to plug in negative values anymore. I don't want to take an input of negative 3. I'm only allowed to plug in positive 3. If I do that, then every single one of these horizontal lines only intersects in one point or no points if you're below the X axis, but we don't care about those.
- 6.26** But all the ones above the X axis only intersect in one point, and now it's one-to-one. Now the function is invertible. Now it makes sense to say, can you give me a function which will take me back? Where is 9 going to get sent to now? It can't get sent to minus 3 because that's not allowed. Where's 9 going to get sent? It's going to get sent to 3. It can't take 1 to negative 1 because that's not allowed. 1 gets sent back to 1, 4 gets sent back to 2, 16 gets sent back to 4, 25 gets sent back to the square root of 25, which is 5. Now it makes sense to invert the function.
- 6.27** But this notion of restricting your domain is the key that we're going to need for our trigonometric functions. That's the key that we're going to need. I need to tell you a couple of things. Number one, I need to give you a notation to indicate what is the inverse function. I need to give you a notation for that. Number two, I have to show you how to find the inverses. Number three, we have to apply these to trigonometric functions. So that's the goal. That's the goal. So let's do that. All right.
- 6.28** Again, I'm going to keep this function up here. This is our motivating example. It's important to have examples. When you're learning math just in general, this is more of a philosophical statement, but when you're learning math in general, you can learn

them from definitions and theorems and proofs and all that stuff, but you need to be able to have a functional knowledge of how to actually apply these things. To do that, it's incredibly helpful to have examples on how to do these.

- 6.29** If I'm trying to understand a concept mathematically of what does it mean to be one-to-one? It's one thing to say, oh, the definition is if X_1 doesn't equal X_2 , then F of X_1 doesn't equal F of X_2 . It's like, that's one thing. It's another thing to be able to say, here's an example, and here's why it's one-to-one. The horizontal line test is passed. I can actually check that it's one-to-one. Having these toy examples to actually understand what these things are is very, very important. Understanding the definitions is one thing, but then being able to apply them is something else.
- 6.30** I joke about this because I always fell into the trap of, as long as I understand the definitions and the theorems and the lemmas and all the statements in the math books, then I'll understand the math. Then I get on an exam and they throw me a question that's, "Hey, do this computation for this concrete example," and I'm like, wait, this isn't abstract. This isn't X_1 's and X_2 's. This is actually $2X$ minus 4. What are you talking about? I don't know how to do it. It's very true.
- 6.31** It's one thing to be able to understand on paper that the Tour de France, arguably the hardest event in the world, is 21 grueling stages of 100-plus miles every single day through the Alps and France and all this stuff and just chaos. It's another entirely to go and ride the event. Those are two very different things. It's one thing to have a theoretical knowledge of it. It's another to have a physical knowledge of it and actually doing it. That's the combination I'm trying to get. It's one thing to understand the theory and the definitions. It's another to be able to look at a concrete example and be able to say, "I can apply that. I can understand this example because I understand various aspects to it."
- 6.32** Here's the notation. If F is invertible, we call its inverse F^{-1} . Now, its notation is literally just that. It's a name. That's it. It's a name. F^{-1} is indicating the inverse of the function F . This notation, this is not, and I know it's confusing, I know it's confusing, but this is not 1 over F . That's not it. That's not it. This F^{-1} , I know negative 1 exponents are typically a 1 over something, but this is not saying that. The negative 1 is just an indicator saying, this is the function I'm giving the name F to the upper minus 1. F to the upper minus 1.
- 6.33** The property that it has, the property is the following. Look, if I have the domain of F , so this is the domain of F , and over here I have the range of F . So F goes from here to here. That's what it does. If it takes an X here over to, let's call it Y , which we're going to denote F of X here, then what should F^{-1} do? F^{-1} needs to undo F . So F^{-1} should go this way. F^{-1} should go this way. So in particular, the domain of F^{-1} should be the range of F , and the range of F^{-1} should be the domain of F .

- 6.34 And what should the property be? Well, what happens if I feed Y into F inverse? I should say, if Y is equal to F of X , then F inverse of Y should equal X . If F takes X over here to Y , then F inverse should take Y back to X . That's all I'm saying. It's a natural notion of inverse. If inverse goes, the function F takes you from left to right, then F inverse should say, it takes you from right to left. It should undo the left-to-right motion. What does that mean? It means that I have either F inverse of F of X is X , and F of F inverse of Y is equal to Y .
- 6.35 If you start with an X on this side and you first apply F and then apply F inverse, you didn't go anywhere. Vice versa. If I first start with Y and I first apply F inverse and then apply F , I didn't go anywhere. That's the notion of inverse. You're undoing that particular operation. That's the notion of inverse. Let's figure out how in the world do you find them? Now that I have a notation, that's one thing. But how do I actually find them? That's another thing altogether. How do we do that?
- 6.36 Well, the method is surprisingly simple for "elementary" functions, in quotes. The real world is not elementary by any stretch of the imagination. The functions that you run into in a math class, if you don't have some computer coding aspect to that math class, the functions that you're going to run into are going to be more often than not toy examples, small numbers, small number of variables, things like that.
- 6.37 If you want to look at something that's like an AI system, you want to look at an AI system, whatever that is—they're popping up all over the place now. But you look at an AI system, you want to say, OK, how does that work mathematically? Well, mathematically, these things have millions of variables associated to them. Every function we've written down in this course has maybe had two terms, three terms, maybe, max. When we're talking about our AI systems that are telling us where we want to eat, how we want to get from A to B , what's the, what's the current sentiment on fill-in-the-blank topic or whatever the case is?
- 6.38 These things have millions of variables associated to them. Unbelievably complicated objects. To invert functions, the way that we're going to do that, it's somewhat superficial to do by hand. It's somewhat superficial to do by hand. But the functions that we have at our disposal right now are these toy examples, these little, like, MX plus B 's and things like that. I want to illustrate that with these, so how to find F inverse. Find F inverse.
- 6.39 OK, here's how you do it. Here's your method. Usually, I'm not a fan of just memorizing a procedure or a method because it, I don't know, in my opinion, I think it takes away the critical thinking nature of mathematics. But in this specific case, I think it's very important to have a methodology for how do I actually approach finding the inverse of a function. Again, what's the motivation for it? You give me an X on this side, that's going to map to a Y , which is going to be F of X , which we call $2X$ minus 4 on this side.

- 6.40** The game is, I want the function which is going to take Y and move it back to X . I want X to somehow be F inverse of Y . That's what I want to be true. That's what I want to be true. OK, so think about that. If this is what I want to be true, well, what do I know? I know that Y is equal to F of X . I know that Y is equal to $2X$ minus 4. Here's the point. This function takes X to Y . In order to get a function which takes Y to X , all I need to do is take the defining equation for F and solve for X . Solve for X .
- 6.41** After all, that's what I want, right? X is equal to something. X is equal to something. OK, so let's solve for X . How do I do that? I'm going to add 4, so Y plus 4 is equal to $2X$, divide by 2. I'm going to write X on the left because that's the proper thing to do. X is equal to Y plus 4 over 2. Therefore, I'm going to define my function F inverse of Y to be that. After all, what am I trying to find? I'm trying to find a function F inverse, such that F inverse of Y equals X . I now have X equals something, so why not just set that equal to F inverse of Y ? That's what I'm trying to do. It turns out that's precisely the right thing to do. That's my inverse.
- 6.42** Now, let's check it and actually see, does this make sense? Does this actually make sense? Remember the two properties that have to be true. If I take F of F inverse of Y , that needs to be Y . If I take F inverse of F of X , that has to be X . Those two properties have to be true. Let's just check them. Let's just check them. Let's see what happens. All right. Here we go. Let's do this one first. What does this mean? F of this. It means, find the formula for F inverse of Y and plug it in right there. OK, so F inverse of Y was what? We said that was this guy, Y plus 4 over 2. So I'm going to plug in Y plus 4 over 2.
- 6.43** What does that equal to? What that means is, it says, take this input, whatever this input is, and plug it into the function of F . So what was F ? F was this guy. So take this input and plug it in for that variable. I know it's Y and this is X and all that. It doesn't matter. It's the fundamental principle. What is the function? It takes an input and gives an output. It doesn't matter what you call the input. If I plug that in, what do I get? I'm going to get 2 times the input, and then I'm going to subtract 4. How do I simplify this? What's 2 times this? 2 divided by 2 cancels. I get Y plus 4 minus 4. Plus 4 minus 4. Fantastic. This principle is true.
- 6.44** If I first start with Y , I move it over by F inverse, and then I apply F back, I haven't gone anywhere. I'm still at Y . What about this property? What about this property? What is F inverse of F of X ? Do I have a formula for F of X ? Sure. F of X is what? $2X$ minus 4. Let's plug that in. F inverse of $2X$ minus 4. What does F inverse say to do? F inverse says, take whatever your input is, so that's my Y , add 4 to it and divide it by 2. Here's my input. I need to take my input, add 4 to it, and divide by 2.
- 6.45** OK, so what do I get? Negative 4 plus 4, cancel, cancel. I'm left with $2X$ over 2, cancel, cancel. They're indeed inverses to each other. They're indeed inverses to each other. This is the methodology for finding it. What do you do? If you have a concrete function, you have a concrete definition for what F of X is, $2X$ minus 4, you set it equal to Y , and

then solve for X . Whatever the relationship is that you find at the end, X is equal to something. That is your inverse function.

- 6.46** This is very misleading. This is very misleading for the following reason. I'm implicitly assuming that you can do this. I'm telling you, what's the method? Set it equal to Y , solve for X . What am I implicitly assuming? That you can solve for X . What if you can't? What if you can't solve for X ? This is a very, very—how do you say? It's a very fickle thing. You can do it in certain cases, but you certainly can't do it in all cases. Let's go back to the X squared example when we restricted the domain. Let's go back to the X squared example.
- 6.47** I think math is hilarious because the more you learn, the more you realize you don't know anything. It's just like, the more I learn about math, the more I realize that I just don't know anything when it comes to the math at all, which I think is beautiful, to be honest. I think it speaks to the complexity of the world and complexity of everything. I think that's a cool thing. I don't think I'd be very satisfied if, at some point, I was like, "Oh, I've learned all of math." I think that would just be the biggest letdown. It'd be the biggest letdown ever because it's like, "Oh, there's nothing else to do. I guess we just go home. That's great."
- 6.48** All right. Let's go back to the example of H of X is equal to X squared on the restricted domain given by zero to infinity. I'm restricting its domain. I'm only allowed to consider inputs that are in the range from zero to infinity now. That's it. That's it. Here's X , here's Y , and I draw the part of the graph that I can draw. I'm only allowed from zero to infinity. Well, let's just run our method. See if I can actually make the method work. What's our method? Method for finding H inverse. Remember, what did we say? We said, take the formula for H , set it equal to Y , and solve for X . That was the game.
- 6.49** I'm going to say, Y is equal to X squared, solve for X . I can say, all right, well, let's just take square roots on both sides. Let's take square roots on both sides. If I take the square root of Y , that's equal to the square root of X squared. This is one of my biggest pet peeves about things like this. Because it's so often, so often we just say, oh, the square root of X squared, it's just X . Because that's what we're told, right? That's what we're told. Oh, the square root of square, it's just X . The answer is, well, in this case, yes. In general, no.
- 6.50** In general, what is the square root of X squared? It's the absolute value of X . I'm going to go on a two-minute tangent here for two seconds to just introduce absolute value and explain to you what these things are. Absolute value. This is very important because this is a mistake that is made far too often, and it is far too easy to explain for this mistake to continually keep happening. The absolute value is what? The absolute value function, absolute value of X , always gives an answer which is greater than or equal to zero. That's what the absolute value is.

- 6.51** If I say, what's the absolute value of 2? You say, 2 is positive, the absolute value of 2, it's 2. What's the absolute value of 5? Oh, 5 is positive, absolute value of 5, it's 5. What's the absolute value of negative numbers? If I said what's the absolute value of negative 1? You're like, negative 1 is negative. I need to make it positive, so what do I do? I just forget about the negative. So absolute value of negative 1 is 1. I say, what's the absolute value of negative 6? And you're like, negative 6 is negative. I got to make it positive, so I just get rid of the negative. And absolute value of negative 6 is positive 6.
- 6.52** How do you think about this? The absolute value of 5 is equal to 5. Why? Because 5 is positive. The absolute value of zero is zero. That's fine. Why? Because well, zero is zero. It's neither positive nor negative. It's literally the differentiation between positive and negative. It's the thing sitting in the middle ground. But if I give you something like, what's the absolute value of negative 2? We'd say, oh, it needs to be positive. The way that we teach it is, we say, oh, just forget about the negative, which is misleading.
- 6.53** Because what the absolute value of negative 2 is, is, it says, look, take what you're given, negative 2, how do you make it negative? Or how do you make it positive? You multiply it by negative 1. That gives you 2. If the number you feed me is negative, I just multiply it by negative 1, and it'll make it positive now. The absolute value function has the following property. Its graph looks like this, which means that the absolute value of X looks like X , if X is greater than or equal to zero, and minus X if X is negative. That's what it is.
- 6.54** Let's go back to our definition down here. What's the absolute value of 5? X is 5. 5 is positive. Therefore, the absolute value of 5 is 5. Solid. Negative 2 is negative. X is negative 2. Negative 2 is negative. So the absolute value of negative 2 should be minus minus 2, which is positive 2. Here's the point. When I take the square root of X squared, why do I get the absolute value? Why do I not just get this? Why is this wrong? Well, let's think about it.
- 6.55** If I take the square root of 3 squared, that's the square root of 9, which is 3, that's great. But the square root of minus 3 squared is the square root of 9, which, well, that's 3. If you claim that this is true, this is a counter example that breaks it. In this example, X is 3, and I get the square root of X squared is X . In this example, X is minus 3, and I get the square root of X squared is not X . So how do I fix it?
- 6.56** That's how you fix it. The square root of X squared is the absolute value of X . It's not equal to X . But now here's the point. What was my restricted domain? Everything bigger than or equal to zero. If everything is bigger than or equal to zero, what is the absolute value of X ? It's equal to X . This is because of the restricted domain. That's why. In particular, now we've solved for X as a function of Y , and I can now define the inverse function of H . H inverse of Y is equal to the square root of Y . There he is.
- 6.57** Do you see how nuanced this is and how frustrating this is? There's so many details that are just so intricately, just all of these little things. It's like when you're trying to, I

don't know, you first start learning to cook or something, and somebody gives you this recipe, and then you realize that somebody else can make a way better quesadilla than you can because they actually understand all these little minute details of all these little things. It's absolutely insane.

- 6.58** The amount of details that just went into this, simply to say that square root is the inverse of the X squared function, what did we have to do? We had to restrict our domain because our function was not one-to-one. OK, we restrict the domain. It's one-to-one now. Great. We then use our method to say Y is equal to X squared. Fantastic. OK, I take a square root on both sides. Well, then I get stuck because the absolute value is here. I need to understand that the square root of X squared is equal to the absolute value. But then I can say, well, that's OK because I'm on a restricted domain. So the absolute value of X is actually equal to X .
- 6.59** Now I have X is equal to this, and I have my inverse function is equal to that, square root of Y . Let's check it. Let's check to see what the properties are. Do the properties hold? Remember the two properties that we have to have for inverse functions. We need what? H inverse of H of X . We need this to equal X . What is H of X ? This is H inverse of H of X , which is X squared. H inverse is take whatever the input is and take its square root. This is the square root of X squared. That's the absolute value of X .
- 6.60** But remember, the absolute value of X is X in this case because of my restricted domain. Because of my restricted domain. So this is equal to X because of the restricted domain. So I get half of the picture. What's the other half? I have to reverse the order. I have to do H of H inverse. What happens there? What happens there? We have H of H inverse of Y . What was H inverse of Y ? That's my square root of Y . This is H of root Y . What does H do? It says take an input and square it. This says take a square root of Y and square the result. And squaring a square root undoes it. And so I actually get Y on the nose.
- 6.61** So H and H inverse are actually inverses to each other, but only when I restrict my domain. If I include the entirety of the domain X squared, this function is not invertible on its entire domain. If I include negative values of X , I'm not allowed to invert it. It's not going to work. Let's talk about inverse trig functions. Let's talk about inverse the trig functions.
- 6.62** Think about the graphs that we had of the trig functions to begin with, the sine and cosine, for example. They had this wave behavior, and they just kept repeating over and over and over again. And ask yourself the question, do those satisfy the horizontal line test? Do those satisfy the horizontal line test? Let's look at inverse trig functions. Inverse trig functions. Let's start with the function F of X is equal to sine of X . Let's look at that. Actually, let's use θ because we've been, let's be consistent. Let's use θ . Let's look at that particular function.

- 6.63** What did its graph look like? Its graph was just this, roughly. This is why Desmos is better. That was roughly what that looks like. Now, we know that a function is invertible if its graph satisfies the horizontal line test. Now draw a whole family of horizontal lines and tell me, do they intersect the graph in two or more points? The answer is, look, if you literally draw a line right here, Y is equal to, say, one-half, it intersects the graph in infinitely many points. That's like as bad as it can get. Not only does it intersect the graph in two or more points, but it intersects the graph in infinitely many points.
- 6.64** This function is not one-to-one on its entire domain. What we want to do is, we want to say, can I restrict its domain so that I still get the full range, I still get the full range, but now I have one-to-one? Well, let's think about it. What is the range here? What is the peak? The peak is 1, the trough is minus 1. Can I still get that somehow? The answer is, well, let's look at it. Look, it bottoms out where? Right here at minus π over 2. Bottoms out at minus π over 2, and it tops out at positive π over 2. If that's what gives me my full range, why don't I just say, OK, let's just consider it there. Let's forget about this part of it, and let's forget about that part of it. What if I only look at it in that range?
- 6.65** I look at that piece. It turns out that's the right thing to do. We're going to restrict the domain to minus π over 2 to π over 2. That's what its domain is going to be. Its range is still the same. It's still minus 1 to 1. The lowest they can get is negative 1, the highest they can get is positive 1. We still have that. But now, if I draw horizontal lines, they only intersect in one point. So my function is invertible. What is the inverse function going to be called? Here's the definition of it.
- 6.66** The inverse of sine on the restricted domain given by minus π over 2 to π over 2 is F inverse of θ , which is called arcsine of, let's say, our F inverse of Y , which is arcsine of Y . It's just its name. That's all it is. If you want, you can denote it by this. There's other ways to denote this. It's sine upper minus 1 of Y . That's totally fine. That's another way to denote it. It's fine. Again, the upper minus 1 does not mean this is not 1 over sine. 1 over sine is cosecant. I'm not saying that the inverse of the function is cosecant. That's not what I'm saying. Be very careful about this.
- 6.67** I steer away from this notation, personally, because the upper minus 1, it's very easy to fall into this trap in inverting it, sticking it in the denominator. So don't do that. Don't do that. I like to stick with this notation right here, arcsine of Y . Now we can go back and we can solve the problem that we had initially at the beginning of class, which said, what is the value of θ for which sine of θ is equal to 1 over the square root of 6? We can now answer that question. Let's go back to the example that we had earlier.
- 6.68** Solve for sine of θ is equal to 1 over the square root of 6. I want my algebraic approach, just like we started with, add 4, divide by 2, solve for X , kind of thing. X equals 5, whatever it is. I want to be able to somehow eliminate the sine and get θ equals blank. θ equals blank. How do I do that? Well, remember, what is this saying? This is saying F of θ is equal to 1 over the square root of 6. What I would

like to do is, I'd like to say, OK, apply F^{-1} on both sides. That's our algebra. Whatever we do to one side, we have to do to the other.

- 6.69** If I apply F^{-1} on both sides, I get F^{-1} of F of θ is F^{-1} of $1/\sqrt{6}$. Yeah? I apply F^{-1} on both sides. What was the property that we had of F^{-1} of F ? These cancel, and I'm left with $\theta = F^{-1}$ of $1/\sqrt{6}$. Well, what is F^{-1} ? F^{-1} of $1/\sqrt{6}$ is $\arcsin$ of $1/\sqrt{6}$. What is that? No clue. Desmos would, though. Desmos would know what that is, because Desmos is an awesome calculator.
- 6.70** But at the very least, I can say, look, now I have a systematic way for saying what θ equals. It equals $\arcsin$ of $1/\sqrt{6}$. Here's your approach. Here's your approach. We can either ask questions such as trigonometric function equals number, or I could ask you a question that says this. Here's another example. What is, let's say, $\sin^{-1}$ of, I don't know, one-half? Let's ask that question. I'm going to give you the input one-half and ask you to plug it into the sine inverse function.
- 6.71** I'm breaking my rule here, saying sine inverse. Really, I'm saying, what is $\arcsin$ of one-half? I need to give you both notations because both notations are very common. I personally don't like this notation, sine inverse. I personally like the arcsine, but it doesn't matter. You need to be able to use both. How do you approach this question? It's exactly the same as what we did with logarithms and exponentials in algebra. I ask you to find the log of a certain number, I can translate that into an exponential question and vice versa.
- 6.72** Here's what I do. I say, well, I don't know I don't know what this is. I have no clue what this is, so I'm just going to give it a name. I'm just going to give it a name. I'm going to set $\theta = \arcsin$ of one-half. Whenever you have something involving an inverse function, you have the corresponding equivalent with the original function. This is equivalent to saying, apply sine on both sides. Those cancel each other out. It's equivalent to saying, when is the sine of θ equal to one-half? I claim that this is how you think about these types of questions every single time.
- 6.73** If somebody asks you, "What is the arcsine of something?", you turn it into the corresponding, oh, I'm saying, "What is the sign of something equals a half?" Then ask the question, what is the angle θ ? Because the angle θ is arcsine. That's how we defined it. That's how we defined it. Now you have to ask yourself, well, on the unit circle, where does this occur? I've got the unit circle. Where is sine equal to one-half? I look at the line $y = 1/2$, and I see two points. I see this point, and I see this point.
- 6.74** Now, remember, θ has to live, it's got to live here. In particular, $-\pi/2$ is here, and $\pi/2$ is here. So it's got to live somewhere in this range. So is this going to be my angle from out to here? No. What is this angle here? You consult the unit circle, you get $\pi/6$. You get $\pi/6$. So that means that $\theta = \pi/6$.

over 6. So what does that mean? It means that arcsine of one-half is equal to π over 6. That's what it means.

- 6.75** Now, there's two other inverse functions. I'm just going to give you their definitions and their corresponding domains. Technically, you can invert all six of them. The ones that you run into the most are sine inverse or arcsine and arc tangent. Those are the two that you run into the most. Sometimes you'll run into arc cosine, but very seldom do you run into secant inverse, cosecant inverse, and cotangent inverse. I'm going to give you the tangent version of this.
- 6.76** For tangent, tangent's graph looked like this, and we had this minus π over 2 asymptote and this π over 2 asymptote. Then we had these repeats. I had this little repeats over and over and over and over again. Well, look, if I draw any horizontal line, it's going to intersect in infinitely many places. What am I going to restrict my domain to? Negative π over 2 to π over 2. For F of θ equals tangent, the domain of F is going to be restricted to minus π over 2 to π over 2 in order to define its inverse.
- 6.77** What is its inverse? F inverse of θ , or F inverse of Y , will be arctan of Y . That's it. If I give you cosine, we'll do cosine up here. If I give you cosine, cosine looks like this. This is Y equals cosine, or F of, let's call this one G . G of θ equals cosine of θ . How do I restrict its domain? I restrict its domain to be zero to π . The restricted domain for G is going to be, domain will be from zero to π , and then the inverse, G inverse of Y , will be arc cosine of Y .
- 6.78** Basically saying, look, if I just forget about this part of the function, now it's one-to-one, and it makes sense to be able to invert it. Being able to invert it then it gives me a systematic way for giving, given in equations such as, let's say the tangent of θ is equal to one, what is θ ? OK. θ is going to be arctan of 1, which just happens to be π -fourths. It gives you a method for solving equations involving trigonometric functions systematically.

Lecture 7: Laws of Sines and Cosines

At a glance — The laws of sines and cosines.

- 7.1 OK, now that we have inverse functions for trigonometric functions, we want to understand, OK, well, where have we been? Let's recap. Where have we been? We've come from right triangles. That's where the sine, cosine, tangent came from. The reciprocals gave us the secant, cosecant, cotangent. OK, so we had this right triangle business. Then we said that there was a relationship between the right triangles and the unit circles.
- 7.2 We tied in right triangles into the circle, and we saw that values of our trigonometric functions could be determined based upon these special values like zero, pi-sixths, pi-fourth, pi-thirds, pi-halves, and so forth and so forth around the unit circle. Then we realized, well, what if we can view these trigonometric functions as actual functions? What if we can view them as just functions in their own right? If we can do that, if we can make sense of that, which we did, can I make sense of how to invert them?
- 7.3 Let me just briefly remind you about inverting. Inverting. How do you invert a particular function? How do you invert a trigonometric function? Again, let me motivate this with an example. If I asked you, how would I say, evaluate, let's say, evaluate arcsin of, let's say, I don't know, square root of 3 over 2. I then want to say arc, let's say, cosine of minus a half. And then let's do arc tangent of, let's do square root of 3. I have to give you values that you can actually do. I have to give you values that you can actually do. I can't simply just say something like, what is the arcsin of, I don't know, 1 over pi or something along these lines.
- 7.4 I have to give you something that you can use the unit circle to actually evaluate. We don't have a calculator that we can just plug this into. We can't just run to Desmos, plug it in, get the value that we're looking for. We need to understand how to be able to do this by hand. How do we approach each and every one of these? My solution, let's say this is number 1, this is number 2, and this is number 3. All right? Let's just approach each one, one by one. If I look at the first one, again, we want to take advantage of the fact that inverse functions can be viewed in terms of the original functions in a very precise way.
- 7.5 What is that precise way? I don't know what this quantity is. That's what I'm trying to figure out. I give it a name. I call it theta, and that's going to be the arcsin of the square root of 3 over 2, and I say that that's equivalent to saying, OK, if theta is equal to arcsin of root 3 over 2, that's equivalent to saying that the sine of theta is equal to the square root of 3 over 2. It's equivalent to that. I'm happy with that because the square root of 3 over 2 reminds me of certain unit circle values. Then what I can do is, go to my unit circle and say, all right, well, let's go to the unit circle. Let's put that up here.

- 7.6 Then let's look at where the Y coordinate, sine of theta is the Y coordinate, where the Y coordinate is the square root of 3 over 2. The square root of 3 over 2 is maybe right up here. That's Y equals the square root of 3 over 2. I'm looking for what angle theta gives me the sine of that angle is equal to the square root of 3 over 2. That's the idea. That's the goal. All right? OK.
- 7.7 I start searching the values, and I have to say, well, be careful. There's a point here and there's a point here. It looks like there's going to be two values, this angle of theta and this angle of theta. But which one do I choose? Well, remember, we had to restrict our domain in order to make sense of the inverse functions because we need them to be one-to-one. Remember that on the restricted domain, this was restricted. So sine is restricted to be from minus pi over 2 to pi over 2. I'm only allowed to consider from minus pi over 2 down here all the way around the unit circle up until pi over 2 up here.
- 7.8 In particular, this point is not going to play a role for us in the second quadrant. We're only going to take the one that's in the first. OK? So I scan through the values and I say, OK, sine of theta is equal to the square root of 3 over 2, and that's going to tell me that theta is equal to pi over 3. In particular, the arc sine of root 3 over 2, this is equal to pi over 3. We get that one taken care of. OK. Well, let's look at the next one. The second one, we're going to just do the exact same approach. We're going to say, OK, arccosine of negative one-half, we want to understand what that is.
- 7.9 OK, well, arccosine of negative one-half, we're going to call an angle theta. That's arccosine of negative a half. Again, what's the relationship between these? The relationship is that theta's arccosine of negative one-half is equivalent to saying cosine of theta is equal to negative one-half. So cosine of theta equals minus one-half. Again, I go and I look at the unit circle and I say, all right, now I have a new angle of theta. Now I have a new angle of theta, so that's going to change this up a little bit. What am I going to get? Well, I look at my X coordinate now because cosine is the X.
- 7.10 Since cosine is equal to X, I'm saying, what angle of theta gives me an X coordinate of negative a half? Well, we know this distance is negative 1, so it's somewhere in the middle. There's my X equals negative a half. I say, all right, well, there's this. That should probably be my angle of theta. OK. Then I have to ask myself, well, look, if I have an intersection point here, I also have an intersection point here. Which angle am I talking about? Am I talking about this angle theta, or am I talking about all the way around down to here? Am I talking about this angle of theta?
- 7.11 I have to go back and say, oh, wait, wait, wait. I remember I had to do a restricted domain. I also had to do a restricted domain for cosine. Remember, cosine has restricted domain given by zero to pi. I'm only allowed to go from zero to pi in order to invert the function. Oh, OK. Zero to pi is where? Zero theta is here all the way around to pi theta is here. So this point is not going to matter for me in the third quadrant. I'm only going to be interested in the second quadrant point that I can get. OK, great.

- 7.12** Then again, I go through my unit circle and I try and figure out the values for which cosine is equal to negative a half. Well, remember, if this is negative a half here, minus one-half, comma root 3 over 2, then the mirror image of that angle here is going to be what? This angle has to be one-half the square root of 3 over 2. That comes from pi-thirds, and I'm going to need to go an additional pi-thirds to get to this point. What am I going to get here? I'm going to get theta is equal to 2 pi-thirds. 2 pi-thirds. OK. All right.
- 7.13** Last one. Arctangent is the square root of 3. This one's a little tricky. But the setup is exactly the same. Again, we're going to set theta equal to arctangent of the square root of 3. That's equivalent to tangent of theta is equal to the square root of 3. But now you have to be a little bit careful, because tangent, we don't really think about the tangent values on the unit circle. The unit circle we think of as an X coordinate, the one-half, the Y coordinate, root 3 over 2, or the X coordinate, 1 over root 2, Y coordinate, negative 1 over root 2, or 0, 1 or something like that.
- 7.14** We don't think about tangent as being like sine divided by cosine. I don't have another table over here that says, oh, here's cosine, here's sine, and here's what tangent is. So one way we can view this is, we can say, well, look, we know that tangent is equal to sine of theta over cosine of theta. That's supposed to equal the square root of 3. That's supposed to equal the square root of 3. Now, I don't see a sine or a cosine. It'd be awesome if I could just look at this and say, oh, the sine of theta is equal to the square root of 3 somewhere and cosine of theta is equal to 1. That would be nice because I could think about this as the square root of 3 over 1.
- 7.15** If I could say that sine is square root of 3 and cosine is 1, I win. Unfortunately, that's never true. What I can do, though, is I can do this little algebraic trick, which seems like cheating, but I'm going to prove to you that this is actually legit, is that I can divide numerator and denominator by the same factor. I can divide the numerator by 2, I can divide the denominator by 2, and I haven't changed the ratio. This is the weird thing about fractions, but if I give you the fraction 5 over 6, that's the same thing as if I multiply top and bottom by 2, 10 over 12.
- 7.16** It's the same as if I multiply top and bottom by negative 2, so negative 10 over negative 12. Same if I multiply top and bottom by 3, so 15 over 18, and so on. As long as I do whatever I do to the numerator and the denominator, I don't change the fraction. In particular, if I take this whole thing and I multiply by one-half, five-halves over six-halves, I still haven't changed it. If I think about the square root of 3 over 1 that way, and I multiply the numerator by one-half, and I multiply the denominator by one-half, then I haven't changed the fraction. I haven't changed the fraction at all.
- 7.17** But now I can say, oh, cool. Now I'm looking for, where is the sine of theta equal to the square root of 3 over 2? Sine of theta is the square root of 3 over 2 when theta is equal to pi-thirds. I'm happy about that. Does that happen to be the same place where cosine of theta is equal to one-half? Yes, it is. Here's cosine. What does that mean? That

means that tangent of theta is equal to the square root of 3, is equal to the square root of 3, precisely when theta is equal to pi over 3.

- 7.18** That's a very common trick, is to try and represent whatever the tangent is in terms of sines and cosines, and then trying to make this match values on the unit circle that you already know. Square root of 3 over 2, 1 over the square root of 2, one-half, things like that. That's just a brief recap of inverse trigonometric functions, how to evaluate them, and so forth. What we want to do now is, we want to say, look, we've been dealing with right triangles this entire time for sine and cosine. That's where they came from. They all came from looking at right triangles.
- 7.19** So I had this angle, theta. I said, this is my opposite, this is my adjacent, this is my hypotenuse. That's great. But the fact of the matter is, is that, well, there are not a lot of right triangles in the world. For example, that's not necessarily a right triangle. I could give you one that looks even worse. That's not a right triangle. So, do sine, cosine, and tangent make sense there? That's what we want to explore now. We want to explore that. OK, so let's see what we can do. Maybe we could do nothing. Maybe we can do nothing. Maybe we can do everything. I don't know. That's part of mathematics. It's just exploring the unknown.
- 7.20** All right. Let's see what we can do. We know how to handle right triangles, so we don't want to abandon all of that. But we want to say, what if I just give you an arbitrary triangle, what can I say? If I give you an arbitrary triangle, let's do something like this. If I give you something that comes up here, down here, and over here. What I want to do is give these side lengths and angles names. Let's say, for example, I give this an alpha. That's the angle measure there. Corresponding the opposite side of alpha, I'm going to give the side length of A.
- 7.21** Then let's say, I don't know, I choose beta here, side length B, and here will be side length gamma, or angle measure gamma, and side length C. I just give you an arbitrary triangle like this. I don't know what alpha, beta, and gamma are. The only thing that I can say for sure is that alpha plus beta plus gamma is equal to 180 degrees. That's the only thing I can say for sure, is that I know that their interior angles add up to what they should be.
- 7.22** Now, here's the trick. Pick two angles. I don't care what they are. Pick two angles and throw the third one away. If we're thinking about Greek alphabet, like alpha, beta, gamma, like alpha and beta, the first two letters, then we're going to go with alpha and beta first. We're going to throw gamma away for the time being. We're just going to pretend that gamma doesn't exist. We're going to pretend it doesn't exist. We're going to bring it back in just a second. We're going to bring it back in just a second. But let's pretend that gamma doesn't exist.
- 7.23** And imagine yourself standing at this particular point. You're physically standing at that particular point. However you want to imagine this, I don't care. But let's say you're

standing at this point and you have this particular line of sight, or maybe this is a wall or something like that. You can think about it as the corner of a wall, maybe, something along those lines. You're standing here. And what you want to understand is, if you imagine standing here with a laser pointer, you've just got a laser pointer, and you imagine pointing it in the direction of this wall, whatever that is.

- 7.24** Then you imagine sweeping that out throughout the angle that we originally called gamma. You imagine pointing. It's like, I'm pointing a laser pointer in this part of the room, and then I'm just going to move it along the line. That's what I'm going to do. Then the laser pointer is going to do what? It's going to trace out along the wall C. Do we agree? I'm just going to have this laser pointer pointing out along this wall C. Now, I want you to convince yourself that there has to be some point—I don't know where it is—but there has to be some point at which, when the laser pointer goes from where I'm standing to across the room, it forms a right angle.
- 7.25** Does that seem plausible, at least? What am I saying? If I shine the laser pointer, maybe it's right here. Maybe it's right there. That looks actually pretty reasonable. Or that's now a right angle. In particular, this is going to be a right angle as well. Both of those are going to be right angles. You can imagine yourself doing this. You're just in a room, you point the laser pointer in one way, you start moving it across the room, and at some point in that range, there's going to be like, ahh, right there. That's where I get my 90 degrees.
- 7.26** Why is that useful? Well, again, remember, sines and cosines come from right triangles. If I just gave you the alpha, beta, gamma triangle, I can't simply just say, oh, sine of alpha is A over B. It's like, what does that even mean? It's like, well, there's no hypotenuse here. There's no 90 degrees. There's no, what are you talking about? But maybe I can get something out of it. The only way that I can use it is if I have a right triangle somewhere. So I manufacture one. I create one.
- 7.27** Well, I don't know what this distance is, from this point up here to this point down here. I have no idea what the distance is, but at the very least, I can say that it's maybe X. I can give it a name. It's a variable. Now what I can do is, I can say, look, I've got two right triangles now. I've got this right triangle that looks like what? This distance, this distance up, that's X. This distance to the side is B, and this angle is alpha. Do we agree? That's this triangle here. Let's call this triangle one. So this is a triangle one.
- 7.28** I have no idea what this distance is down here, because all I do know is that this distance is C, from this point over here to this point over here. That distance is C, but I have no idea what this distance is. It's like, I know that the back wall is 10 meters long, but when I'm sweeping the laser pointer across, it's like, I don't know how far from one side the laser pointer is. Maybe this distance is 8 meters, if this is 10. Maybe this is 2 meters, if this is 10. I don't know. I can't label this side, but I can say that this is a right triangle.

- 7.29** Then I want to ask myself the question, what trigonometric function can I use that relates alpha, X, and B? I say, look, if alpha is the angle, then the opposite side of alpha is X and the hypotenuse is B. Which trigonometric function is the opposite over the hypotenuse? That's sine. It makes sense to say, what is the sine of alpha? That's going to be X divided by B. In particular, I can solve for B if I want to, or I can solve for X if I want to. Just multiply both sides by B. In particular, X is equal to B times the sine of theta. OK, that's great.
- 7.30** If I knew B and I knew alpha, then I would know the height of the triangle. I would know this distance here. I would know the distance from me to the wall, which is given by this laser pointer distance, which makes this 90 degrees. I would know that. But I also have another triangle sitting over here. I've got this triangle, which looks like this, where this angle is now beta. This is still 90 degrees. This is X, and now this side is A. I want to ask myself the question, what trigonometric function will relate beta, X, and A?
- 7.31** Again, I look at the angle beta, and I say opposite of it and hypotenuse are known. I can take sine of beta. If I take sine of beta, that's equal to what? X divided by A. I can now solve for X. X is equal to A sine beta. Then you stop for a second, you're like, wait a second. You just told me that X was equal to B sine alpha. I'm like, that's true. Now you're telling me that X is equal to A sine beta, and I'm like, that's also true. Then what does that mean?
- 7.32** Well, if X on the one hand is equal to this, and X on the other hand is equal to this, then that means those two things have to be equal to each other. It's like I'm saying X equals 5, but I'm writing 5 in a goofy way over here, and then I'm saying X equals 5, and I'm writing 5 in a different goofy way, well, that means those two goofy ways have to be equivalent. Those two things have to be the same. They're both equal to 5. I'm just writing them differently.
- 7.33** This side I'm writing it as B sine alpha. This side I'm writing as A sine beta. In particular, B sine alpha is equal to A sine beta. The left-hand side is what? That's X. That's equal to X. The right-hand side is equal to X. Well, and last time I checked, X is equal to X. That's fine. It's like saying 4 equals 4. It's like, well, that's facts. That's true. But what do I notice? Alpha, the opposite side of alpha is A. The opposite side of beta is B.
- 7.34** Maybe it would be better to divide both sides by A, and then divide both sides by B, so that I can write this as sine of alpha over A is equal to sine of beta over B, so that the alpha and the A are somehow related and the beta and the B are somehow related on opposite sides of the equality. There's nothing wrong with writing it this way. That's completely fine. It's just, it would kind of be nice to have a formula that involves the angle and its opposite side. The angle and its opposite side. I can couch it like this.
- 7.35** But why did I get rid of gamma? Why did I get rid of gamma? Well, I just made a choice. I had to choose which angle to get rid of because whichever angle you get rid of, that's where you're going to stand and shine your laser pointer across the wall. That's what

you're going to do. Why did I choose to get rid of gamma? Why didn't I get rid of, say, alpha, for example? OK, well, let's do the computation. Let's change the problem. If I have the same triangle, I'm going to do my best to try and reproduce this. But now this is my gamma, this is my beta, here's A, here's B, here's C.

7.36 But now I'm going to get rid of alpha instead. Again, same triangle, same triangle here. Well, what did I do over here? What was my process? I got rid of the angle gamma. I fixed myself physically here with a laser pointer, and I just started shining it across the room to find out where I got a perpendicular line on the opposite side. What am I going to do here? I eliminated alpha, so now I'm going to situate myself standing here, point my laser pointer in this part of the room, and start sweeping it out until I find, oh, there it is. There's the perpendicular.

7.37 Where is that going to occur? It's hard to say. It looks like maybe something like that. No, I don't know. OK. Something like that. OK. Let's do the exact same procedure we did here. This has some distance associated to it. I don't know what that distance is. Let's call it Y. I'm not going to call it X because I called X this distance here. I'm going to call it a different variable. Let's call it Y. But now I can play the same game. I can say, look, there's two triangles here. There's triangle one and there's triangle two.

7.38 This triangle one looks like what? It looks like, it's hard to visualize because it's tilted on its side. We haven't really looked at triangles in this orientation. All of our triangles have been nice, perpendicular or parallel to the axes of the whiteboard. But still fine. Same principles apply. Here's C. What do I get? Here's beta and here's Y on that side. I ask, what trigonometric function relates beta, Y, and C? I say, well, Y is opposite beta, and C is the hypotenuse. I can say, oh, OK, well, sine of beta is equal to Y over C, which means that Y is equal to C sine of beta.

7.39 OK? All right. What can I say from there? Well, that was triangle one. What about triangle two? Triangle two looks like, OK, this one's a little harder to draw, but it looks something like this. There's my 90 degree. What do I know about it? I know this is Y, I know this is B, and I know that is gamma, yes? Again, I ask the same question, what relates gamma, Y, and B? That's sine. I can say, sine of gamma is equal to Y over B. In particular, Y is equal to B sine gamma.

7.40 But look, that's the exact same setup I had over here. Over here, I said X was equal to something, and then I said X was equal to something different. Well, I have Y is equal to something, and then Y is equal to something different. Those two things have to equal each other. But what does that give me? That tells me, C sine of beta is equal to B sine of gamma. Again, just like I did over here, let's get the angles and their corresponding opposite sides together. I have gamma and its opposite side C, and beta and its opposite side B. What if I divide both sides by B and C?

7.41 What do I end up getting? The C's cancel, B's cancel, and I'm left with what? Sine of beta over B is equal to the sine of gamma over C. Here's the whole beauty. Pair this

with what you have in the other one. The other one told you that sine of alpha over A is equal to sine of beta over B, but this one's telling you sine of beta over B equals sine of gamma over C. You can now write this as sine of gamma over C, and call that the law of sines. That's a big result in trigonometry.

7.42 It tells you that if I give you an arbitrary triangle, an arbitrary triangle, we can now get rid of the X and go back to the original triangle that we were talking about. If I get rid of the X and this little perpendicular line, so I get rid of this, and I keep the same angle that I had up here, which was gamma, it tells me the sine of alpha divided by its side length is equal to the sine of beta divided by its side length, which is equal to the sine of gamma divided by its side length. That's what's called the law of sines.

7.43 Let's see an example. Why is this useful? Why is this useful? Well, think about this. This triangle doesn't necessarily need to be a right triangle. It doesn't need to be. It's telling me that, if you give me, say, the angle alpha and, say, the side lengths C and A, I should be able to find gamma. Once I know alpha and gamma, I can find beta. Once I know beta, I can find B. You can actually fill out what this triangle is. Let's do an example of this. Let's do an example. I'll keep the law of sines on there.

7.44 What we just did was the mathematical proof of the law of sines. I think mathematical proofs are scary, especially when you first start learning them because they seem very abstract. They seem very, I don't know, "where did that come from" kind of thing. That's fine, which is why I didn't want to give you the statement of the law of sines first. I wanted to just walk through, basically, what? We were just exploring. We're saying, OK, drop a line down, what can I say? Drop a line over, what can I say? I made some observations. I came up with a relationship between these things, and oh, that's a theorem. That's great. That's super useful.

7.45 OK, so let's figure this out. How can I use this? That's my law of sines. Let's do this. Fill in the missing sides and angles of the given triangle. I'm going to give you this triangle. Let's say it does something like that. I don't know. I'm going to say, maybe this angle here is 70 degrees. That's 70 degrees. Maybe this angle here is 75 degrees. What do I want to say? Maybe this side over here is length 4. I want you to fill in the remaining sides and the remaining angles. How do we do that? How would I get this angle? Do I even need the law of sines for that angle?

7.46 What's the defining characteristic of the interior angles of any triangle whatsoever? Well, they have to add up to 180 degrees. If I know that this is 70, if I take 70 plus 75, that's what, 145? Which means that this angle needs to be what? 180 minus 145, which is 35, something like that? I know that this angle is 35 degrees. Well, that's great. I didn't need to use the law of sines there, but that's OK. Now, if I don't know what this angle is here or this side length is here, let's call that A, and let's call this side B. I know that this side is 4. That's fine. I know that this side is 4.

- 7.47** Let's say we want to go after the side length of B. What could I use? Well, the law of sines tells me, if I take the angle and the opposite side, and then, for example, this angle and its opposite side, those two things have to satisfy that relationship up there. So by the law of sines, the law of sines says what? It says, if I take the sine of 70 degrees, that's the angle, and I divide by its side length, so sine of 70 divided by 4, that has to equal the sine of this angle divided by its side length, B. Sine of 35, 35 degrees, divided by B. That's what that's saying.
- 7.48** And then the question is, well, OK, solve for B. How do I do that? Well, if I multiply B over, multiply by B on both sides, what do I get? B times sine of 70 degrees, divided by 4, is equal to sine of 35 degrees. What am I going to do? Multiply by 4 on both sides and divide by sine of 70. That's going to tell me that B is equal to 4 times the sine of 35, divided by the sine of 70. Now, I run to my favorite calculator and I plug those in, and it tells me, that's my side length. Now I know what B is.
- 7.49** But why did I choose B? Why didn't I choose A? I just as easily could have said what? Sine of 70, divided by the side length 4, is equal to the sine of 75, divided by its side length A, and then did the algebra to get that A is also 4 sine of 75, divided by sine of 70. Now I know what all of the side lengths of A, B, and C are. I know what all of the lengths of A, B, and C are. You can see how this is useful. Now I have triangles that don't necessarily have right angles, but I have tools to be able to solve for side lengths or angle measures accordingly.
- 7.50** The question, a natural question would be, well, look, if there's a law of sines, is there a law of cosines? Is there a law of tangents? Is there a law of secants and all that? There is a law of cosines. What does that say? Let's figure out how we can, I don't know, stumble upon that one, if you will. Let's see what the law of cosines tells us. Again, we're just going to discover this. I could give you the formula, but it's not really necessarily motivated. Let's see how we go about this. All right. Here's what we're going to get.
- 7.51** I'm going to take a triangle that looks like this. What I'm going to do is, I'm going to do the same thing that I had before. Let's suppose this is alpha, on the opposite side of this is A. I'm going to assume this is beta. Let's call this beta. On the opposite side is B, and this side is gamma, and on the opposite side of that is C. We're going to do the following thing. We're going to say, take a line from here, stand at this point, drop the line down, and make that my 90 degree, just like we did in the law of sines, because I need to somehow extract sines and cosines which come from right triangles.
- 7.52** Now, I don't know what that is, but it's a height of something, so I'm just going to call it H. It's the height of my triangle. It's the height of my triangle. Now, what I can do is, I can say, look, this distance, I don't know what it is, but I'm going to call it X. This distance is length B. In particular, if this is length X, then this has to be length what? That has to be length B minus X, whatever it is. If the side wall is 10 meters long and

this is 2 meters long, then this has to be 10 minus 2, which is 8 meters long. This has to be B meters long. This is X. This is B minus X.

7.53 What can I say? Well, let's take the cosine of this angle alpha and see what we get. Cosine of alpha is equal to what? That's equal to the adjacent, which is X, divided by the hypotenuse. This is X over C. Do we agree? OK. I can solve for X. That tells me that X is equal to C times cosine of alpha. It's annoying because the coefficient is a C, and I also have the C coming from the cosine. But then I can also say what the sine of alpha is. I can say that the sine of alpha is equal to the opposite over the hypotenuse. That's going to be an H divided by C. That allows me to say that H is equal to C times sine of alpha. I can say that.

7.54 What can I say? Well, I know that this is a right triangle. One of the first things that we talked about in trigonometry is one of the defining theorems of all of this, is the Pythagorean theorem. The Pythagorean theorem relates the sum of the squares of the sides of triangles to the square of the hypotenuse. In particular, the Pythagorean theorem tells me what? It tells me that this side here, B minus X, when I square that, should equal or plus, excuse me, plus this side here, which is my height, that's H squared, should equal what? Well, that should be A squared on this side. That's what that tells me.

7.55 But what is B minus X? Well, B minus X, look, H is equal to C sine of alpha. We're going to plug that in. I'm going to get C sine of alpha, the whole thing squared, plus what is B minus X? Well, X is C cosine. This is B minus C cosine of alpha, the whole thing squared, and that should equal A squared. Now we have to do the algebra. How do we do this? This is B minus C cosine of alpha times B minus C cosine of alpha. Plus, OK, how do I square this? Well, this is just a product, so what do I get? I get C squared, sine squared. And that's equal to A squared.

7.56 What happens here? B times B, I get B squared. B times minus C cosine is minus B C cosine of alpha. I'm going to get a minus another one, so minus B C cosine of alpha, and then I'm going to get a plus C squared cosine squared of alpha plus C squared sine squared of alpha. OK? Bear with me. We're almost there. We're almost there. What do I notice here? What's common to both of those terms? I have a C squared and a C squared. That's common to both. So if I factor out C squared, what am I left with? Cosine squared of an angle plus the sine squared of that same angle.

7.57 What do we know about that? That's a Pythagorean identity. That's equal to 1. That's equal to 1, by Pythagorean identity. So that's just a C squared term. These two things combine to give me a minus 2 BC cosine of alpha, and then I have B squared sitting off on the side. What did we just find? We found that A squared is equal to what? A squared is on the right. It's equal to B squared plus C squared, this guy and this guy, minus 2 BC cosine of alpha. This is the law of cosines. That's what this is. That's the corresponding law of cosines.

- 7.58** What does it do? It says, look, it relates the length of the opposite side of alpha squared to the square of this side of the triangle, this side of the triangle, minus some term in the middle times cosine of alpha. That's what it does. How do you use this? How do you use this? Again, this is the mathematical proof of what the law of cosines is, but big deal. Basically, what we're just doing is just goofing around with these triangles, creating right triangles and saying, what can we say from that? What can we say from it? How would we do an example of this?
- 7.59** First off, let me give you motivation as to why we should care. Why should we care about this in the first place? Think about it in the following way. If you have an airplane, for example, OK? If you have an airplane, for example, last I checked, you need wind in order to actually fly, right? You need that pressure difference of the airfoil in order to actually get lift on the wings to get your aircraft off the ground, OK? So you need that flow of air over the wings in order to get things moving.
- 7.60** Well, when you get up into the sky, you want to know, OK, I want to fly from, I don't know, let's say Los Angeles, California, to, I don't know, the Czech Republic, let's say. I want to fly from LA to the Czech Republic. OK, cool. What direction do you point your plane in? It's an important question. You can't just be sitting there and take off and be like, "I think the Czech Republic is that way, so I'm just going to point the plane in that direction." You have to be very careful about these things. You have to be very careful about these things.
- 7.61** Even if you know you're heading, as to where you're going, you might encounter crosswinds associated to that. You might be traveling in this particular direction and say, oh, OK, I need to go in this direction, but I have a very strong crosswind that's hitting me from this direction. So what does that mean? Well, that doesn't mean that you can just fly in this particular direction if the wind is hitting you in this direction. What do you need to do? You need to angle yourself off of that a little bit, so that your air speed versus the wind speed against you balances out, so that you can actually swim in that direction.
- 7.62** Endurance swimmers do this all the time. Endurance swimmers. If they want to swim from—I think one of the most famous examples is swimming from an island in Hawaii to another island in Hawaii. Let's say these are Hawaiian islands. You want to swim from here to here. It seems natural. It's like, OK, just point yourself in that direction and you're great. If I want to swim in that direction at however fast you can swim, the problem with this is that you have a massive current that's flying or that's coming in this direction from the Pacific Ocean.
- 7.63** What does that mean? That means if you actually want to start from here and end up here, if you just take the line-of-sight distance, then this is going to do what? You're going to get midway through. And what's going to happen? I'm going to start here and start here and start here and start here. I'm going to miss the island. Because the ocean is pushing me down in the way that's not allowing me to go there.

So what do I need to do? Well, I can't just take line-of-sight distance then, just like the airplane. You can't just take the line-of-sight distance if you have a massive crosswind.

- 7.64 So what do you need to do? You need to say, all right, well, rather than starting off at the line of sight, I'm going to start swimming in this direction. Because if I swim in that direction, the current is going to push me down to here. Then once I get to here, I'm going to swim off in that direction because then the current is going to push me down to here and swim off in that direction. It'll push me here and so on. It'll keep pushing me here until eventually it pushes me back down into the island.
- 7.65 What does this have to do with anything? This is saying, look, if I want to be able to fly line of sight from here to here, and I've got some crosswind associated to it pushing down on me, then I really need to angle myself at this particular point, so that over the short run, angling myself here gets me into the direction that I want to go. Well, there's my angle. If I know the distance here, we typically know the distances between these places. Somebody knows the distance between Los Angeles and the Czech Republic. I certainly don't, but somebody does. We know that distance for sure.
- 7.66 We could say that, in the short run, we know what this distance is. We're going to travel in this distance maybe for one mile, and that's going to reset me back down, and I can say, all right, that's fine. What angle do I need to offset myself off of from my target in order to be able to actually fly in that direction so that I reach my destination? Well, that looks like the law of cosines. I've got this side, that's A. I've got this side, that's B. I have, let's see, let's call this C, to stick with the notation that we had over here. Stick with this side.
- 7.67 Opposite my angle, I know that A^2 is going to be equal to B^2 plus C^2 minus $2BC$ times cosine of α . Well, I know B. That's the distance from where I'm starting to where I'm going. So that's known. C is fixed. So I know C. So I know B and C, therefore I know those two things. Based on the strength of the wind, I can say, I know what α is going to be. What angle do I need to offset myself off of, so that I can fly in accordance to the wind pushing me off my original destination? That's going to tell me, what's the distance that I'm going to travel along that final leg?
- 7.68 Why is that relevant? Why is that relevant? Now, are pilots doing this? I don't know. Probably not. They probably have these amazing instruments. I mean, intuitively, they probably are. You have the crosswinds. You know how strong it is. You know you need to angle yourself into it in order to do it. You've got to adjust your air speed. You've got to do all these different things. But you probably have a million instruments on board that's also helping you do these things. But why is that relevant?
- 7.69 Well, look, you're traveling distances, for goodness' sakes. You're traveling distances. So if you're having to move off of your course because of some crazy crosswind, well, that's going to consume fuel, which is kind of necessary to stay in the air. You need to know what the distance that you're going to travel because you need to understand,

how is this going to affect my fuel reserves? It all has its basis here. It all has its basis here.

- 7.70** So yes, we're looking at triangles, we're drawing things on the board, we're saying A and B are 17 and 12 or whatever it is. But these instruments that are built into these technological devices, airplanes, things along those lines, telling you, "Hey, go along this trajectory because you need to be able to get to your destination. By the way, you're going to have to travel this extra distance of A, so be careful about that. That's going to consume extra fuel," all of these things are built into these systems, and it all starts from trigonometry, which is really cool.
- 7.71** Let's do an example. Let's finish up with an example of the law of cosines. I think my favorite example of the endurance swimmer who was swimming through the Hawaiian Islands, he said something to the effect of, "I knew that I could do this in roughly—if I didn't have any current, if I didn't have anything pushing me down the side or anything like that, if I could just swim in a recreational pool, if it were just a recreational pool, no currents, no nothing," he knew that he could swim the distance from island to island in something like 11 hours.
- 7.72** How somebody can swim for 11 hours is beyond me. That's very impressive. But he knew that just straight-line distance in a non-turbulent pool, he could get there in 11 hours. Took him 17. Took him 17 hours to get to the other side. Those six additional hours came from what? He was angling himself against the current. The current was pulling against his speed in the water. So rather than swimming at X amount of speed, maybe it was half of that.
- 7.73** It's like trying to bike into a headwind. If you're biking at 20 miles an hour and you're going into a headwind, the more surface area you put towards the wind, the more resistance you have, the more it's going to push you back. So maybe normally I'm traveling at 20 miles an hour, but now because I'm going into a stiff headwind, I'm only going 16 miles an hour. Well, it's going to take me longer to get to my destination. I'm going to have to burn more energy to get there, more fuel to get to my destination. These things are relevant.
- 7.74** That particular endurance swimmer, if you think about it, you have to consume calories to get there. You can't do this activity for 17 hours straight and just assume that you're good on nothing, an empty stomach. That's not a thing. Knowing that it's going to take an additional six hours allows this swimmer's team to be like, "OK, shoot, we're going to have to actually provide calories for an additional six hours for this person to actually survive." That's a relevant question. It's a relevant question.
- 7.75** OK, so let's do an example. Let's do an example. Let's take one of these really thinned-out triangles because I motivated that from the pilot example. Let's say this is 100. We're going to say it's 100 miles. Why not? We're going to say that, let's say this distance over here is, I don't know. Let's say this is going to be 95 miles. I want to know,

what is this distance over here? What is D ? What is the distance of the other side of the triangle? I'll tell you what this angle is. Let's say this angle is relatively, relatively small. The angle doesn't have to be massive to have some incredible effect.

- 7.76** I think one of my favorite quotes was from Dieter Uchtdorf, who was a former pilot. And he had this quote that he said, look, in aviation, he said, if you're off by a degree, a single degree, one degree, which I'm willing to bet that our eyes wouldn't even be able to detect, a one-degree difference, over the short term, it doesn't really matter. One-degree difference doesn't really matter. If I'm walking from this side of the room to the other side of the room and I deflect by one degree, I'm basically going to end up at the exact same place by the time I get to the end of the room. But over a 400-mile journey, over a 2,000-mile journey, that one degree of deflection suddenly accounts for a massive distance.
- 7.77** You've got to be careful about these things. Let's say this is a small angle. Let's just say, for the sake of argument, it's five degrees. Tiny, incredibly small angle. The example is, find D . Find D . You're not allowed to just simply circle it and be like, "There it is." No, that's not allowed. You can't just be like, "Oh, here's D . It's right here. I found it." No, I'm saying you actually have to compute it. You actually have to compute it. What I notice is that it's set up perfectly for the law of cosines, because what do I have? I have two neighboring sides of the triangle, and I have an angle between the two.
- 7.78** In particular, by the law of cosines, I can say what? I can say the opposite side of this angle—I'm not saying opposite in the sense of sine, cosine, tangent, opposite over hypotenuse, things like that. I'm literally just saying the side which is directly opposite this angle. I can say what? I can say D^2 is equal to the side lengths. Here's my 100 squared plus 95 squared minus 2 times 95 times 100 times the cosine of the angle between the two. Times the cosine of the angle between the two.
- 7.79** Now, step back and think about this for a second. Look, 100 squared, you know what that is. What is that, 10,000? Something like that? 95 squared, you know what that is. 2 times 95 times 100, you know what that is. Desmos will tell you what that is, cosine of 5 degrees. In particular, you know the right-hand side as a number. That's great. Then I'm saying, D^2 is equal to that number. In particular, D is equal to 100 squared plus 95 squared minus 2 times 95 times 100 times the cosine of the angle 5 degrees square rooted.
- 7.80** Now, if these are miles, 100 miles, 95 miles, this D , whatever it is, is going to give me the miles that I go from this particular location to that particular location. Again, mathematics is all about the critical thinking side of things and taking these elementary principles, things like this, law of cosines, law of sines, and being able to say, look, how is this relevant? Why would this be useful? I hope the aviation examples and the endurance swimming examples gave you reasons as to why this is relevant.

Lecture 8: Useful Applications

At a glance — Useful applications.

- 8.1** All right. Now, we come to the application side of things. We've gone through trigonometric functions from triangles to circles to functions, to inverting them, to law of cosines, law of sines. Now we want to understand, OK, how can I put this into practice? What can I do with these things? Maybe it's just these parlor tricks for mathematics. Maybe it's actually something relevant. We'll see.
- 8.2** In order to get to that, I need to give you a notation. It's a mathematical notation. It's founded in a very basic idea, in a very basic idea. But notationally, it looks very intimidating. It looks very intimidating. The idea is the following. If I ask you to add numbers, if I ask you to add numbers, I could ask you the following question. I could say, all right, I want you to compute the following. Compute 1 plus 2. You're like, OK, 3, fine. All right. OK. But then I want to ask, OK, I'm going to add another one. Then I'm going to add another one and another and another and another until I keep going until I get out to 1,000.
- 8.3** Suddenly, the problem is not as tractable as it was when we started. 1 plus 2, you're like, ah, it's 3. That's fine. OK, add 3. OK, that's 6. Great. Add 4 to that. OK, that's 10. Great. Add 5 to that. That's 15. Add 6, 21. Add 7, 28. And suddenly you're like, add 967, add 998, add 1,000. You're like, all right, I've lost count. I have no idea. The problem is very simple. Just add up the numbers. That's it. But how do I actually do it? I want a number. I want a number. And more than that, I don't want to write dot, dot, dot all the way up to 1,000.
- 8.4** In fact, I don't want to write this expression at all. I don't want to write this. I mean, you imagine a book, some math book or something like that, where every single one of these things is written out in gory detail, from 1 up to 1,000. No, that's terrible. It's going to take up too much space. I want a compact way of representing this notation. I want a compact way. I want something that I can write in something just right here to represent this entire thing, from 1 to a 1,000.
- 8.5** Can I do that? Well, why not? This is what's called summation notation. Summation notation. I want you to think the entire time, what is summation notation? What is it doing? All these symbols, all these indicators, all these things are confusing. What is it really? It is literally just a notation to talk about how do you add numbers. That's it. Here's how it goes. Summation notation. You have the capital Greek letter sigma. You have big sigma. That's what it looks like. Then you do the following.
- 8.6** You have a counter. You declare a letter, a variable, to count for you. It's going to start at a certain place, and it's just going to count one up each time you move to the next one. I'm going to call that k . They're typically denoted by k or m or n or things like that, i , j . I'm going to call it k , and I'm going to declare it to start somewhere. I'm going to say

it just starts from 1. It's just going to start from 1. I'm going to say, OK, K, you're going to start from 1, and then you're going to move to 2, and then you're going to move to 3, and then you're going to move to 4, until I tell you to stop moving.

8.7 I'm going to declare that it stops up here at, let's say, N, where I get full control over N. I get full control over N. Then what's going to go right here is going to go some function that involves my counter K, some function that involves K. What is this notation telling me? This notation is telling me the following. It's saying, start from K is equal to 1, plug it into my function. What do I get? I get F of 1. I don't know what that is because I don't know what F is, but at the very least, I can write this as F of 1. Then the sigma is simply indicating then add what? Move K to 2. Move K to 2. I get F of 2. Keep going.

8.8 Now add. K is 3. And so on. When do I stop? You stop when K gets up to N. You stop when K gets up to N. So this is F of N. So that's the notation. Now I have a compact notation for writing things that have goodness knows how many terms. That's the game. OK, well, let's think about that for a second. Let's go back to the original question. I want you to add up 1 plus 2 plus 3 plus 4 plus 5, all the way up to 1,000. The first question I want to know is, how could I put that in terms of this notation here? How do I represent that?

8.9 OK. So represent 1 plus 2 plus all the way up to 1,000 in sigma notation. So how do I do that? Now, remember, sigma notation says the following. The Greek letter is just fixed. It is what it is. You get to choose what K is, where it starts, where it ends, and then you have to somehow match F of K to match the terms that you get up here. Let's think about that. How would I do this? I have sigma. Let's say K starts from 1. That's usually a safe bet. K starts from 1. Then I want to go to a particular ending point.

8.10 Well, I don't know what that is, and it's going to depend on what my function is here. Let's suppose that my function is equal to K. What I'm saying here is that F of K is just equal to K. That's all I'm saying here. F of K is equal to K. Why is that useful? Because what is F of 1? That's 1. What's F of 2? That's 2. What's F of 3, and so forth? You can see, what am I getting? I'm getting precisely 1, 2, 3, 4, 5. I'm getting those. What do I notice? If K starts from 1, and I plug 1 in, I get a 1. Then K is going to go to 2. I plug K is equal to 2 in, I get a 2. Now K is 3, I plug it in, I get a 3. OK.

8.11 How far does K need to go? What's my last term? My last term is 1,000. So K needs to go from 1 up to 1,000. This is the representation. So my second to last term is 999, and then I get 1,000 from there. I get the sum K equals 1 to a 1,000 of K is equal to 1 plus 2 plus 3 all the way up to 1,000. Now I have a concrete way of representing that number. I don't have to write it in this long string of numbers, pluses, dot, dot, dot, all of that. I have sigma notation to write it in a nice compact form, albeit terrifying. It looks terrifying, right?

8.12 You look at something like this, you're like, I don't even know what I'm reading. Well, that's fine. That's why mathematics is a language. You actually have to understand the

symbols. You have to understand the letters. You have to understand what the meaning is behind things before you actually understand what the mathematics is trying to say. Well, that's great. OK, so that's my nice compact form for doing that. The question still remains, what is the answer? What is it equal to? What is it equal to? How could I even figure this out? Well, let's start small. Let's start small.

- 8.13** I want to convince you that you can actually find the formula for this simply by using algebra, basic algebra. Let's figure this out. What if I did the following thing? Rather than adding up all the way to 1,000, what if I say I only want to add up to 5? 1 plus 2 plus 3 plus 4 plus 5. Let's start small. Let's do some base cases, if you will. I want to take 1 plus 2 plus 3 plus 4 plus 5. Maybe that's too small because maybe your approach is like, oh, OK, 1 and 2 is 3, 6, 10, 15. OK, my answer is 15. Fair. But remember what the goal is. You're trying to get it up to 1,000. What you literally did was just add the numbers up. You haven't come up with a systematic way to try and generalize it to something that goes all the way to 1,000. Do we agree?
- 8.14** What I could notice is, I could say, look, think back to when we were doing inverse trig, when I said, what's the arcsine of one-half? We said, I don't know, but I'm going to give it a variable name. I'm going to call it theta, and I'm going to use the fact that that means that sine of theta is one-half to actually prove the result. The same thing holds here. I don't know what this is, but I'm going to call it X. I'm going to call it X. Now, would you agree with me that I could say that X is equal to, rather than starting from 1 and adding 2 and adding 3 and adding 4 and adding 5, I can start with 5, add 4, add 3, add 2, add 1.
- 8.15** I'm going to get the same answer. Do we agree? I could say, OK, I take 5 plus 4 plus 3 plus 2 plus 1. Do we agree? That seems a little ridiculous. If I take the string of numbers and add them this way versus a string of numbers and add them backwards, that it's the same. But now what do I notice? If I add these up, what do I get? Well, it's X plus X. That's two copies of X. What do I get when I add these individually? 1 plus 5, 6. 2 plus 4, 6. 3 plus 3, 6. 4 plus 2, 6. 5 plus 1, 6. So I'm getting a 6 plus a 6 plus a 6 plus a 6 plus a 6. How many sixes do I have? Well, I have as many as I have numbers up here. So I have 1, 2, 3, 4, 5 numbers. So I'm going to have 1, 2, 3, 4, 5 sixes.
- 8.16** But if I take 6 and I add it to itself 5 times, this is really equal to what? This is just 5 copies of 6. Do we agree? So I literally just used basic algebra and said, oh, well, now I know how to solve for X, I just have to divide by 2. If I divide by 2, then I get X is equal to 5 times 6 all over 2, which is what? 5 times 6 is 30, 30 divided by 2 is 15. That's exactly what we got with 1 plus 2 is 3, 3 plus 3 is 6, 6 plus 4 is 10, 10 plus 5, 15.
- 8.17** But now I can say, OK, well, maybe this was a fluke. Maybe this was a fluke. Well, how would I have written this in sigma notation? How would I have written that in sigma notation? Well, just like I did over here, I want to write 1 plus 2 plus 3 plus 4, all the way up to 1,000, all the way up to 1,000. How would I get 1 plus 2 plus 3 plus 4 plus 5? The only thing I have to change is how far out I go. I don't want to go all the way to

1,000. I just want to go to 5. This notation right here, for 1 plus 2 plus 3 plus 4 plus 5, I can write in sigma notation, how?

- 8.18** I can say that 1 plus 2 plus 3 plus 4 plus 5 is equal to the sum of K , where K starts at 1 and it goes to 5. Do we agree? K is going to start at 1. I plug it in, I get a 1, and I'm going to add because it's a sigma plus K is now 2. I plug it in, that's a 2, 3, 4, 5, and now I stop because I end up there. I can say, well, I know what the answer is now. It's 5 times 6 all over 2. It's 5 times 6 over 2. OK. Well, maybe that's a general statement. Maybe that's actually a general fact, that if I want to add up from 1 up to 5 of K , the answer is whatever the upper limit is, 5, times 1 more than the upper limit, 5 plus 1, that's 6, divided by 2. Maybe that's the answer.
- 8.19** Maybe that would allow me to figure out what this summation is by simply saying, oh, it should be 1,000 times 1,001, divided by 2. Maybe that's the right answer. I don't know. I only have evidence to suggest that it works in this one case. I don't have evidence to suggest that it works in other cases. OK, so what if I did something else? What if I said, all right, I want you to do, 1 plus 2 plus 3 plus 4 plus 5 plus 6 plus 7 plus 8. I want you to go all the way up to 8. Well, again, you can do this just by adding them up. That's 3, 6, 10, 15, 21, 28, 36.
- 8.20** But maybe this approach still works. Maybe this approach still works. Maybe I could say, let's call it a different variable. Let's call it Y . That's 1 plus 2 plus 3 plus 4 and so on. Rather than adding it 1 plus 2 plus 3 plus 4, why don't I add it 8 plus 7 plus 6 plus 5 plus 4, all the way down? Then, would you agree that Y is also equal to 8 plus 7 plus 6 plus 5 plus 4, 3, 2, and 1? If I added up one way, it doesn't matter if I added up the other way. That seems reasonable.
- 8.21** Well, what happens when I do that? If I now add, I get Y plus Y on the left, which is a $2Y$. And what am I going to get here? 1 plus 8 is 9, 2 plus 7, 9. 3, 6, 9. 4, 5, 9. 5, 4, 9. 9, 9, 9, all the way down. How many copies of 9 do I get? Well, I have as many copies of 9 as I have numbers, and I have 8 numbers. I'm going to get 8 times 9, which tells me that Y is equal to 8 times 9 all over 2. Well, what was the sigma notation for this? The sigma notation for this was the sum, K goes from 1 up to 8 of K . My formula seems to track.
- 8.22** I take the upper limit, I plug it in there, I take one more of it, that gives me a 9, and then I'm going to get a 2. It turns out that that's a general fact. It turns out that that's a general fact. If I wanted to get this value, this number is 1,000 times 1,001 all over 2, simply because it's equal to the summation K goes from 1 up to 1,000 of K . Our anecdotal evidence—we don't have a mathematical proof of this, but we at least have evidence to suggest that adding up things this way, I'd simply take the upper limit, add 1 to the upper limit, and divide by two. That is a true fact, by the way. That is a true fact. That's our summation notation.
- 8.23** Now, why would you ever do something like this? Why is this notation interesting? Well, there's a lot of ancient questions associated to this. They come in the following

way. They come in the following way. All right? Let's suppose I give you the following question. I want to walk a certain distance, and I'm going to choose to walk it in a very weird way. I have a set distance that I want to walk. Let's say that that distance is length 1. I'm going to start from zero and I'm going to go from length 1. I start at zero and I want to walk to 1, whatever 1 represents.

- 8.24** I could say I walk from zero to D, where D is the distance that I'm trying to travel, that's fine. But I'm simply just going to couch it as saying zero to 1. But I'm going to do it in a really goofy way. I'm going to say, OK, if I want to go from zero to 1, I'm first going to go half the distance to 1. I'm going to go half the distance to 1. At first, I'm going to go from zero to a half. I first travel this distance, half of the way there. Then once I make it to one half, I'm going to go half of the way between one half and 1.
- 8.25** What's the distance between one half and 1? That's distance one half. So, I'm going to go half of that, which is one quarter. I'm going to travel this distance, which is one quarter. Once I make it to this location, I'm going to say I'm going to go half the distance to 1, which is one eighth. Exactly. I'm going to go one eighth, and then I'm going to go one-sixteenth, and then I'm going to go so forth and so on. That's the game. It's very easy to explain. Why anyone would ever move this way, I don't know. But very easy problem to explain. Just go half the distance every single time between me and whatever it is.
- 8.26** The question that's asked, the question is, if you do this infinitely many times, do you make it to 1? Do you make it to 1? Our brains instantly say what? No. Of course not. You're never going to make it to 1. Why? Because our brains can't comprehend doing something infinitely many times. They can't. Our brains only understand how to do a finite number of things. Our brains know that eventually, if I were to get to that particular wall—let's say I'm going from zero is my position right here, and 1 is the wall across the room, then my brain would say, oh, if you ever got to 1, then you must have gotten there in a finite amount of time, and therefore, there must have been a finite amount of steps that you took to get there.
- 8.27** Therefore, if you do this infinitely many times, there's no way that you can get there. Our brains do not understand infinity as a concept, doing something infinitely many times. The answer to the question is yes, you do get there, which is very weird when you think about it. But that's what infinity does. Infinity breaks our intuition in a lot of ways. It's an incredibly useful concept, but our brains do not understand that. If you think about this, for example, if you go to a sporting event, you go to a sporting event and you go to the stadium, whatever it is, and they say the stadium's capacity is 120,000 people.
- 8.28** 120,000 people. And you go to the event and you're like, oh, my gosh, this is amazing. Look how many people there are. There's 120,000. Your brain can't even fathom that. There's so many people. And then you realize that 120,000 is just a drop in the bucket when you're doing something infinitely many times. 120,000 doesn't mean anything in

terms of infinity. Our brains have a very difficult time of doing something infinitely often. Now, if you say, can you do this practically? Of course not. You can't do this practically. You can't do anything infinitely many times, practically. You don't have an infinite amount of time to do it. So you can't do that.

- 8.29** It takes a finite amount of time to move. You don't have an infinite amount of time to actually do that. But if we throw all of that kind of aside, we're really saying what? Add up the distances that you travel. What is the distance that you travel? Let's just add it up. So this is the distance traveled. Well, the first distance I cover is one half. The first distance I cover is one half. All right. The next distance I cover is what? That's one fourth. The next distance is one eighth. The next distance is one over 16. Next distance is going to be one over what? Two times that is 32. One over 64, one over 128, and so on.
- 8.30** I'm going to keep doing this infinitely many times. The distance I'm traveling each time gets smaller and smaller and smaller and smaller. That's going to be my total distance. The question is, if that distance is equal to 1, then I've traveled the distance from zero to 1. Do we agree? If that distance is 1, then I've reached 1. I've gotten to 1. How could I write this in sigma notation? Summation notation? Well, I have to find a pattern between this number and this number, this number and this one, this one, this one, this one, this one. What am I doing?
- 8.31** At each stage, I'm taking the denominator and I'm simply multiplying by 2. Multiply by 2, multiply by 2, multiply by 2. That's my pattern. In particular, what do I have? I can say this is equal to one half, plus 1 over 2 squared, plus 1 over 2 cubed, plus 1 over 2 to the fourth, plus 1 over 2 to the fifth, and so on, so forth and so on. Do we agree? It's just multiply the denominator by 2 each time over and over again. All right. Well, this is a one half. This is a one half squared. This is a one half cubed. This is a one half to the fourth. This is a one half to the fifth, and so on and so on.
- 8.32** Properties of exponents. That's what we covered in algebra. If I have a fraction, A over B , and I raise that to a power, that's A to the power divided by B to that power. Since 1 squared is 1, 1 cubed is 1, 1 to the fourth is 1, 1 to the fifth is 1, I can write this like this. Now I have a pattern that I can use to write this in sigma notation. If I write this in sigma notation, what do I get? I'm going to say K starts from 1, and it's going to go to, I don't know yet. But what do I notice? I notice that everything looks like one half raised to the K power. Do we agree? Everything looks like that.
- 8.33** How would I start this? K equals 1, I plug it in, one half to the first. That's this guy. K is now 2, one half squared, this one. K is 3, one half cubed. K is 4, one half to the fourth, and so on. When does it end? Well, it doesn't. I don't have an upper bound here of, say, 1,000, or, say, 8 or 5. It goes on infinitely many times. It goes off to infinity. I just keep doing this over and over and over again. This notion is what's called a series. Mathematically, that's what these things are called. It's called a series. It's a notion that's covered in, say, pre-calculus or calculus course or something along those lines.

- 8.34** But the question I want to ask is, what is this? What is this? Now, the general fact is that, if I give you a series that looks like this, say, K goes from 1 off to infinity, of something that looks like A times some ratio, R to the K minus 1, let's say, if I start working this out, what is this? This is A plus AR plus AR^2 plus so forth and so on. This series has a special name. It's called a geometric series. It literally says your next term is gotten by your previous term simply by multiplying by the common ratio R over and over and over again, just like I have here.
- 8.35** My next term is gotten from my previous term by multiplying by the common ratio of one half. Then I multiply by half again and I get this. I multiply by half again and I get this. I just keep doing that over and over and over. It says the following thing, that this will equal the first term of your series. This will be A divided by 1 minus R . It will equal that number, provided R is small enough, meaning that, in absolute value, R is less than 1 . That's a requirement. That's a requirement, is that R has to be less than 1 in absolute value. If that is the case, then this infinite sum of numbers is equal to A divided by 1 minus R .
- 8.36** I can say, sweet. Now I can apply that result because what is R here? R is equal to one half, which has absolute value less than 1 . Then what am I going to get? I'm going to get my first term. That's one half. This will be one half divided by—what does my formula tell me? 1 minus one half. One half stays the same upstairs. Downstairs is 1 minus a half. That's a half. It looks like you traveled the whole distance. If you go half, then half, then half, then half, and you do that infinitely many times, you will cover the entire distance between zero and 1 .
- 8.37** Now, can you do this practically? Of course not. You can't do something infinitely many times. Mathematically speaking, does it make sense? Yes. Let me give you another ridiculous example. Let me give you another ridiculous example. It's the following. Let's suppose you and three of your friends, you and three of your friends have a particular item that you're going to share for lunch, let's say. Let's suppose you and three friends, I don't care what the item is, want to split, I don't care, an apple equally.
- 8.38** Let's do this, actually. You and two friends. We'll make it a little simpler. You and two friends want to split an apple equally. Well, what would you do? You say, OK, it's me and two friends, so three people total. What would I do? I'd say, all right, I'm going to, I don't know, cut it into something maybe like this. That I give to person one, this I give to person two, this I give to person three. I cut the apple into thirds, and I say, "Here's your third of the apple, here's your third of the apple, here's your third of the apple." That seems like what we would do normally. This is intuitive. That's the intuitive notion of what we would do.
- 8.39** Well, mathematicians don't do that. Mathematicians ask questions like, how can I make things as hard as possible? Here's what we're going to do. Here's what we're going to do. We're going to say, no, I don't want to do that. I'm going to take the apple, which I'm going to think of as just a circle, and I'm going to cut it into fourths. I'm going to cut

it into fourths. I'm going to cut it into fourths. I'm going to cut it down this way and cut it down this way. I'm going to give this part to person one, this part to person two, and this part to person three.

8.40 How much does each person have at this point? Well, they have a quarter of the apple. Do we agree? There's a quarter of the apple remaining. There's this part. I'm going to cut that into fourths. I take this part of the apple out, and I'm going to cut it into four equal pieces. I don't really know how to subdivide that. I don't know. We'll just say it looks like this. It's obviously not equal pieces, but use your imagination.

8.41 I'm going to then take the remaining part of the apple. I'm going to give that to person one, that to person two, that to person three, and the remaining part of the apple, I'm going to cut into fourths, and I'm going to give one fourth of it to person one, one fourth to person two, one fourth to person three. I'm going to do this in perpetuity. I'm going to keep doing this over and over and over again. Now, can you do this? If you've ever cut an apple, you know, no, there is a stopping point. I mean, this is going to become so paper-thin, you can't subdivide it any further.

8.42 But mathematically, just run with it and see what happens. See what happens. How much does each person get? How much does each person get? Well, it doesn't matter which person you ask because they're all getting an equal amount. Do we agree? They're all getting an equal amount. So what we want to say is, OK, so let's just say person one, so P1, person one. On the first pass, how much does person one get? Well, they get one quarter. Do they agree? They get one quarter.

8.43 Then after those one quarters have been distributed, so now this part's gone to person one, this part's gone to person two, this part's person three. The remaining quarter left, I cut into quarters, and I give a quarter of that to person one. What am I really doing? I'm giving them one quarter squared. Then that part goes to person one. Then I subdivide this one further, and I give this part to person one. What else am I going to do? You can kind of see the pattern here. I'm going to take one fourth of the remaining, one fourth of the remaining, so forth and so on. I'm going to do this infinitely many times. I'm going to do this infinitely many times.

8.44 That's how much person one gets. In particular, that's how much person two and person three gets, because they're getting the same amounts. Well, what is that? That's a summation. K goes from 1 up to infinity. It goes on forever. Of what? $\frac{1}{4}$, raised to the K power. When K is 1, I get one fourth. When K is 2, I get one fourth squared, one fourth cubed, one fourth to the fourth, and so on. Well, this is a geometric series. What's my common ratio? How did I go from here to here? I multiplied by one fourth. How did I go from here to here? I multiplied by one fourth, one fourth, one fourth, and so on.

8.45 I'm multiplying by one fourth every single time. So my R is equal to one fourth, and that's less than 1 in absolute value, which means that the formula I had for this in the

previous one reigns supreme. What is this going to give me? This is equal to the first term of my series, one fourth, that goes in the numerator, divided by 1 minus my common ratio, 1 minus a fourth. What do I get? This is equal to 1 over 4, divided by, OK, I have one full thing and I subtract one quarter of it, then I've got three quarters left. So I've got three quarters.

- 8.46** How do I take fraction divided by fraction? This is the numerator fraction divided by, take the bottom fraction and flip it. That's four thirds. Four cancels, then I get one third, which means that person one got one third of the apple, which is exactly what we wanted them to have, if we take the apple and we want to split it equally. Our intuition said, just split it into thirds. Each person gets a third. And we said, no, no, no, no. The mathematician said, no, I've got a better way. Cut it into fourths, and then take the remaining fourth, cut that into fourths, take the remaining fourth, cut that into fourths, and do that. Then each person is going to have, well, a third of the apple.
- 8.47** That's great. That's why you don't invite mathematicians to parties. You don't do stuff like that because they do annoying stuff like this. But that's why these things are relevant. These things are relevant because they give us answers that we're like, why would you ever do that? But then it turns out to be something that's actually quite profound and actually recovers the answer that we would get intuitively. What does this have to do with the trigonometry side of things? What does this have to do with trigonometry? Here's what it has to do with it.
- 8.48** These things, these objects that we're talking about, are called infinite series. They're called series. Again, you'd run into this in a pre-calculus or a calculus course or things like that. These are the things that I was mentioning at the very beginning when I said, why is trigonometry relevant? Well, let's think about it for a second. If I were to give you a wave, whatever that is—the words coming out of my mouth, they vibrate the air particles, they create a wave. You can see on a particular machine, you can see the wave coming out as I'm speaking. So I have some particular wave, whatever that is.
- 8.49** Maybe it looks something like this. I don't know. This looks a little EKG, like a heartbeat thing. Well, hopefully, that's not a heartbeat. That's erratic and scary. But I have a waveform that looks like this. Now, if that waveform is a sound form, it's coming from sound, then I could say something like, OK, well, have you ever listened to, for example, a vinyl record? You listen to a piece of music on a vinyl record, an actual record player. When you listen to it, you don't necessarily hear perfectly clear background. You hear this crackling a little bit. You hear, maybe you would classify it as static or some ambient noise associated to it.
- 8.50** Music today, what can we do? We can run it through a software program and we can say, OK, great. I hear some static in the background or some ambient noise in the background, and what can I do? I can say, well, I want to filter that out. I want to filter that out because I want the audio to just be as clean as possible. I only want to hear the

audio of what I'm hearing, the singer, the drummer, the guitar player. That's the only thing I want to hear. I don't want to hear the background sound at all whatsoever.

- 8.51** What can I do? When I have these waves, I can try and say, look, these little waves in here, those little waves, those are coming up from static. That's the background music. That's the background sound. I don't want that. I want this wave for sure. I want this wave. I want this wave. I want this wave and all of that. That's what I care about. In some sense, I want to cancel these out. I want to cancel these out. This is the basis of noise-cancelling headphones. It's the basis of noise-cancelling headphones. You put your headphones on, you're in a noisy, maybe you're in an airport, you're in a store, something like that. It's really noisy. You put your headphones on, and suddenly you feel like it's just you.
- 8.52** Maybe you can hear a little bit, but it's significantly decreased. It's the basis of this. It's the basis of this. For example, what's a real-world example of this? Let's do this. The World Cup. World Cup. Years and years ago, when World Cup was being played, they still allowed the vuvuzelas. It's the horn that people blow at these games. Sounds like just this beehive sound to it a little bit. It just has this buzzing sound. Well, as they're recording the matches, well, obviously, they're picking up the audio. They're picking up the audio.
- 8.53** So as they broadcast this, it was hilarious. You'd go to a pub or a diner or something like that, and you see all the matches on the TV screen. And the second you open the door, it sounds like you're walking into a beehive. It's just this buzzing sound because every single one of them have all these vuvuzelas being played. What happens now? Well, you could say, OK, we're just not going to allow vuvuzelas going forward for any more World Cup matches and things like that.
- 8.54** Or we can say, well, the audio that we actually recover by recording this in the broadcast, we can say, rather than just broadcasting that live, we're going to run it through a really quick filter and broadcast it with a two- or three-second delay. At the end of the day, does it really matter? If you're in, say, Panama and you're watching a match that's taking place in South Africa, does it really matter if the broadcast to you gets to you two or three seconds later? Not really. You're still going to be able to watch that broadcast. OK, fine.
- 8.55** You record the match. It's got maybe this annoying sound or there's background sound. What do you do? You run it through a filter, eliminate that sound, and then broadcast that. That way, you don't have that ambient sound anymore of the stadium. That's what's done. That's what's done. How is it performed? Well, here's how it's done. It's done with trigonometry, surprisingly. It's done with trigonometry. So here you go.
- 8.56** So it's done with what's called Fourier series. Now, you're going to have an entire course on Fourier series, what's called harmonic analysis. That's fine. But what I want to do is the following. I'm going to give you a function, and I'm going to make it a very

simple function. I'm going to make it a very simple function. I'm going to make it a sequence of taps. I'm going to make it a sequence of taps. What do I mean by that? I mean that, in a period of time, it turns on to a constant value, and then in another period of time, it goes down to a negative value.

- 8.57** Then it's just going to tap back and forth between those two values. That's all it's going to do. It's going to do something like this. It's going to be a 1 in between here. Once it hits here, it's going to jump down and go to negative 1. Then it's going to jump up, 1, negative 1, 1, negative 1. That's all it is. It's just a sequence of taps. You can imagine, it taps up to 1, it taps down to negative 1, taps up to 1, taps down to negative 1. However you want to think about this modeling something, whatever that is, it's totally up to you.
- 8.58** What is this going to do? I'm going to tell you it's going to go up to here, it's π , over here, it's minus π . Does it have to be π ? No. But that's OK. You can always scale things differently. I'm going to have minus 2π and so forth and so on. That's what it's going to do. If I wanted to write this function out, what is it? I'm going to say $f(t)$ is equal to what? I'm going to say it's equal to 1 in the range zero less than t less than π . Then I'm going to say, it's minus 1 in the range from π less than t less than 2π . When t is in between π and 2π , so π and 2π , my function is negative 1. When t is in between zero and π , my function is positive 1. That's it.
- 8.59** You can have more complicated things than this. That's completely fine. You could think about this as, I don't know, what's a good example for this? You could think about this maybe like alternating current in an electrical switch. You have the alternating current, it goes into this state, and then this state, this state, this state, this state, this state. It's just bouncing back and forth. Just this alternating current. That's an example. Choose your favorite. It's totally fine.
- 8.60** My point in all of this is that series plus trig can approximate, it can approximate this. I want to show you why that's the case. Well, not why that's the case, but I want to at least provide plausible evidence as to, that this is actually true, that this is actually true. Well, think about it for a second. What am I actually doing here? I'm claiming that a function that is literally horizontal bar, horizontal bar, horizontal bar, is just a sequence of taps up and down, can be approximated by things that do this. That seems impossible. That seems completely impossible. That sines and cosines can somehow approximate this particular gadget.
- 8.61** And the answer is, it's true. So let's see how that's true. Let's see how that's true. All right. So again, we're going to go back to our trusty Desmos. So we're over here. We're going to define our functions. I have this function, which is just this sequence of taps, plus 1, minus 1, plus 1, minus 1. I claim that I can take wavy functions, sines and cosines, and actually approximate this thing. How do I do that? Well, I turn to Desmos, because there's no way I'm going to do this by hand.

- 8.62** So, Desmos, what do I need to do? I need to define the function. I'm going to say F of X . That's how we define our function. F of X equals something. Now, let's say I just say, is equal to 1. If I say it's equal to 1, I just get a constant function equals 1. But I don't want it to be constant everywhere. I only want it to be constant 1 in between zero and π , 2π and 3π , 4π and 5π , and so on. Now what I can do is, with Desmos, I can say, look, put a set brace, put a set brace, and then I'm just going to define the chunks of X for which it's equal to 1. I'm only going to define a couple of chunks of this.
- 8.63** We're going to say zero is less than X , less than π . There's one chunk of it, comma. Now I'm going to have to put in the rest. Let me zoom out of this a little bit. I can give you a feeling for that. All right. Let's see for zero to π , and then I want what? I want 2π , I want 2π less than X less than 3π . There's the other branch. Then let's do one more. We want 4π to 5π . So 4π less than X less than 5π . Then I'll close the bracket. I'll close the brace.
- 8.64** What's actually happening? There are my three little chunks of the function. I've got this piece, this piece, and that piece. But now I need to define the negative pieces. I need to define the pieces that are negative 1 in between π and 2π , 3π and 4π , negative π and zero, and so on. OK, so let's do that. Let's define a new function down here. Let's call it G of X , and we're going to define it to be negative 1, but we don't want it to be negative 1 everywhere. We only want it to be negative 1 in this window, this window, that window, and so on.
- 8.65** Again, I do my set brace, and let's say minus π , less than X , less than zero. There's that piece. Then what do I want? π to 2π , so π less than X less than 2π , there's that piece. Then I want 3π to 4π , so 3π less than X less than 4π , and I have that. Now, if I want to adjust my window to allow this to—I don't really need the vertical space. I only need the space from maybe 2, from 2 up and maybe negative 2 down. I don't need that. I only need in between negative 2 and 2.
- 8.66** How do I do that? Well, you just come up here to this little wrench and click on that, and it'll tell you, look, for your X axis, what range do you want? And for your Y axis, what range do you want? Well, right now, I see my Y axis is saying go all the way down to negative 14. That's way too far. I don't need that. I really only need to go to negative 2. Then my Y axis, maybe I only go to 2. Now they're spaced out a little bit. You can see how it's rescaled my Y axis accordingly, and now I have these little dashes and these little dashes here. OK? All right.
- 8.67** Now, my X axis, let's see. How far did I go over? I went over to negative π to zero, so negative π is 3.14159, so maybe I go negative 4. OK, so my X axis, maybe I should go negative 4. And my Y axis, I go all the way out to what? We went all the way out to 5π . π is 3, 5 times π is like 15 and change. Maybe 17. Let's say I go out to 17. We're just trying to fit it in a nice little window. There it is. There's my function. I've got these three little tick marks down here, and I've got these three little tick marks up here. It's in a nice little viewing window, and I can go from there.

- 8.68** Now, here's the part that I have to give you on blind faith. I have to give you this on blind faith. This is where you would take a pre-calculus course, a calculus course, things along that line, to actually be able to compute the object that I'm going to give you. To compute the object that I'm going to give you. The object I'm going to give you has a name. It's called a Fourier series. That's how you say this. Fourier series. Fourier was a French mathematician.
- 8.69** Allegedly, he came up with this when he was actually fighting in a war in France, which I'm like, that seems kind of crazy. You're doing, in the middle of a war, and you're doing math. That seems odd. But Fourier series, it says the following thing. It says that this function, this F of T , as I extend this out periodically, really what I'm saying is that this function is 2π periodic. I'm saying F of T plus 2π is equal to F of T . I can see that every 2π , what happens? I see this little window, and then the next 2π , I see the same window. The next 2π , I see the same window. It's just like sine and cosine, it's 2π periodic.
- 8.70** The function that I get, just like my series, let's go back to our answer with series. How do I input series in Desmos? This is what I love about this calculator. I want to compute those sums. Let's go back to the sum that we had at the beginning. We had the sum K equals 1 to 1,000 of K . We said, oh, that should be 1,000 times 1,001 all over 2. That's what we said it should be. If you just type S-U-M, it immediately populates sigma. Now, Desmos doesn't like the fact that you have a choice of what you're going to label your counter. We've been calling it K . Desmos doesn't like that. Desmos says, no, I'm going to call it N . You don't get a say on that.
- 8.71** Well, that's fine. I just type S-U-M, and it populates sigma immediately. I'm willing to cede that ground to Desmos, that they're allowed to choose what N is. That's fine. If I go from N equals 1 up to 1,000, and I tab forward and I just hit N , it says, oh, it's 500,500. Tells me exactly what the answer is. I go back down and I say, oh, but I thought the answer was 1,000 times 1,001, divided by 2. It's like, yes, we're saying the same thing. 1,000 times 1,001 over 2.
- 8.72** Here's the point. Here's what Fourier series will do, and I think this is a really beautiful way to cap off trigonometry. I want to look at the sum. I'm going to start it from 1, and I'm going to go up to A , where I get to control A . Well, let's say we call it M , where I get to control M , whatever that is. Now, I'm going to give you an expression that's going to seem like total magic. Again, you would do this in a calculus course. But I'm going to give you the following expression. It's 4 divided by $2N$ minus 1 times π in the denominator. Then our trusty trig function, sine of X , comes in. Again, a $2N$ minus 1 times X .
- 8.73** It's going to ask me if I want to—again, Desmos is going to say this. You've defined a variable here, M , that I don't know. It's going to ask me, do you want M to be a slider? Do you want it to be something that you can control? It's a parameter that you get to control. The answer is, well, yeah, of course I do. I'm going to let it be what it is. Now,

this is a sum. N goes from 1 up to M , which means it starts at N equals 1, it goes to N equals 2, it goes to N equals 3. It doesn't make sense for this to just be a continuous slider where M is, say, the square root of 7. That doesn't make sense.

- 8.74** What I'm going to do is, I'm going to click on this little negative 10 here, and it's going to say, where do you want it to go? I'm going to say, I want it to start at M is 1, and I want it to go to, say, 100, and I want it to step by ones. Now it repopulates that slider. It says it starts from 1, it goes up to 100, and now it's ready to roll. It's ready to roll. Now, if I look at this, I say this is a pretty awful approximation to my function, which is just these little taps. Do we agree? I'm saying, hey, let's approximate red and blue by this black wavy curve. It's like it's not even close. Do we agree? It's not even close. OK?
- 8.75** But what happens when I start moving this forward? That's M is 2. What happens when I go to M is 3, or 4, or 5, or 8, or if I just let it run. What does it do? Completely approximates these things. If I run it all the way to the end, yes, I have some weirdness here. I have, around here, I have these weird squiggles associated to it. But more or less in here, it's doing a pretty fantastic job. It's literally just adding sines together. That's all it's doing. That's all it's doing.
- 8.76** It's starting from something that looks like this, which is a terrible approximation to it, with M is equal to 1. By the time M is equal to 9, it's pretty clear that this thing is actually doing a pretty good job of approximating these. If I let it run, you can see by the time I get to M is equal to 100, M is equal to 99 in this case. Look at how close that is to approximating these particular things. That's it. That's it. They're trig functions, they're sines and cosines, they come from our little right triangles, our SOHCAHTOA, our unit circle, all of that, and it's coming from sines and cosines.
- 8.77** This is the beginning of what's called harmonic analysis, which has everything to do with noise-cancelling headphones and things like that. Being able to take a given function, whatever it is, this periodic function, which is ones and minus ones, up and down and up and down, whatever that's representing, this is a very simple example, just constant, constant, constant, constant. What if I had put a sound wave in here? Could I do the same thing? The answer is yes.
- 8.78** This has application to security as well, which is concerning a little bit. For example, what is this doing? This is approximating a very simple wave. Constant, constant, constant, constant, up, down, up, down. Just approximating that. Sound waves are, again, waves. Maybe I could do this to approximate waves. Or maybe I could use this to somehow reverse-engineer what somebody says, or make it sound like somebody said something if they didn't actually say it.
- 8.79** Maybe I could use this to detect a fake. Somebody creates an audio signal and says, "Hey, this is somebody saying this particular thing," and I can analyze the signal and say, "Actually, it's not, because here's the mathematics that proves otherwise." It's a very valuable thing, and it all has its genesis in trigonometry.

